



e-Hospital Business Documents

Robotic Surgery – Final Report

BMG 5109

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Abstract

The e-Hospital system is a unified platform that aims to optimize the delivery of healthcare services through the use of Artificial Intelligence (AI) and a multi-portal infrastructure. This report compiles the e-Hospital website's available resources, features, functionalities, and project history into a document that can help future developers and designers. The e-Hospital system has a total of eight portals: Admin, Patient, Doctor, Hospital, Clinical Staff, Pharmaceutical Office, E-Pharmacy, and Pharmacy Admin. Each portal is designed for a specific type of user. The system utilizes React.js for frontend development, allowing users to interact with the system effortlessly. Insights into the system's robust architecture are contained in the backend and database documentation. Some of the salient features include automated appointment scheduling, Patient records management, AI-powered tools (such as chatbots and calendar organizers for doctors), and API integrations for live data access. The efficiency of the Doctor Portal is remarkable; it is fully customizable and saves users time with various filtering options on dashboards. This document describes the automation of existing functions to facilitate subsequent developments in redesigning processes and timelines.

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1. Introduction

This document serves as an e-Hospital business document, compiling all resources, features, functionalities, and past projects from the e-Hospital website into a single file to facilitate information retrieval. By doing so, all relevant information would be gathered, which can help future designers understand what is currently available on the website, what still needs to be completed, and what has not been addressed yet. The features and functionalities of all portals, dashboards, and sections will be thoroughly documented.

2. Background

The e-Hospital is a rapidly growing medical website with the goal of maximizing efficiency and providing the best support to patients and healthcare professionals. While some other websites currently exist and are being actively used, the eHospital works to integrate AI in many aspects of its design, whether it be to help organize a doctor's calendar or utilize a chatbot to reduce the time and effort doctors have to invest in finding information.

The e-Hospital utilizes information from multiple APIs and software to access all the necessary data. It is also set up with separate logins for patients and healthcare workers. There are eight different portals listed in the order they appear on the e-Hospital platform: Admin, Patient, Doctor, Hospital, Clinical Staff, Pharmaceutical Office, E-Pharmacy, and Pharmacy Admin.

To facilitate the understanding of the different parts of the e-Hospital, this document will be divided into frontend documentation, backend documentation, and database documentation. All the information included is taken from past final reports submitted by previous students and observations made by logging into the various e-Hospital portals.

3. Frontend Documentation

The frontend is the user interface portion of the e-Hospital website. Many teams worked tirelessly to create and improve the visual and functional requirements of the website. It is essential to note that React.js is the coding language used to create frontend code that is compatible with the rest of the site.

Each portal has multiple sections within them that will be discussed in detail. Some of the more important portals, such as the doctor and patient portals, were worked on extensively and, therefore, have more sections to discuss. The majority of this section was completed by reviewing the e-Hospital website itself and noting all visible information on the various portals. Most of the information was later verified using past reports.

3.1 Doctor Portal

This portal is dedicated to doctors, and all the projects on the website are designed to enhance their lives and ensure that their time is used efficiently. This can be achieved by automating the task or simply assisting them with it. Below are the six main sections of the doctor portal, including all necessary features and functionalities.

3.1.1 Dashboard [1]

The doctor dashboard features several important elements, starting with the doctor's greeting, which displays the doctor's name at the top of the page in a large, dark blue rectangle. It is important to note that this section is also not editable.

The next thing shown would be the patient appointments. Besides the patient appointments, there is a button that allows the doctor to select today, this week, or this month. There is also a button that allows the doctor to filter the results by patient ID, name, age, visit type, scheduled time, and status. This feature includes an arrow with a drop-down button, making it easier for the doctor to find a patient or view their appointments.

Some other features include the diagnose section, located at the top right of the doctor's dashboard, which is editable. When clicked, the doctor can search for the patient using their

phone number, which will then display the patient's basic information, including the patient ID, name, blood group, age, phone number, email address, and home address. At the bottom of that page, it would display patient-specific services based on diagnosis, such as skin cancer diagnostic, brain diagnostic, kidney stone diagnostic, and many more. Once a specific diagnostic is selected, the corresponding medical image, record date, action, and diagnosis appear.

Another feature is the general services, located beside the diagnosis button, which displays health services such as doctor services and patient services. Under doctor services, there are EHR, managing appointments, referral network, integration with diagnostic labs, emergency consultation services, CME courses, billing and invoicing, feedback, and reviews. For patient services, there is medical chatbot assistance, a 24/7 medical helpline, virtual services, and the option to find a hospital or pharmacist. None of these services currently display any information when clicked on.

The next feature is the Radial Health Graph, which displays a circular diagram illustrating the various measurements (BP, stress level, etc.) and activities of a patient. The green section illustrates the phases, measurement, and activities. The red dots (above or below normal) represent critical or abnormal values for the health metrics, such as elevated stress or blood pressure (BP), whereas the black dots indicate normal or expected values within a healthy range. One can hover over the dots, and they provide further information on things the patient is doing well and ways they can improve.

Next is the calendar, where the doctor can select a specific month, year, and date, and this is an editable section. The upcoming tasks are completed once the doctor selects a specific date from the calendar. To create a new task, the doctor must select the date and then create the task. When adding an event, it displays the category, doctor, start date and time, end date and time, and a description for the event. The doctor can then click 'OK' to save it to the calendar. The notification button displays all the notifications that the doctor has received; the blue ones indicate all notifications, and the red ones indicate unread notifications. It also displays new features added to the platform, messages from patients and doctors, orders placed, meeting reminders, and new tasks assigned, among other updates. If "View All" is clicked, it goes to a separate page, which is the e-Hospital Smart Digital Medicine Solutions. That has a lot of

information, including Revenue Generated, Recent Transactions, Campaign, Sales Quantity, and Geography-Based Traffic

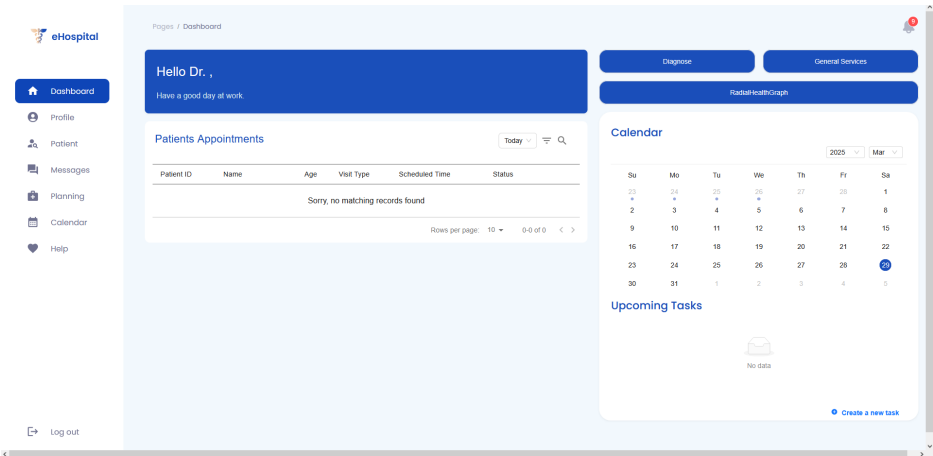


Figure 1. Frontend Doctor Portal - Dashboard Section - of the e-Hospital

3.1.2 Profile [1]

This section is straightforward, comprising only three sections: profile, signature, and security. For the profile, there are editable fields for personal information, including First Name, Last Name, Email Address, Birthday, Mobile Number, Address Line 1, Address Line 2, City, Zip, and Specialization. The signature section is simply an automatic first and last name signature, accompanied by a “Save Settings” button. Lastly, the security section is designed to facilitate password updates, allowing users to input their current password, new password, and confirm the new password. It also has a timeout duration, which allows you to select a time after which the site will log you out (5 minutes, 10 minutes, 15 minutes, 30 minutes, or 1 hour). Additionally, it features a “Save Settings” button.

Doctor Personal Information

Profile

Signature

Security

First Name: Adam

Last Name: Wayne

Email Address: doc1@test.com

birthday: 2023-04-06

Mobile: 6131231001

Address Line 1: 75 Laurier Ave. E

Address Line 2: Room 1101

City: Ottawa

Zip: K1N 6N5

Specialization: not a doctor

Medical Identification:

Medical License Number: 123456789-ABC

Billing Number:

Save Settings

Figure 2. Frontend Doctor Portal - Profile Section - of the e-Hospital

3.1.3 Patient [2]

There are three main sections in the patient tab: patient information, the dashboard, and the filter table with a search function. Starting with the patient information which shows the total number of patients as well as the Patient’s ID, Name, Age, Gender (Male or Female), Status (On treatment, recovered, inactive, or awaiting surgery), Last appointment, Next appointment, and options (which takes us to patient file when pressed).

Total Patients (164)

Dashboards

ID	Name	Age	Gender	Status	Last Appointment	Next Appointment	Options
382	3333 333 333	33	Male	Awaiting surgery	/	/	...
306	ab ab ab	18	male	On treatment	/	/	...
224	Abigail Ethan Stewart	30	Female	Recovered	2023-10-28	/	...
294	Aeg Hsh Isis	23	Male	Recovered	/	/	...
168	Aiden Juan Green	48	Male	Inactive	2023-10-21	/	...
157	Aiden Nicholas Baker	46	Male	On treatment	2023-10-27	/	...
143	Alexander Michael Lewis	52	Female	Awaiting surgery	2023-10-30	/	...
153	Alexander Thomas Thompson	62	Male	Awaiting surgery	2023-10-26	/	...
177	Amelia Jerry Green	42	Male	On treatment	2023-10-22	/	...
216	Aria Ethan Bell	46	Female	On treatment	2023-10-27	/	...

Rows per page: 101-10 of 164

Figure 3. Frontend Doctor Portal - Patient Section - of the e-Hospital

For the dashboards, there are four subsections: General, Lifestyles, Personalized, and Lung Cancer. These subsections are all unclickable.

General

- Asthma, kidney disease, skin cancer vs. heart disease plot
- Diabetic vs heart disease
- Heatmap for heart disease vs. numerical features
- Stroke vs. heart disease

Lifestyle

- Heart disease vs. BMI
- Smoking vs. heart disease
- Alcohol vs. heart disease
- Physical activity vs. sleep time for heart disease

- Sleep time and heart disease
- Heart disease vs. general health
- Physical activity vs. heart disease

Personalized

- Heart disease vs. age
- Heart disease vs. BMI
- Heart disease vs. sex

Lung Cancer

- Chronic disease and allergy vs. lung cancer
- Symptoms vs. lung cancer
- Gender vs lung cancer
- Smoking vs age and gender
- Smoking vs. lung cancer
- Rates of symptoms vs. lung cancer

The last section in the patient section is the filter table where there is a button that the doctor can filter out using patient ID, Name, age, visit type, scheduled time, status where there is an arrow with a drop button so that it would make it easier for the doctor to find the patient. There is also the search bar button that the doctor can use to search for a patient's name and information.

3.1.4 Messages

This section is straightforward, consisting of three simple subsections. Firstly, Send New Message allows doctors to send new messages and search through their message history. They can send messages to patients and clinic staff. It contains features such as a subject line and a section to enter text, along with all the typical formatting elements like font, font sizes, bullet

points, etc. There is a send button to allow messages to be sent, as well as a back button to return to the previous page. The Messages section lists all the messages, allowing the doctor to search for specific messages or simply click on a message they want. Lastly, the Message display shows the message that was clicked, allowing the doctor to view and read it.

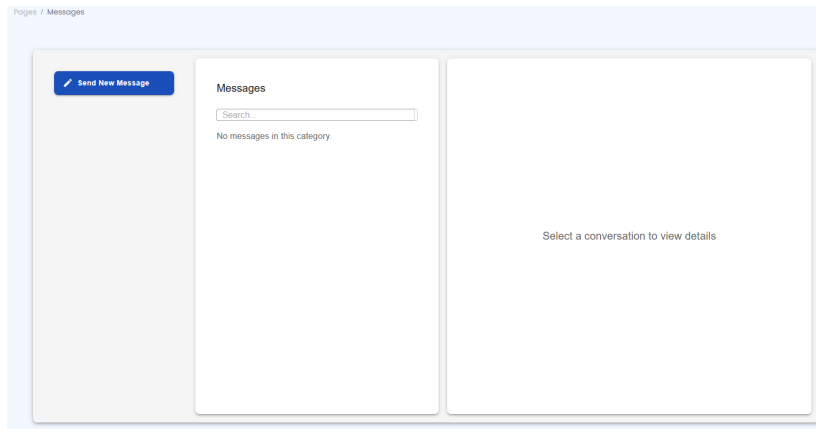


Figure 4. Frontend Doctor Portal - Messages Section - of the e-Hospital

3.1.5 Planning

There are seven types of plans in this section; some are interrelated, while others are standalone planning projects.

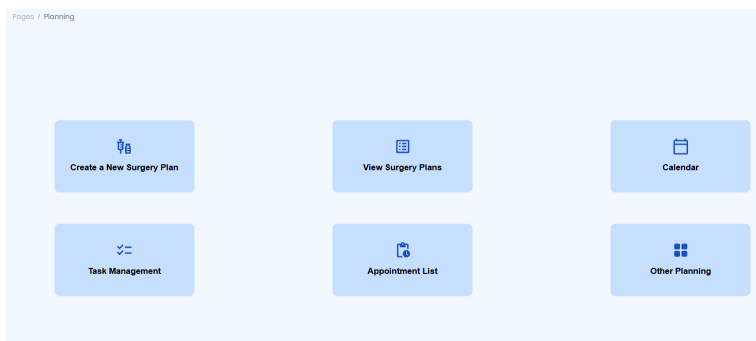


Figure 5. Frontend Doctor Portal - Planning Section - of the e-Hospital

The first is to create a New Surgery Plan. When creating a surgery plan, there is a search and selection of the patient, surgery type, surgery date, pre-surgery consultation details, risk assessment details, and post-operative care plan. Then, the doctor can save the surgery plan. At the bottom right, there is a chat subtitle button.

Next is the View Surgery Plans, which displays all the surgery plans, each one showing the following information: surgery type, date of surgery, patient ID, and view details. When clicking on view with the associated patient, it shows surgery type, patient ID, name, age, gender, weight, consultation details, risk assessment details, post-operative care plan, surgery date, plan created on date and time

The Calendar is where the doctor can switch between different views, such as weeks and months, and includes options for adding new events. The calendar includes tabs for various types of activities, such as surgery, general consultation, lab testing, task, W/P state, and pending approval, each corresponding to a different color. These activities are represented by different colors to help differentiate them, making it easier for the doctor to organize and identify each one quickly

In Time Management, a doctor can select to see task for either today or one week. The doctor can add a task. Doctors can edit and delete tasks

The Appointment List displays all available, approved, pending approval, and unbooked appointments. Requesting an appointment requires the approval of a doctor. Upon clicking on the appointment, it displays requests with the patient's name, the status of the appointment (approved or pending), category, and description, such as a dental checkup. Connect to the calendar where you can approve bookings. Can edit appointments. We can choose the date we want to review.

Other Planning currently shows nothing and is not editable.

In Surgical Planning, the doctor can add information, including patient name, patient age, gender, medical history, diagnostics, upload diagnostic files, surgery type (ex: cardiothoracic

surgery, plastic surgery, neurosurgery, etc.), and procedure name; then the doctor can click on generate plan.

3.1.6 Calendar [1]

The calendar features a Week option, which displays a weekly calendar that shows all appointments, tasks, and other events. The calendar also features a Month option, which displays a monthly calendar that shows all appointments, tasks, and other events. The Date Selection feature allows for precise date selection, narrowing it down to the day, week, month, or year. Lastly, Add new event allows the doctor to add an event and fill out the following sections:

- *Category*
 - o *Available Slot*
 - o *Taks*
- *Doctor*
 - o *Enter the doctor's name*
 - *Start*
 - o *Start date and time of the slot or task*
 - *End*
 - o *End date and time of the slot or task*
 - *Description*
- o *Provide a description of the task or slot to help the doctor understand the appointment.*
 - *Click ok to save and add the event or cancel to exit the popup page.*

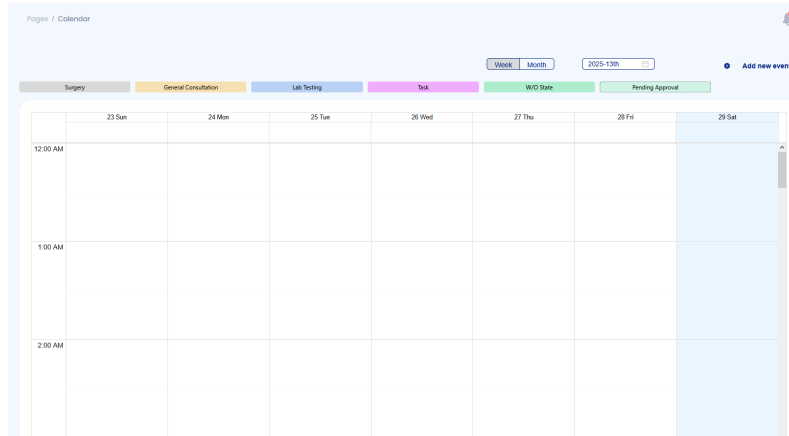


Figure 6. Frontend Doctor Portal - Calendar Section - of the e-Hospital

3.2 Patient Portal

This section is similar in structure to the doctor portal, with functionalities that are more patient-focused. It is designed to improve and facilitate patient lives by offering more information, patient records, and test results. Below are the seven main sections of the patient portal, including all necessary features and functionalities

3.2.1 Dashboard [1]

As shown in the Doctor portal's dashboard section, it is obvious that it contains many important and helpful subsections for its users. Starting with the Patient greeting, it shows a greeting for the patient, "hello," in a large blue rectangle at the top of the page. Next, we have the Current medication section, which is located under the greeting; it shows a list of medications that the patient is taking. Showing the number of pills should be taken with the duration. (the number of pills and time). The Health overview is where the patient can select the measurement they want to see change over time: weight, BP, or glucose. This feature on the patient e-hospital platform allows the patient/user to track and view changes in weight, BP, and glucose levels over time. It provides a visual representation that can help users monitor their health. The Live text chat opens a window where the patient can send a message (it is editable). The Video call section opens a new window where the user can have a virtual meeting with a specialist. Next, the

medical chatbot opens a new window where it provides the user with a virtual assistant to help with medical inquiries, access to medical records, or basic health advice. The patient can type or start a speech regarding the inquiry.

Following that, the Medical records take the patient to the health report, where they can monitor and track glucose, blood pressure, and weight by visualizing the trend, allowing them to better understand their health and concerns with their healthcare providers. The patient can view their medical health records including MRI (patient can view spine, brain), CT scan (patient can view abdomen, brain, breast), X-rays (chest, feet, lung, skin) blood test (patient can view their blood results such as totalbilirubin, alkine phsophates, etc.) signal analysis.

The Calendar allows the patient to select a month, year, or specific dates on the calendar. The Upcoming Tasks is the section that appears when a specific date is selected; it would show the tasks or appointments that the patient has. Lastly, the most recent medical records allow patients to view their most recent medical records, such as x-rays, instead of searching for them in the health report section; they can be easily found under the dashboard.

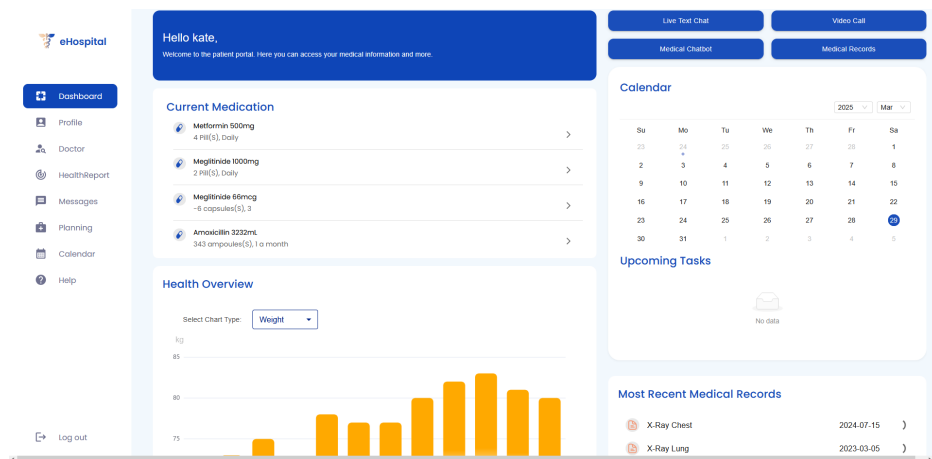


Figure 7. Frontend Patient Portal - Dashboard Section - of the e-Hospital

3.2.2 Profile [3]

The patient portal has two major subsections in the profile section. The first is the profile itself, which has editable personal information. This includes details such as First Name, Last Name, Email Address, Birthday, Mobile Number, Address Line 1, Address Line 2, City, and Zip Code. It also features an editable Healthcare Information section, which includes details such as Healthcare Card Number, Patient Name, and Age, along with a functional “Save Settings” button. The second is security, where you can change the password by entering the Current Password, New Password, Confirm New Password, and clicking the “Change Password” button.

Personal Profile

Profile

Security

First Name: kate

Last Name: Evans

Email Address: pat1@test.com

DOB:

2023-02-19

Mobile: 6137778888

Address Line 1: 5670 William McEwen Drive

Address Line 2: Apt. 105

City: Ottawa

Zip: K0A 2E0

Healthcare Information:

Healthcare Card Number: 123456789-ABC

Adam Wayne, 58

Save

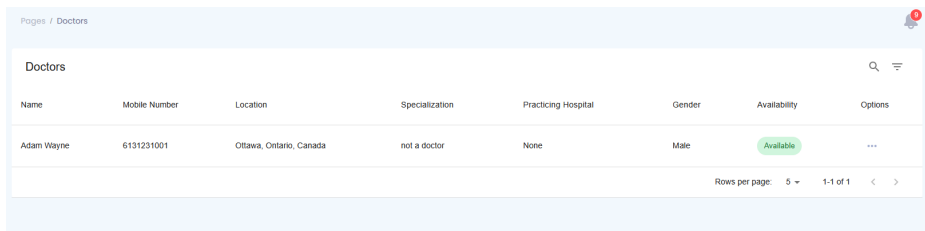
Figure 8. Frontend Patient Portal - Profile Section - of the e-Hospital

3.2.3 Doctor

The doctor section is important as it allows the patient to get access to the doctor’s Name, Mobile Number, Location, Specialization, Practicing Hospital, Gender, Availability, and Options

including Messages (Displays a message saying “you clicked the Message button”), Chatbot (Displays a message saying “you clicked the Chatbot button”), Video Call (Displays a message saying “you clicked the Video Call button”), and Book Appointment (Displays a message saying “you clicked the Book Appointment button”). You can select the number of rows per page you want to display (5, 10, or 20). You have arrows taking you to the next or previous page as well as the number of rows (doctors) available and total pages

There is also a Filter Table, which features a button that allows patients to filter by Location, Specialization, Practicing Hospital, Gender, and Availability. This feature includes an arrow with a drop-down button, making it easier for patients to find a doctor. There is also a reset button in the filtering section. Additionally, there is a search button that allows the doctor to search for a patient's name or information.



Name	Mobile Number	Location	Specialization	Practicing Hospital	Gender	Availability	Options
Adam Wayne	6131231001	Ottawa, Ontario, Canada	not a doctor	None	Male	Available	...

Rows per page: 5 1-1 of 1 < >

Figure 9. Frontend Patient Portal - Doctor Section - of the e-Hospital

3.2.4 HealthReport [3]

One of the two main sections here is the Health Overview, where the patient can select the measurement they want to see change over time: weight, blood pressure (BP), or glucose. This feature on the patient e-hospital platform enables patients/users to track and view changes in weight, blood pressure (BP), and glucose levels over time. It provides a visual representation that can help users monitor their health.

The second one is the Medical Records where the patient can view their medical health records including MRI (patient can view spine, brain), CT scan (patient can view abdomen,

brain, breast), X-rays (chest, feet, lung, skin) blood test (patient can view their blood results such as totalbilirubin, alkine phsophates, etc.) signal analysis.

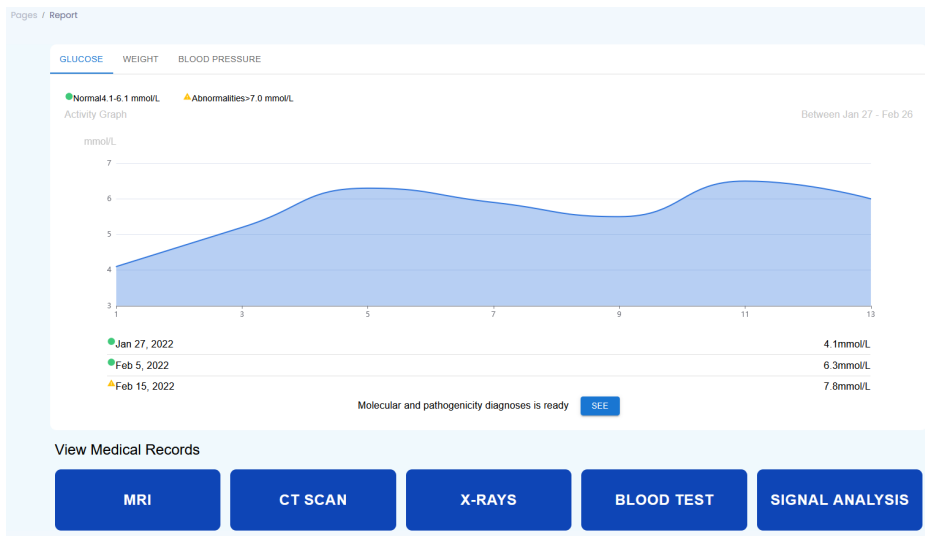


Figure 10. Frontend Patient Portal - HealthReport Section - of the e-Hospital

3.2.5 Messages [3]

This section is straightforward, consisting of three simple subsections. Firstly, Send New Message allows patients to send new messages and search through their message history. They can send messages to doctors and clinic staff. It contains features such as a subject line and a section to enter text, along with all the typical formatting elements like font, font sizes, bullet points, etc. There is a send button to allow messages to be sent, as well as a back button to return to the previous page. The Messages section lists all the messages, allowing the patient to search for specific messages or simply click on a message they want. Lastly, the Message display shows the message that was clicked, allowing the patient to view and read it.

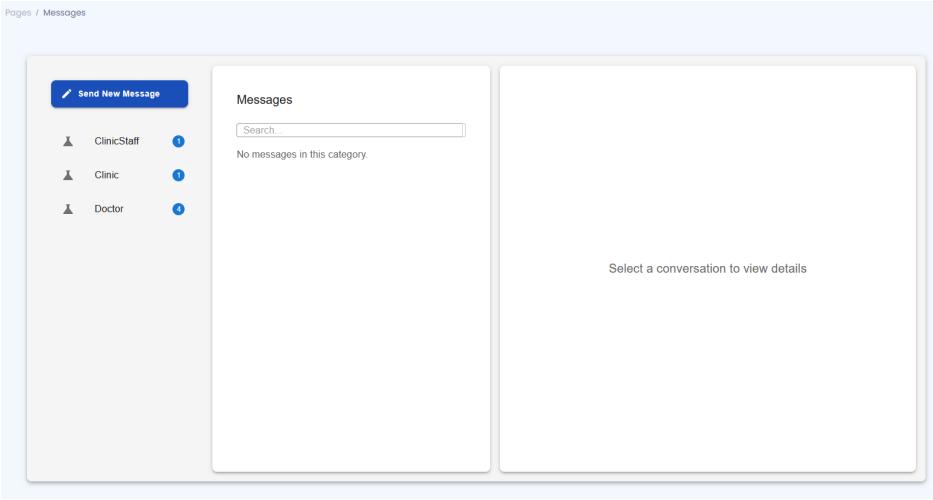


Figure 11. Frontend Patient Portal - Messages Section - of the e-Hospital

3.2.6 Planning

There are five planning methods, some more advanced and better developed than others. The five sections are My surgery plan, Book an appointment, My appointments, My referrals, and Prescription refill.



Figure 12. Frontend Patient Portal - Planning Section - of the e-Hospital

The My surgery plan Displays a table including Surgery Type, Date of Surgery (which can be organized from most or least recent), Patient ID, View Details). You can select how many rows per page you want to display (25, 50, or 100). You have arrows taking you to the next or previous page as well as the number of rows (patients) available and total pages.

For the Book an appointment section is linked to calendar and takes you there once you click on it.

The My appointments is linked to calendar and takes you there once you click on it. My referrals are broken down to Create Referrals and Incoming Referrals. You can select how many rows per page you want to display (25, 50, or 100). You have arrows taking you to the next or previous page as well as the number of rows (patients) available and total pages.

- Created Referrals:

- ID (Sort by DESC, Unsort, Filter, Hide column, Manage columns)
- To Patient (Sort by ASC, Sort by DESC, Filter, Hide column, Manage columns)
- Patient (Sort by ASC, Sort by DESC, Filter, Hide column, Manage columns)
- Created On (Sort by ASC, Sort by DESC, Filter, Hide column, Manage columns)

- Incoming Referrals:

- ID (Sort by ASC, Sort by DESC, Filter, Hide column, Manage columns)
- Patient (Sort by ASC, Sort by DESC, Filter, Hide column, Manage columns)
- From Patient (Sort by ASC, Sort by DESC, Filter, Hide column, Manage columns)
- Created On (Sort by ASC, Sort by DESC, Filter, Hide column, Manage columns)

Lastly, the Prescription refill shows a table containing the Prescription ID, Doctor First Name, Doctor Last Name, Doctor's Phone, Doctor Office Address, Prescription, and Prescription Create Time)

3.2.7 Calendar [3]

The calendar features a Week option, which displays a weekly calendar that shows all appointments, tasks, and other events. The calendar also features a Month option, which displays a monthly calendar that shows all appointments, tasks, and other events. The Date Selection feature allows for precise date selection, narrowing it down to the day, week, month, or year.

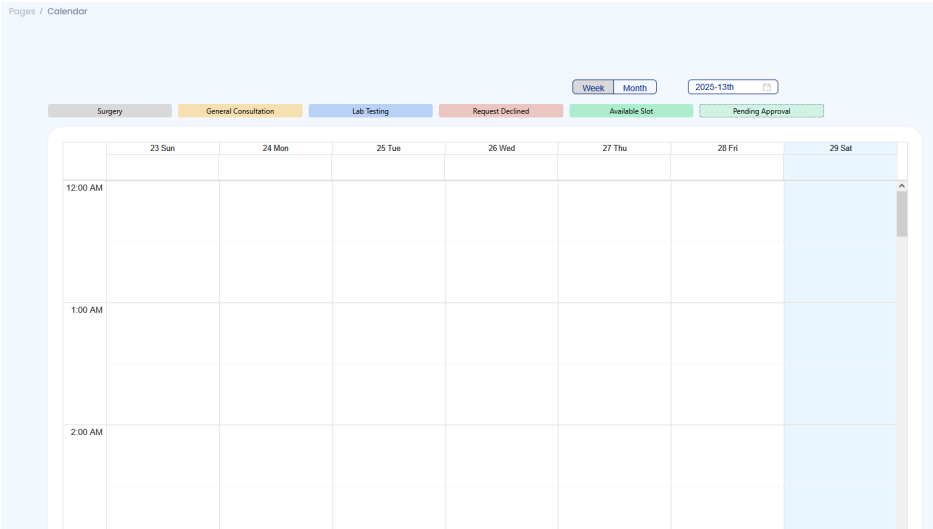


Figure 13. Frontend Patient Portal - Calendar Section - of the e-Hospital

3.3 Pharmaceutical Office Portal

This portal has four main sections: Dashboard, Clinical Trial, Messages, and Management. The portal focuses on medication-based clinical trials, where new prescriptions are tested, and the patient's progress is tracked to ensure the medication's effectiveness.

3.3.1 Dashboard

This subsection is filled with valuable information relating to clinical trials. It begins with a welcoming message in a blue rectangular box, similar to all previous dashboards. Below are three boxes indicating the number of clinical trials currently running, as well as the number of doctors and patients involved. Each of those boxes directs you to the e-Hospital Smart Digital Medicine Solutions, where you can view various information, including alerts, sales obtained, geographic-based traffic, and more. The boxes also show the percentage of active use each of the three categories had in the past seven days. Next, a shortcut to the management tool is visible at the top right corner of the page, where we can see a looping visual demonstrating the auditactions actions, inviteactions actions, and the applyactions actions. Another important aspect of this subsection is the source of patients for clinical trials, which is illustrated by a diagram showing the number of patients invited to participate in the trial versus those who applied to participate. Lastly, a clinical trial list is presented, clearly summarizing the trials currently underway, along with their status (under review, ongoing, complete, or rejected). Like many lists discussed previously, the user can easily search for a specific clinical trial, filter by trial status, and use the three dots, which serve as a shortcut to the clinical trial subsection.

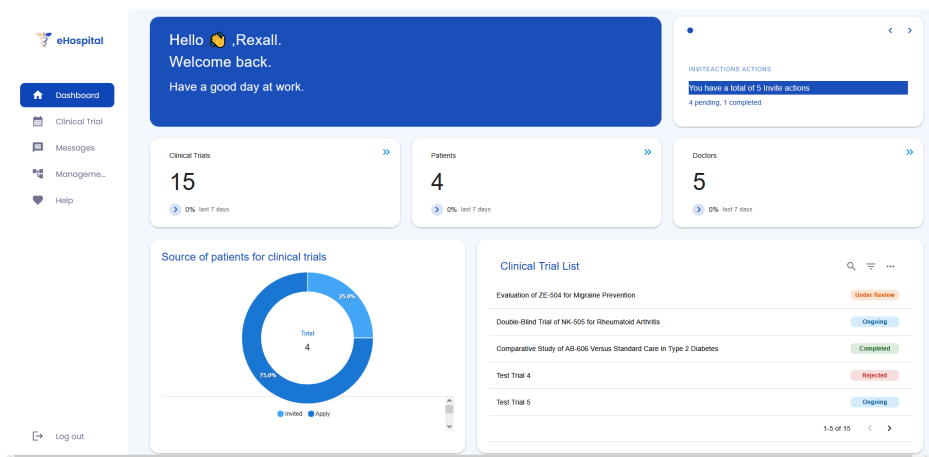


Figure 14. Frontend Pharmaceutical Office Portal - Dashboard Section - of the e-Hospital

3.3.2 Clinical Trial

The main page in this subsection displays a list of clinical trials, including information such as the trial name, ID, conditions, phase (I to IV), type (interventional or observational), location, investigator, sponsor, ethics committee, and status. Once you click on a specific clinical trial, we see information on this trial, including the official title, brief summary, detailed description, start date, detailed information, management, and matched patients.

You can search for a specific trial and filter the list based on each of the different columns of information. It is possible to navigate between pages and select the number of rows to be displayed per page (5 or 10). The last feature in this section is the plus button, which allows you to add a new clinical trial to the list. The following information must be entered to create the trial. We would need to include the contact information, including the first, middle, and last names, as well as a phone number and email address. Next, the basic information, such as the trial name, ID, location, and official title, would need to be entered. A brief summary and detailed description of the trial will be necessary. Trial details are needed such as primary purpose, trial phase, study type, etc. Inclusion criteria require information like related conditions, pathology, gender, and age range. Lastly, the exclusion criteria, including BMI, diseases, surgeries, prior medications, and pregnancy, would be filled out prior to clicking the 'Create Trials' button to finalize the process.

Pages / Clinical Trial

Clinical Trial List

Name	Id	Conditions	Phase	Type	Location	Investigator	Sponsor	Ethics	Status
Evaluation of ZE-504 for Migraine Prevention	1	Hypertension	Phase III	Interventional	New York, USA	Dr. John Smith	XYZ Pharma	Approved by XYZ Ethics Committee	Under Review
Double-Blind Trial of NK-505 for Rheumatoid Arthritis	2	Hypertension	Phase II	Interventional	Los Angeles, USA	Dr. Emily Brown	ABC Biotech	Approved by ABC Ethics Committee	Ongoing
Comparative Study of AB-606 Versus Standard Care in Type 2 Diabetes	3	Type 2 Diabetes	Phase III	Observational	Chicago, USA	Dr. Michael Johnson	DEF Health	Approved by DEF Ethics Committee	Completed
Test Trial 4	4	Type 2 Diabetes	Phase I	Interventional	Boston, USA	Dr. Sarah Williams	GHI Inc.	Approved by GHI Ethics Committee	Rejected
Test Trial 5	5	Alzheimer's Disease	Phase IV	Observational	Houston, USA	Dr. Laura White	JKL Research	Approved by JKL Ethics Committee	Ongoing

Rows per page: 5 1-5 of 15

Figure 15. Frontend Pharmaceutical Office Portal - Clinical Trial Section - of the e-Hospital

3.3.3 Messages

This section is straightforward, consisting of three simple subsections. Firstly, Send New Message allows trial coordinators to send new messages and search through their message history. They can send messages to doctors. It contains features such as a subject line and a section to enter text, along with all the typical formatting elements, including font, font sizes, bullet points, and more. There is a send button to allow messages to be sent, as well as a back button to return to the previous page. The Messages section lists all messages, allowing staff to search for specific messages or simply click on the one they want. Lastly, the Message display shows the message that was clicked, allowing the patient to view and read it.

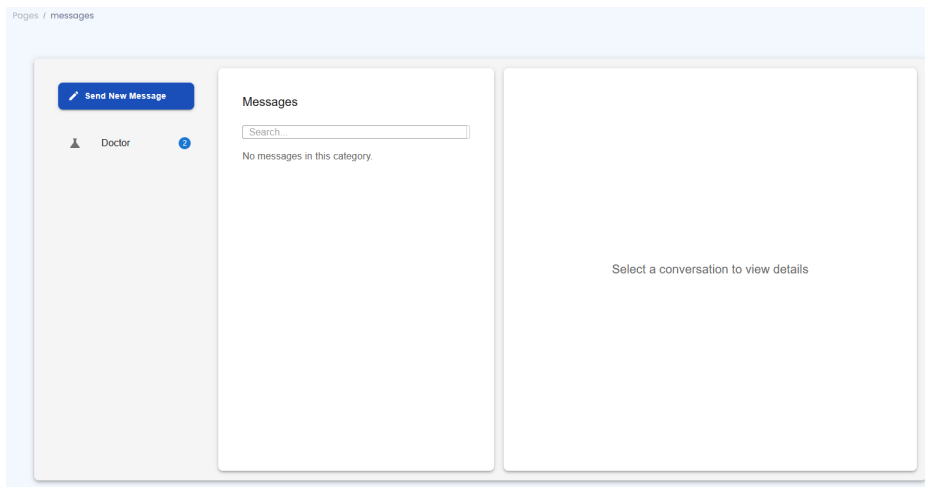


Figure 16. Frontend Pharmaceutical Office Portal - Messages Section - of the e-Hospital

3.3.4 Management

The overall page displays three boxes at the top, labeled Audit, Invitations, and Applications. Each section contains the total number of its relevant action type, as well as the number of unread or unprocessed trials. Below that, you can view a list of trials, which includes details such as the trial name, action type, completion status, and time.

We can obtain detailed information about each action type, along with its corresponding trials, starting with the audit, which displays a list of trials containing information such as trial name, submission date, reviewer, webstaff read status, and audit status. If you choose to expand on any of the specific trials it provides you with the notes the clinical trial staff made as well as the webstaff notes. Moving on to the invitations action, which contains the trial name, recipient name, invitation date, recipient read status, and response on their list of clinical trials. Once you expand the information on a selected clinical trial, it displays both your note and the recipient’s note. Lastly, the application action displays the trial name, applicant name, application date, your read status, and your response, along with the applicant’s note and your note once you expand a specific trial.

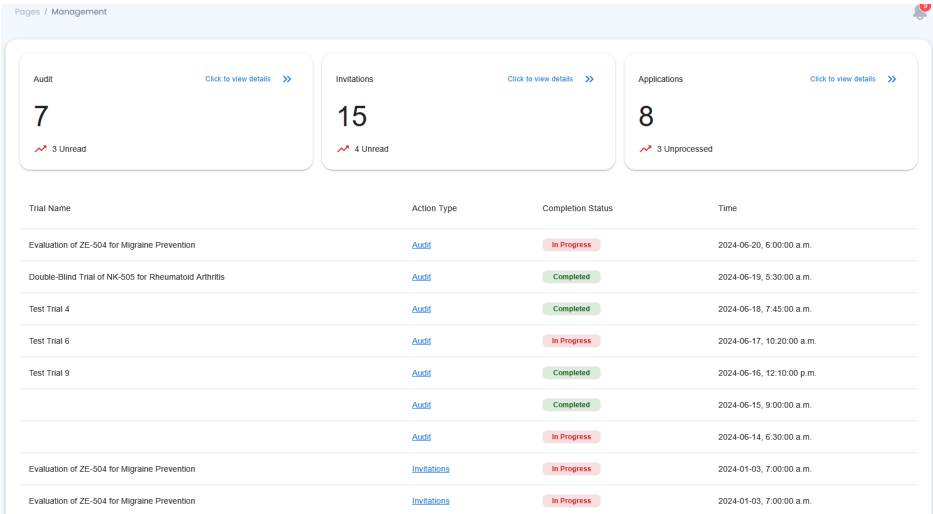


Figure 17. Frontend Pharmaceutical Office Portal - Management Section - of the e-Hospital

3.4 E-Pharmacy Portal

Unfortunately, no login information was provided for this section.

3.5 Admin Portal

This section is an administrative portal that contains subsections like the dashboard, profile, messages, and management. This portal requires further work to provide a functional admin portal.

3.5.1 Dashboard

This section contains four types of information. The first is the welcome, blue, rectangular box that welcomes the user. Next is a list of 207 users, 12 new users, 127 patients, and 6 providers. This number, of course, is changeable based on the accurate number of people per section. Next, we see two types of graphs. The first shows a circular graph illustrating the user base, which includes the percentage of patients, doctors, and clinical staff. The second graph displays a bar chart showing the number of visits made by each of the doctors, patients, and clinical staff. For this graph, the x-axis labels the days of the week, while the y-axis lists numbers ranging from 0 to 1000. The last section displays clickable information, including doctor verification, service requests, consent for monitoring, and potential issues. Clicking on any of the previously listed buttons takes us to the management subsection. The last button is the ML models evaluation, and clicking on it displays a list of ML models, the evaluation report (containing accuracy, cross-validation scores, and F1 scores), and a section for uploading models and data before clicking the 'Evaluate' button.

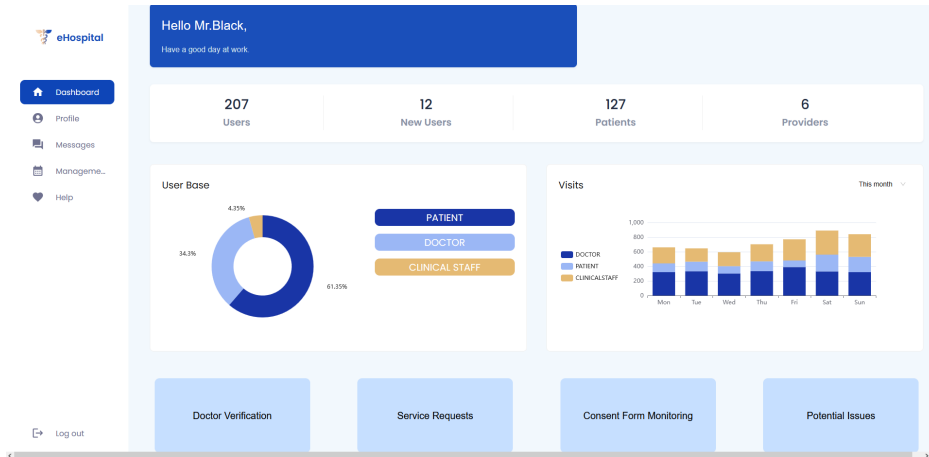


Figure 18. Frontend Admin Portal - Dashboard Section - of the e-Hospital

3.5.2 Profile

This section is straightforward, comprising only three sections: profile, signature, and security. For the profile, there are editable fields for personal information, including First Name, Last Name, Email Address, Birthday, Mobile Number, Address Line 1, Address Line 2, City, Zip, Specialization, and an Admin ID, accompanied by a “Save Settings” button. The signature section is simply a signature box that does not seem editable, accompanied by a “Save Settings” button. Lastly, the security section is designed to facilitate password updates, allowing users to input their current password, new password, and confirm the new password. It also has a timeout duration, which allows you to select a time after which the site will log you out (5 minutes, 10 minutes, 15 minutes, 30 minutes, or 1 hour). Additionally, it features a “Save Settings” button.

Admin Personal Information

Profile

Signature

Security

First Name: Admin

Last Name: User

Email Address: admin@example.com

Birthday: 1980-01-01

Mobile: 1234567890

Address Line 1: Admin Address Line 1

Address Line 2: Admin Address Line 2

City: Admin City

Zip: 00000

Specialization: Administration

Identification:

Admin ID: ADM001

Save Settings

Figure 19. Frontend Admin Portal - Profile Section - of the e-Hospital

3.5.3 Messages

This section is straightforward, consisting of three simple subsections. Firstly, Send New Message allows staff to send new messages and search through their message history. It contains features such as a subject line and a section to enter text, along with all the typical formatting elements, including font, font sizes, bullet points, and more. There is a send button to allow messages to be sent, as well as a back button to return to the previous page. The Messages section lists all messages, allowing staff to search for specific messages or simply click on the

one they want. Lastly, the Message display shows the message that was clicked, allowing the patient to view and read it.

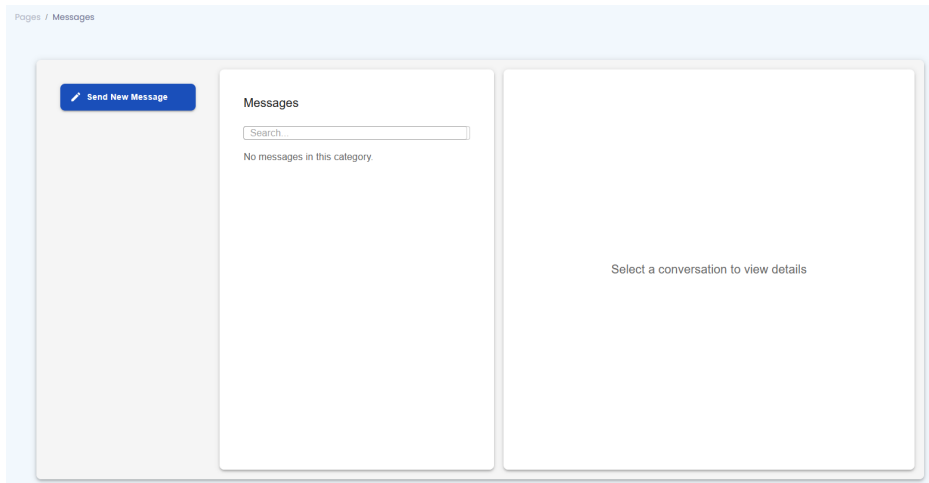


Figure 20. Frontend Admin Portal - Messages Section - of the e-Hospital

3.5.4 Management

This is the most developed section in this portal, containing features such as the patient list, doctor verification, tickets, service requests, and potential issues. The patient list contains information such as their No, Name, Assigned doctor, active date, location, and an edit section. Clicking on the edit section directs us to the patient portal, where we can access the specific patient's record for editing. You are also allowed to click between pages and set the number of rows (10 per page, 20 per page, 50 per page, or 100 per page). Moving on to the doctor verification list, which contains the following information: No., Name, status, activation email, and remaining time. You are also allowed to click between pages and set the number of rows (10 per page, 20 per page, 50 per page, or 100 per page). The ticket list contains the following information: No., Ticket ID, Name, Email, Subject, Category, Status, and Date and Time of Creation.

In the Service Requests section, it displays a list of requests and allows us to create New Tasks. When creating a new task, we are asked to provide the Date, Time, and Description of what we need before clicking 'OK'. Lastly, the potential issues feature displays problems such as unusual visits to medical records, unusual logins, multiple password attempts, and potential data leaks. There is a "View" button next to each issue that appears to do nothing.

PATIENT LIST					
No	Name	Assigned Doctor	Active Date	Location	Edit
25	Emma Charles Thomas			33 Nicholas St	...
132	Katie Ann Evans	Adam Wayne Bob D59 Test D3 Test D4 Test D107		5670 William McEwen Drive	...
133	Daniel John Thomas	Adam Wayne Bob D59 Test D3 Test D4		1907 Sequoia Drive	...
134	Emily George Walker	Bob D59 Test D3 Test D4		109 Amberwood Crescent	...
135	Daniel Kevin White	Adam Wayne Bob D59 Test D3 Test D4		4237 Boundary Road	...
136	Daniel David Jackson	Adam Wayne Bob D59 Test D3 Test D4		5205 Sand Road	...
137	Henry James Wilson	Adam Wayne Test D3 Test D4		2779 Sanfield Road	...
		Adam Wayne			

Figure 21. Frontend Admin Portal - Management Section - of the e-Hospital

3.6 Hospital Portal

This portal features a webpage that provides a wealth of information about the e-Hospital itself. It also conveniently includes a voice button that allows the page's contents to be read aloud, promoting accessibility. The top of the page displays header links that direct us to the desired section of the page when clicked. These sections are the e-Hospital home, the About section, Services, AI Solution, and Contact. At the start of the page, we see a 'Join Us Now' button which takes us to separate page where we have information on the joining process followed by the form to fill out (where you provide the First and Last Names, Email, Phone, Specialty, Working Address, Certificate Number, and any Notes we may want to include) before

submitting the application. Next, we present a summary of the Platform, including its machine learning involvement and user-oriented functionality.

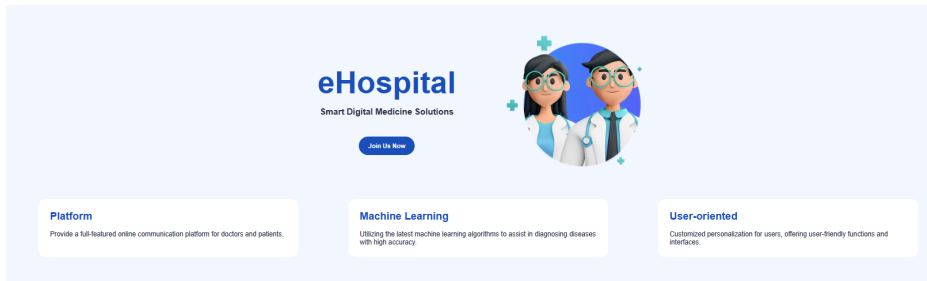


Figure 22. Frontend Hospital Portal - General - of the e-Hospital

The next section discusses the ‘About Us’ section of the e-Hospital, as well as its mission. We then see ‘Services’, which contain specific services for doctors, patients, and clinics. When clicking ‘More Info’ on any of those services, we get taken to the original portal containing the information (for example, the doctor portal for the doctor services). The ‘AI Solution’ section provides information on the role of AI in the e-Hospital and how it can be utilized to provide progressive assistance in diagnostics and patient care. Information on the e-Hospital, including the number of countries, doctors, and patients using it. Lastly, some contact information is provided, including the address and company details of the e-Hospital.

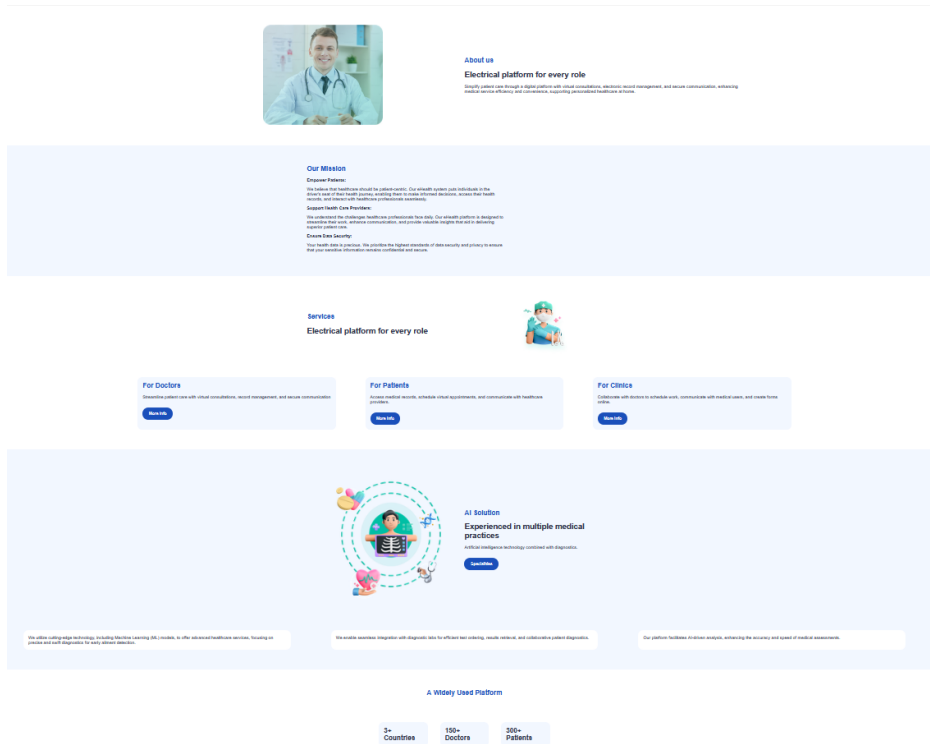


Figure 23. Frontend Hospital Portal - Main Sections - of the e-Hospital

3.7 Clinic Staff Portal

This portal is straightforward, comprising only three sections: the dashboard, calendar, and management. Additional work will be necessary to maximize the portal's effectiveness and better assist clinic staff in the future.

3.7.1 Dashboard

This section begins with a welcoming message in a blue rectangular box, similar to all previous dashboards. We can also view a list table similar to the one found in the patient section of the doctor portal. The patient information displays the total number of patients, as well as their Patient ID, Name, Visit Type, Scheduled Time, and Status. There is a section to select a specific

doctor, which provides a list of their patients. In this table, we can search for patients, download information, print information, select the columns we want to view, and filter the table based on the five columns. We can further filter the list based on patients coming today, for the week, or for the month. The number of rows per page ranges from 10 to 15 to 100 per page.

Next is the calendar, where the clinic staff can select a specific month, year, and date, and this is an editable section. The upcoming tasks are completed once the staff selects a specific date from the calendar. To create a new task, staff must select the date and time, include a description, and then create the task.

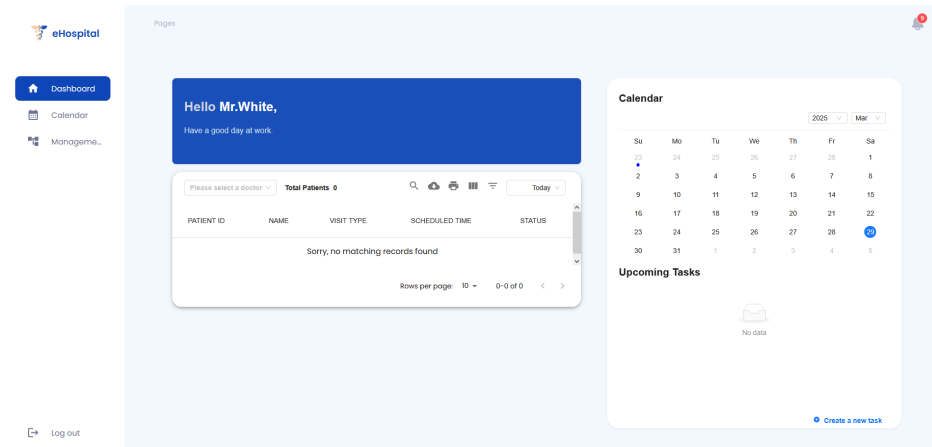


Figure 24. Frontend Clinic Staff Portal - Dashboard Section - of the e-Hospital

3.7.2 Calendar

The calendar features a Week option, which displays a weekly calendar that shows all appointments, tasks, and other events. The calendar also features a Month option, which displays a monthly calendar that shows all appointments, tasks, and other events. The Date Selection feature enables precise date selection, allowing you to narrow it down to the day, week, month, or year. Lastly, the "Add New Event" feature allows doctors to add an event and fill out the Date, Time, and Description.

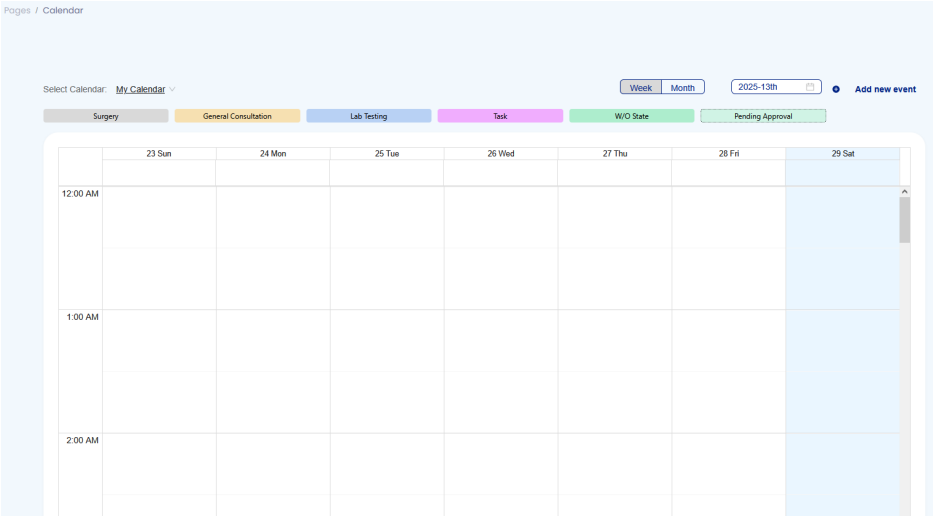


Figure 25. Frontend Clinic Staff Portal - Calendar Section - of the e-Hospital

3.7.3 Management

This section is similar to the one seen on the Clinical Trial portal, specifically, the application action. It displays the trial name, applicant name, application date, your read status, your response, response date, and the applicant’s read status. When expanded, we can view the applicant's notes.

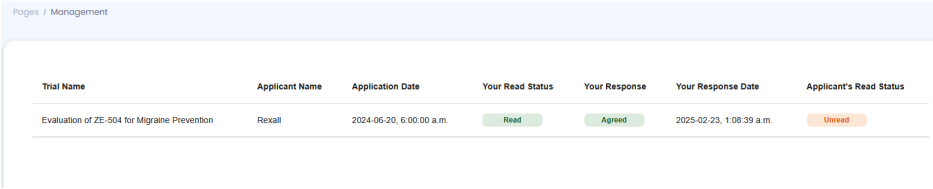


Figure 26. Frontend Clinic Staff Portal - Management Section - of the e-Hospital

3.8 Pharmacy-Admin Portal

This section takes us directly to the e-Hospital Smart Digital Medicine Solutions. This is a substantial site with numerous distinct sections. This includes the dashboard, the Team (which contains the Manage team, Contacts Details, and Invoices Balances), Reminders (Profile Form, Calendar, FAQ Page), and Analysis (Bar Chart, Pie Chart, Line Chart, and Geography Chart).

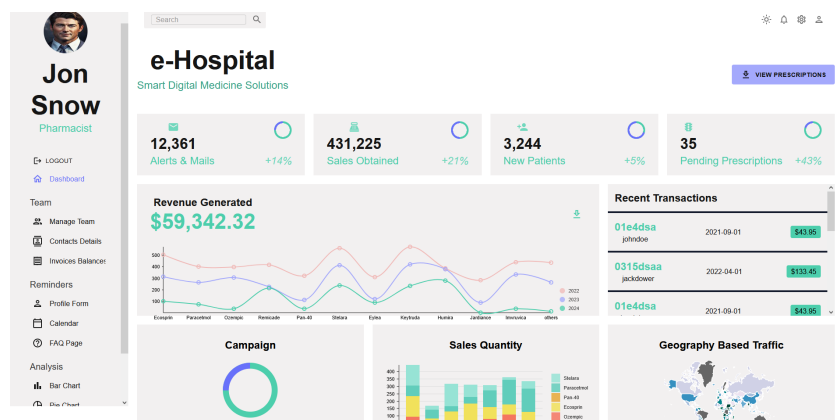


Figure 27. Frontend Pharmacy-Admin Portal of the e-Hospital

Starting with the Dashboard, it shows at the top of the dashboard page “e-Hospital” smart digital medicine solutions, which is not editable. On the top right of the dashboard page, there is a feature for viewing prescriptions and it is editable, when clicking on it shows prescription list (e.g., prescription 1, prescription 2, etc.) where the prescription has information which includes: date and time it was received on, doctor’s name and phone number, patient’s name and phone number, and the medication prescribed with the appropriate dose, frequency route, duration, refill, and description. Then, there is a button that allows the user to mark the prescription as completed when they have finished with it. Additionally, there are non-editable buttons located in the top middle of the dashboard page, which display various information, including alerts and emails, sales obtained, new patients, and pending prescriptions. In the middle-left corner, a line graph displays the revenue generated, showing the total revenue and highlighting financial performance. For the middle-right corner, there is a recent transactions tool that lists the latest transactions, including order IDs and amounts, for real-time monitoring. In the bottom-left

corner, a campaign performance tool is illustrated with a pie chart. This feature displays the percentage of revenue contributed by different campaigns. In the bottom-middle section, a sales trends graph is displayed, which is a bar graph illustrating the monthly trends in medication dispensation. This allows administrators to analyze usage patterns and manage inventory effectively. In the bottom-right corner, there is geography-based traffic, which illustrates the distribution of pharmaceutical resources or medication demand across geographic regions.

Next, we will discuss the team section, starting with the 'Manage Team' section. This section likely displays a contact directory for the marketing team within the pharmacy administration portal of the e-hospital platform. It provides essential details such as names, roles, ages, phone numbers, and email addresses for team coordination and communication management. The contact details section is likely designed to provide quick access to essential contact information for staff, suppliers, or affiliated professionals. This feature facilitates communication and coordination within the hospital's pharmacy operations. Additionally, the invoice balances feature indicates a system for managing client invoices, tracking outstanding balances, and maintaining contact information for follow-up or billing purposes.

The Reminders section and the profile form feature under the pharmacy-admin portal likely indicate a function for creating and managing user profiles specific to the pharmacy department within the hospital's digital platform. It allows administrators to add new users, potentially for managing staff access or patient information related to pharmacy services, including the user's first and last name, email, contact number, and address. The calendar feature likely demonstrates the integration of event scheduling and time management within the pharmacy-admin portal of the e-hospital platform, allowing users to view, manage, and possibly set reminders for important dates and tasks. Lastly, the FAQ page under the pharmacy-admin portal for the e-hospital platform provides guidance on handling common issues with the portal. It includes instructions on logging into the portal, with guidance on what to do if it doesn't work well, as well as how to view prescriptions, search for prescriptions and patients, track prescription progress, and resolve issues if prescriptions are not appearing.

Lastly, the Analysis section features four types of charts, which are also displayed in the dashboard section: bar, pie, line, and geography charts. The bar chart in the analysis section

shows the monthly medication dispensation, with the graph set to display Medicines vs. country. The pie chart also shows the medication category distribution, which includes medications such as blood pressure Medications, painkillers, antibiotics, antidepressants, vitamins, and cholesterol medications. The line chart illustrates the cost of medicines for the years 2022, 2023, and 2024. Lastly, the geography chart is the same as the one explained in the dashboard section.

3.9 Modern e-Hospital App Front-End Using React Native [4]

The e-Hospital mobile application front-end was fully rebuilt to create an integrated, modern user experience on both iOS and Android. The process took advantage of the React Native and Expo frameworks allowing for a cross platform so that the application can be developed in one set of code with the same UI/UX user experience. The process started with setting up the project through Expo's CLI, which provided the basic structure to create a multi-platform application development cycle while also standardizing some of the routine development functions (e.g., debugging, hot reload, device testing) using Visual Studio Code and default emulators. The design was implemented using a modular feature approach, whereby each area of functionality (e.g., authentication, dashboards, chatbots, etc.) was rolled out as a reusable React component approach which aligned with the best practice React architecture.

A prominent element of the new front end is its robust role-based authorization and navigation. Upon application start, the user is greeted with a login screen which includes a role picker for choosing their account (Admin, Doctor, Patient, or Clinic Staff). After the user selects their role and does a successful back-end authentication, they begin the role based dashboards (via React Navigation). For instance, doctors never leave their dashboard which includes Home, Patients, Messages, and Help, with each tab loaded with consumption utility including, search patient list (with call and chat abilities), patient profiles, messaging, appointments, a calendar to support approvals for slots, and a chatbot floating button to allow users to easily create availability slots and communicate to support workflow.

Likewise, the patient dashboard is well-designed, including a home page with welcome messages, medication updates, appointment scheduling through an interactive calendar, and a chatbot to obtain assistance. Patients can move around the sections of their medical records, their profile information, and streamlined appointment booking, as well as messaging with their care

providers. There is also a comprehensive sign-up process that allows new users to sign-up directly in the application, with stringent form validation to preserve data integrity and security.

Importantly, the user interface incorporates platform-specific enhancements due to literature and official documentation, including secure password-type inputs, and more subtle platform-specific handling of native functionalities such as phone calls to achieve the experience users would expect on each operating system. API communication is secure as error handling during communication follows best practices with try that catch patterns, and the use of React error boundary components means app stability can be maintained in the event of errors occurring in isolated components.

The redevelopment was conducted with a focus on scalability, maintainability, and compliance, as well as meeting Apple App Store privacy and branding needs. Privacy policy and practices, branding, and legal documentation were all given careful consideration in order to receive approval after multiple rounds of review. Extensive testing through manual and TestFlight beta distribution, and continuous integration pipelines were incorporated into the development process to guarantee reliability and polish for the end user. Ultimately, the new launched front end is a game changer in the healthcare app community-with role specific features, seamless navigation, secure functionality, and designed on a modern and sustainable technical architecture.

3.10 UI Improvements and Developments 2025 [5]

Introduction

Problem statement

In Canada, Electronic Healthcare Records (EHR) are vital as they aim to be the bridge that connects healthcare professionals and patients. Despite most of the existing systems primarily pursue to improve patient care through constant iterations of their interfaces; and reduce administrative burden while increasing satisfaction, lack of consistency and usability challenges persists.

Particularly, the E-Hospital platform is designed to serve as an AI-ready and unified EHR solution to support clinical workflows and help patients manage their care. Despite the platform was implemented and functional, usability issues persisted. For doctors, the interface felt overwhelming, with too much information scattered across different sections, prioritizing data logic. Meanwhile, for patients, the experience was similar as simple actions such as booking an appointment, which is a crucial and common workflow found in recent health systems, or checking lab results or medications were unintuitive and confusing.

As a result, the E-Hospital platform for Patients and Doctors was not aligned with how users think, decide or act when they want to manage or provide healthcare services. Therefore, improving interface clarity, emotional ease (comfort), task recognition, AI-perception and overall satisfaction became our main mission. Our team took the opportunity unveiled from this misalignment to simplify and improve the way Doctors and Patients interact and their experience.

Project Scope

Driven by the identified and described problem above, our project primarily focused on creating an interactive prototype by redesigning the E-Hospital experience for two key user groups: Doctor and Patient. For the first one, we aimed to diagnose pain points and improve the usability particularly of the Dashboard, Patient List and Patient Overview. On the other hand, for the Patient side, we also aimed to diagnose pain points and simplify key daily healthcare related tasks, starting from the dashboard, like booking an appointment, review their appointments, checking lab results and requesting prescription refills.

Our redesign also included as part of the scope other activities such as:

- Enhancing the current design system to ensure visual consistency while maintaining the original colors or font styles developed by previous teams.
- Conduct usability testing with real users leveraging interactive Figma prototypes, aligned with UX research practices.
- Propose a more intuitive information architecture for both portals.

- Design an intuitive approach to visually showcase existing and upcoming AI features placed in key pages or workflows within the scoped pages.
- Front-end implementation of redesigned portals and components through pull request reviews.

Objectives

- Evaluate the current state of the platform through the lens of UX principles for the ongoing portal and find usability weak points and/or issues.
- Foster knowledge about other competitors platforms focusing on their approach for Patient and Doctor users.
- Analyze workflows and propose an information architecture that is more logical for Doctor and Patient users.
- Deliver wireframes and complete interactive prototypes using Figma, focusing only on Patient and Doctor portals, and their following pages:
 - Doctor Dashboard, Patient List and Patient Overview pages, including their modals.
 - Patient Dashboard, Appointments and My Health pages, including their modals.
- Simplify navigation paths and adjust clinical/patient language throughout the interface.
- Reorder AI features and group them in an intuitive approach to support Doctor's clinical routines.
- Conduct usability testing with a real doctor and end-users (patients) to measure interface clarity, user comfort, task recognition, AI perception and general satisfaction.
- Compare testing results between the current design and the redesign prototypes.
- Enrich the current design system to standardize and ensure consistency for new components and layout behaviour, including new text and status colors.
- Translate the redesigned pages into code for front-end implementation.

Expected Outcomes

- Improve interface clarity, user comfort, task recognition, AI perception and general satisfaction.

- Enhance task recognition for both portals by streamlining particular workflows for Doctor (patient overview and encounter sections) and Patient (book an appointment).
- Reorder, bring visibility and transparency to AI features through an intuitive approach to support Patient and Doctor's routines.
- Ensure front-end implementation is aligned and consistent with the design of components and overall UX/UI, so it matches what users need and expect.

Benchmarking and Specifications

Key findings

At the early stages of the project, the team recognized the importance of benchmarking against leading digital health platforms to guide the redesign of the E-Hospital experience. However, direct access to platforms like Maple Health, TELUS Health, OSCAR EMR, and Epic was limited due to licensing restrictions and login-based data barriers. To overcome this, we conducted a thorough secondary analysis of public resources such as user manuals, product walkthroughs, official documentation, and demo videos.

From this analysis, we extracted valuable UX and UI principles including:

- Streamlined task flows (TELUS, Maple)
- Predictive search and filtering (OSCAR)
- Role-based dashboards (TELUS)
- Visual consistency and health data transparency (Maple).

To enrich this desk-based benchmarking, our team received strong support from the client, who facilitated interactions with two healthcare professionals:

- Dr. Aziz, a tech-savvy clinician, walked us through OSCAR and TELUS in a live session. He highlighted features that facilitate multitasking, such as customizable dashboards, modular layouts, smart search engines, and quick documentation tools. His emphasis on minimizing cognitive load and supporting real-time clinical decisions directly influenced our approach to the Doctor Dashboard and Patient Overview redesign.
- Dr. Patrick, an experienced physician at the University of Ottawa, participated in two rounds of usability testing (Phase 1 and Phase 2). His feedback refined our layout

prioritization, dashboard clarity, and actionable summaries, especially regarding status indicators, task panels, and encounter documentation.

In parallel, we conducted usability research with patient users, which played an equally critical role in shaping the redesign. Patients were asked to complete typical healthcare tasks such as booking appointments, checking test results, and requesting refills, first on the existing live platform, then on the redesigned wireframes.

Their feedback was clear:

- The original platform was confusing, with too many steps and unclear labels.
- Patients reported feeling uncertain about whether their actions were successful (e.g., not knowing if an appointment was confirmed).
- Some functionality was buried under menus that used technical or administrative language unfamiliar to non-clinical users.

As a result, the team restructured the Patient Dashboard, combining functions like Planning, Calendar, and Doctor View into a single, intuitive **Sessions** section. A new “**My Health**” tab grouped test results, prescriptions, and health history in one accessible location. Navigation was simplified, language was made more user-friendly, and the number of steps required to complete common tasks was reduced significantly.

Through this dual-pronged benchmarking and user-centered research approach, we ensured that both clinician and patient perspectives were integrated into each design iteration. The resulting redesign reflects not only the best practices from industry platforms but also the real-world mental models and preferences of our end users.

Specifications

The development of our redesign was guided by clear specifications derived from both client expectations and UX evaluation metrics. At the project outset, our client requested three primary goals:

1. Improve the usability of both Doctor and Patient portals
2. Ensure visual and functional consistency across pages and workflows
3. Prepare the interface for scalable AI integration, by making current and future AI elements more accessible and intuitive.

Doctor portal specifications:

- Provide a redesigned interface for the Dashboard, Patient List and Patient Overview that reflects clinical logic according to daily tasks and improve the experience during patient visits
- Prioritize the most relevant actions required by a doctor for their daily practice for the Doctor Dashboard
- Improve the efficiency and design of the Patient Overview page with patients' stats, medical history summary, an encounter section and a record hub to review the patient history regarding surgeries, treatments, reports, health trends, referrals and documents.
- Propose a design approach to make existent and deployed AI features more intuitive and easily accessible for doctors
- Provide an encounter section for consultations, including tabs for documentation, prescription, referrals and billing
- Deliver measurable improvements and positive feedback around interface clarity, user comfort, task recognition, AI Perception, General Satisfaction for the specified pages.

Patient portal specifications:

- Provide a more intuitive and accessible portal for booking, reviewing and managing appointments and health records
- Deliver measurable improvements and positive feedback around interface clarity, user comfort, task recognition, AI Perception, General Satisfaction for the specified pages.

For both portals from an implementation perspective:

- Ensure that all implemented components match the visual and interaction guidelines defined in Figma, particularly:
 - Updated buttons, icons, modals
 - Improved layout grids
 - Enhanced status and text color systems
- Implementation was developed within the existing frontend codebase, ensuring compatibility and consistency
- GitHub was used for pull request reviews, and Heroku was used for testing environments
- Iterative development included:
 - Regular client and TA feedback

- Feedback from real doctor walkthroughs (e.g., Dr. Patrick's usability sessions)
- Final visual refinements during beta testing.

Concept Generation and Analysis

Ideation process

Throughout the project, we adopted an agile, iterative approach that allowed us to continuously refine the platform based on user feedback and structured testing. Our process enabled multiple feedback loops across three main design stages: MVP, beta release, and final release.

Our ideation process began with discussions to understand the client's needs, particularly the challenges faced by doctors in navigating the current system and the cognitive burden it placed on both clinicians and patients. The client emphasized the importance of streamlined workflows, accessible information, and AI-ready features that could enhance clinical efficiency without adding complexity.

Then, we conducted a UX audit of the current platform to identify usability issues, confusing navigation, inconsistent design patterns, and friction in completing key tasks. (Figure 28 and Figure 29). To review the UX audit, please refer to the [links and tools](#) section of the report.) This was complemented by a benchmark study of comparable healthcare platforms. This helped us identify common best practices and areas for innovation. Also, we benchmarked the platform against several modern healthcare interfaces and comparable platforms across both domestic and international systems. This competitive review helped us identify industry best practices, such as simplified appointment booking flows, dashboard summaries, and integrated virtual assistants. These benchmarks provided a foundation for our initial design direction.

After collecting insights, the team restructured the information architecture of the platform based on user flows. The original structure of the platform was data-oriented, offering information based on the database. However, some workflows were not intuitive enough for users to understand how to access the main functionalities of the platform. Thus, creating a new information architecture is necessary for improving user experience.

First Iteration: MVP

The first round of ideation culminated in our Minimum Viable Prototype (MVP), where we translated audit and benchmark insights into a functional design concept based on refined information architecture. We used Figma to create low-fidelity prototypes of doctors' and patients' workflows. After incorporating the initial feedback, we prepared the MVP for real-world validation. We gathered feedback through semi-structured interviews with clients and technical advisors. These sessions helped us validate core concepts and uncover context-specific insights.

Second Iteration: Beta Release

Our second iteration built upon all prior feedback and introduced refined concepts. This version was tested against the UX metrics defined in the Introduction section (including clarity, efficiency, task recognition, and emotional comfort). We also conducted interviews with patients and medical professionals, including a practicing doctor who reviewed both our designs and the platforms they're currently using. These expert insights clarified the practical limitations of current systems and emphasized the importance of alignment with real clinical workflows. The Beta prototype allowed us to confirm that our revised structure supports the intended user's goals.

Final Release

In the final stage, we synthesized all previous learnings and incorporated the last round of feedback, primarily from doctors and our clients. We focused on small improvements to make the platform more accessible and useful. For example, in the final stage, we optimized color choices and reconsidered the location of the AI Chatbot button. This led to the development of the Final Release, which brought together validated ideas and final refinements to interaction structure, feature visibility, and UX scalability.

Criteria and evaluation methods

To ensure our redesign efforts were grounded in evidence-based practices, we adopted a multi-step evaluation strategy that combined expert audits, iterative feedback loops, and direct user testing with both doctors and patients. The goal was to assess not only interface improvements but also their alignment with real-world mental models, task efficiency, and emotional comfort.

Audit Approach

The project began with a comprehensive UX audit of the existing E-Hospital platform. This audit focused on identifying usability problems in both the Doctor and Patient portals. Key methods included:

- Walkthroughs of existing workflows (e.g., appointment booking, reviewing patient records)
- Evaluation of navigation patterns, visual hierarchy, and component consistency
- Mapping of user frustrations to specific interface elements.

The audit revealed several critical pain points:

- Redundant or mislabeled navigation tabs
- Task interruptions caused by non-linear flows
- Inconsistent button placements and unclear icons
- Lack of visual feedback when completing actions.

These findings provided a baseline for all future redesigns and were revisited at each iteration to verify improvements.

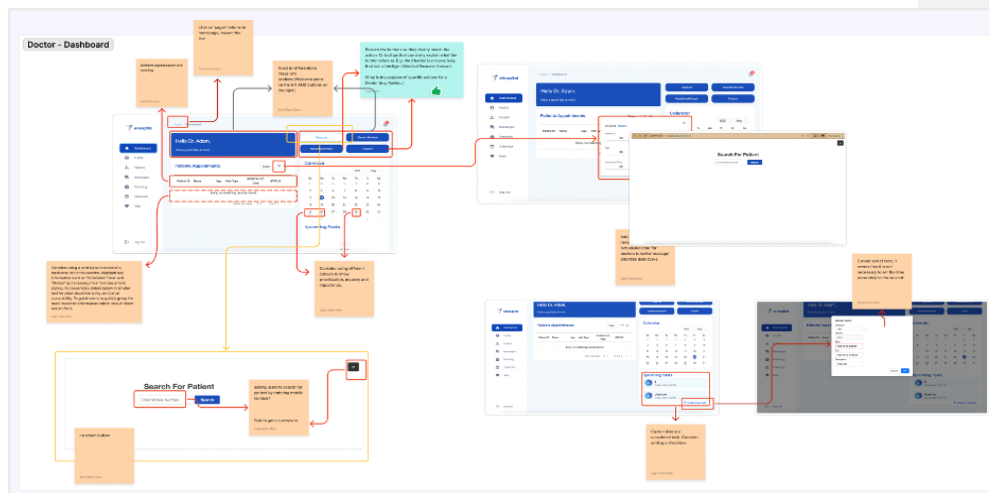


Figure 28. UX audit - Doctor Portal (Dashboard)

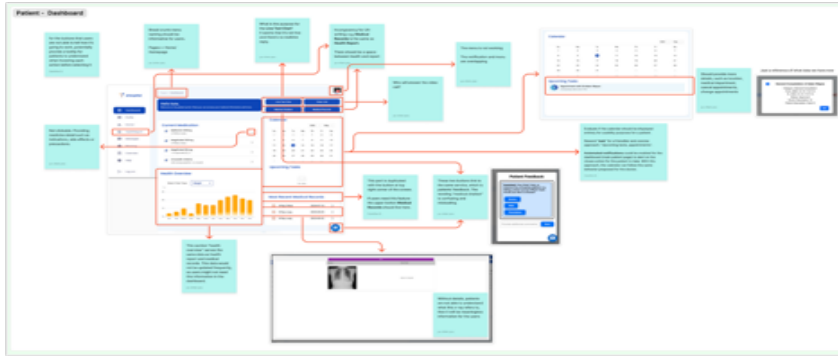


Figure 29. UX audit - Patient Portal (Dashboard)

Redesigning the information architecture

The team initiated the restructuring of the information architecture by identifying and organizing users' primary tasks. This approach aimed to enhance task efficiency and ensure that the structure aligns with actual clinical workflows.

For doctors, the main responsibilities include managing daily appointments, reviewing patient records, communicating with care teams, writing or updating diagnoses, and addressing pending items such as lab results or referrals. To support these workflows, we streamlined the sidebar menu in the doctor portal, reducing it from seven to five categories: Dashboard, Messages, Calendar, Planning, and Help. In addition, we introduced a section labeled "Global Utilities" to host features that are relevant across different user roles, such as profile settings, notifications, and feedback functions. These utilities are shared consistently among the five user types on the platform.

The revised structure helps minimize unnecessary clicks and better reflects real clinical routines. For instance, the Calendar was placed before Planning, since doctors typically review their schedules before organizing tasks. The Dashboard was refined to display only the most relevant and time-sensitive summaries, including appointment overviews with status indicators, previews of tasks and upcoming events, and shortcuts to frequently used tools such as diagnosis input. Key functions like writing diagnoses were also made more accessible by repositioning them in more prominent locations. In contrast, functions considered lower in priority or still in development

were moved further into the navigation structure to reduce clutter and keep the interface focused on daily clinical needs. One example is the ‘Radial Health Graph’ feature, which can now be accessed through a dedicated button within the ‘My Patient’ section.

These changes reflect an effort to logically group related features and simplify the overall structure. The result is a cleaner and more compact layout that improves user satisfaction and presents a more professional interface. This also supports one of our core goals, which is to build a consistent and scalable design foundation for future development.

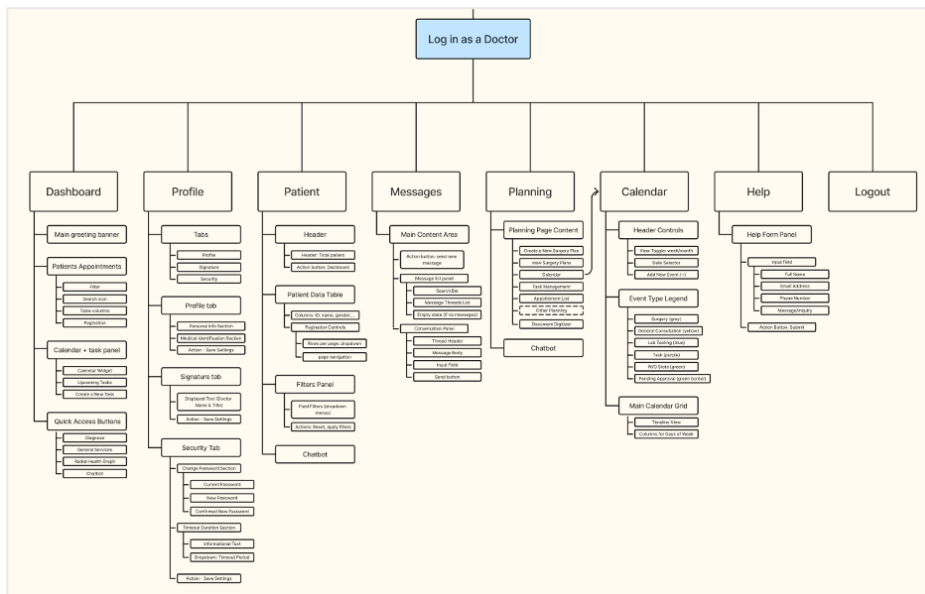


Figure 30. Current - Information architecture of the doctor portal

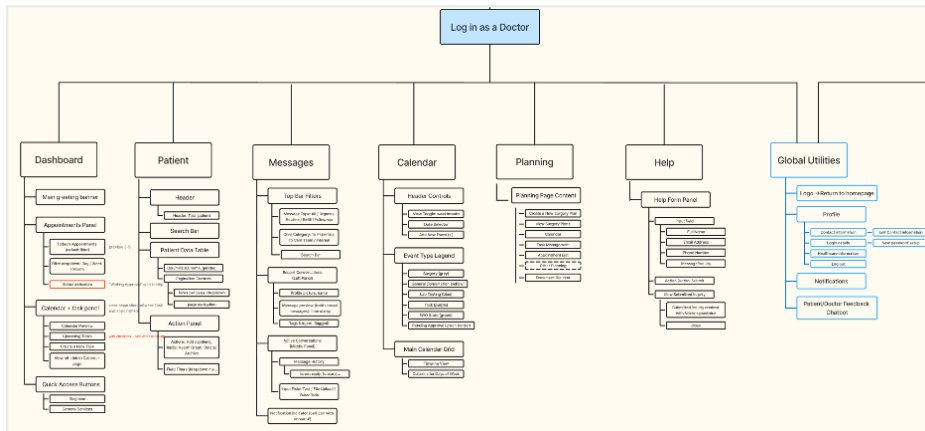


Figure 31. Redesign - Information architecture of the doctor portal

In addition to the global utilities included in the doctor portal, we also made several important changes to the information structure of the patient portal. For patient users, we identified three main tasks they commonly use the platform for:

- Booking, viewing, or managing appointments
- Checking medical test results, viewing medication information, and requesting refills
- Communicating directly with clinical staff or doctors

With these tasks in mind, we redesigned the layout to bring related features closer together, since they were previously spread out across multiple sections. For example, to book an appointment, patients first had to go to the “Planning” section, which displayed five different buttons. Selecting “Book an Appointment” would then redirect them to the “Calendar” page, where they could choose a time based on a doctor’s availability. However, this process lacked clear instructions and action buttons to guide users through each step. In addition, while a calendar view works well for doctors who use the system regularly, it is not as practical for patients who only log in occasionally.

To make the process more user-friendly, we combined the Planning, Calendar, and Doctor sections into one streamlined section called “Appointments.” This new section focuses on what patients need and allows them to manage everything related to appointments in one place.

We also improved how patients access and manage their health information. In the previous version, refill requests were placed under Planning, while test results and medication details were in separate sections. We grouped these features into a new section called “My Health,” which serves as a central area for viewing medical records and taking related actions. Unlike Appointments, which supports short-term tasks, My Health is designed to help patients keep track of their health over time.

By organizing the platform around patient goals rather than internal system categories, we created a layout that is easier to navigate and more focused on real user needs. This update helps patients complete their tasks more smoothly and gives them greater confidence in using the platform.

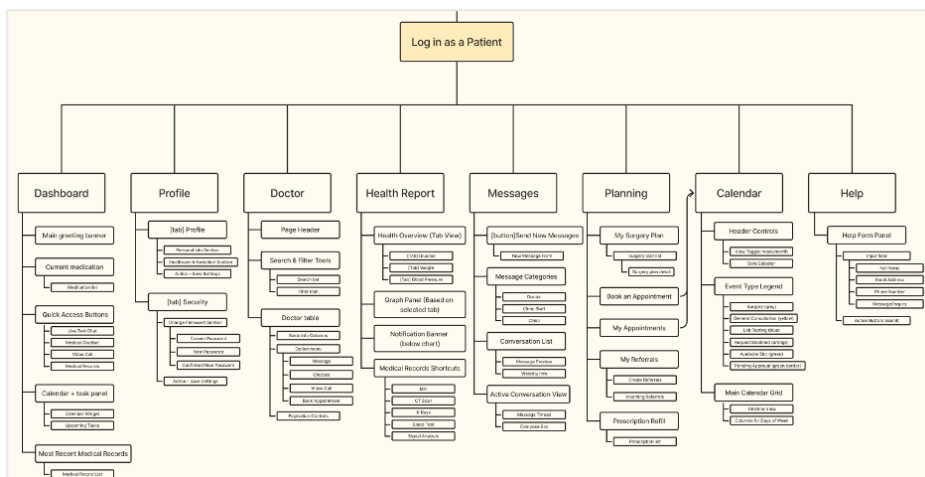


Figure 32. Current - Information architecture of the patient portal

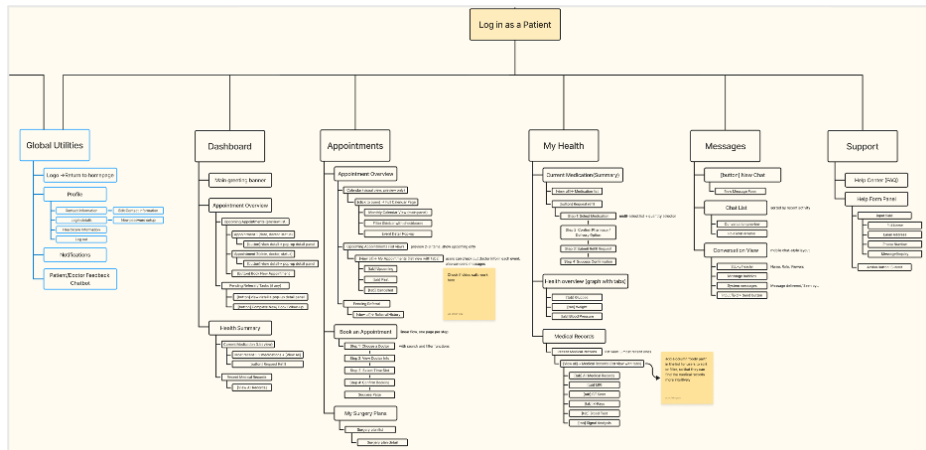


Figure 33. Redesign - Information architecture of the patient portal

Feedback from the Client (Subject Matter Expert)

Throughout the project, we engaged regularly with **Professor Ali Abbas**, our client and a recognized expert in healthcare informatics and **Professor Ismaeel Al Ridhawi**, the Technical Advisor. Their inputs were essential to:

- Clarify clinical workflows
- Ensure modular design logic and AI readiness
- Set clear priorities around patient autonomy and doctor efficiency.

Professor Abbas encouraged the team to "think beyond just portals", pushing us to include conceptual groundwork for future features like chatbots, voice assistants, and modular AI components, which we then prototyped visually in our interface.

The user research was conducted as part of the UX/UI redesign initiative for the E-Hospital platform (Module B). The objective was to evaluate the redesigned wireframes and assess whether the updates addressed key pain points identified in the original platform, such as unclear navigation, cognitive overload, and low task efficiency.

The user research was structured using tailored methodologies for each user group:

For doctors, research was conducted in two distinct phases:

- Phase 1 focused on early wireframes and alignment with clinical workflows.

- Phase 2 involved feedback on interactive prototypes with refined layouts, visual hierarchy, and embedded AI suggestions.

For patients, a comparative pre-post study was conducted:

- Participants first navigated the current live version of the E-Hospital platform and completed a questionnaire evaluating mainly usability and comfort.
- They then explored the redesigned wireframes and answered the same set of questions, allowing a direct comparison of perceptions before and after the redesign.

Both quantitative and qualitative methods were used to gather robust insights. This mixed-method approach ensured a deep understanding of users' cognitive expectations, emotional comfort, and perceived usability of the redesigned interfaces.

Research approach for patients

The patient user group in this research phase represents individuals who rely on the E-Hospital platform to manage various aspects of their healthcare independently. Typical use cases include:

- Booking and managing medical appointments,
- Accessing lab test results,
- Reviewing medication history and requesting prescription refills,
- Viewing personal medical records.

Three participants were recruited to represent the patient user segment. While no specific clinical background was required, all had basic digital literacy. This allowed the team to evaluate the platform from the perspective of everyday users, particularly those who may not interact with healthcare portals frequently but rely on them during critical moments of care.

The patient research followed a before-and-after comparative design:

- Participants were first asked to complete a series of common health management tasks on the existing E-Hospital platform. Their initial impressions were gathered.
- They were then introduced to the redesigned wireframes representing the same user journeys but with improved structure, layout, and interactions. Participants completed the same set of tasks, enabling side-by-side comparison of usability outcomes.

This approach allowed the research team to directly measure the impact of the redesign on perceived usability and comfort.

Research flow included:

- Task-based walkthroughs of redesigned wireframes (e.g., book appointment, check medication),
- Observation of navigation behavior and pain points,
- Short post-test questionnaire using a 1–6 Likert scale to measure:
 - o Interface clarity
 - o Emotional ease (comfort)
 - o Task recognition
 - o AI perception add-on
 - o Overall satisfaction
- Open-ended questions to gather insights about perceived control, confusion, or positive elements.

By combining these insights, the team was able to determine whether the redesigned wireframes truly enhanced user experience, particularly in terms of accessibility, decision-making confidence, and efficiency in completing essential healthcare tasks.

Research approach for doctors

The doctor user group includes clinicians managing tasks such as:

- o Reviewing patient records,
- o Approving appointments,
- o Writing notes and prescriptions.

Participants:

Dr. Patrick contributed through two structured evaluation sessions (Phase 1 and Phase 2), providing critical insights into workflow alignment, interface clarity, and clinical usability from an experienced practitioner’s perspective.

Dr. Aziz, a digitally proficient clinician, participated in a strategic exploratory session where he navigated TELUS and OSCAR platforms. His expert feedback focused on identifying best-in-class features and interaction patterns, offering design-savvy recommendations that informed several key elements of the E-Hospital redesign.

Approach included:

A multi-method qualitative and quantitative strategy was employed to capture nuanced insights from clinicians.

In Phase 1, a card sorting activity was conducted to uncover how Dr. Patrick naturally organizes patient-related information based on his clinical mental models. This exercise revealed key preferences for grouping, prioritization, and information hierarchy relevant to real-world decision-making.

In Phase 2, Dr. Patrick participated in an interactive prototype walkthrough, where he engaged with the updated wireframes of the doctor's dashboard and patient overview. The session focused on evaluating layout clarity, task alignment with clinical workflows, and visual hierarchy to support rapid comprehension and action.

Following each session, a post-test survey using a 1–6 Likert scale captured Dr. Patrick's feedback on:

- o Interface clarity
- o Emotional ease (comfort)
- o Task recognition
- o AI perception add-on
- o Overall satisfaction.

Additionally, open-ended feedback was collected during both sessions, offering deeper insight into design strengths, usability concerns, and improvement opportunities from a clinician's point of view.

This dual-phase approach ensured that the evolving wireframes were not only visually optimized but also clinically intuitive and workflow-aligned, as validated by a practitioner with hands-on expertise.

The data collection protocol and tools are listed in the [tools and links](#) section.

Detailed Design and First Prototype

Overview

Our Minimum Viable Product (MVP) aimed to deliver a foundational version of the platform that addressed the most immediate needs identified through our initial UX audit and client

discussions. This version was used to gather feedback from the teaching team and set the stage for testing with real users. We used Figma and FigJam as the primary tools for designing wireframes and prototypes. The design process followed an agile workflow with iterative check-ins to assess feasibility, design quality, and user alignment.

Doctor Portal

Key features

In the first prototype, the doctor portal retained the core structure of the original platform while enhancing it with improved features and a cleaner, more intuitive layout. The objective was to support real clinical workflows with minimal cognitive load, enabling doctors to manage appointments, patients, and tasks more efficiently.

The **Doctor Dashboard** was restructured to surface key information without overwhelming the user (Figure 33). The Appointment Overview was divided into two segments: Upcoming Appointments, which allowed doctors to view patient details or begin a diagnosis; and Pending Appointments, supporting better coordination with staff by indicating which appointments required confirmation or preparation. Doctors can view, edit, or cancel an appointment through the dashboard buttons. This helped streamline the consultation process and reduce ambiguity. The New Upcoming Tasks Panel now allows doctors to mark completed tasks. It could show completed tasks as well, to offer flexibility in how users tracked progress throughout the day. We also incorporated a mini calendar for quick date reference and appointment navigation, improving temporal awareness and planning.

The **Patient List View** in the MVP displayed basic but essential patient information: name, age, gender, appointment status, and last visit (Figure 7). Moreover, doctors could use quick action buttons to perform core actions such as:

- Adding new patients
- Viewing a Radial Health Graph across all patients (a conceptual visualization of overall health distribution)
- Filtering, deleting, or archiving patient records.

These features were included to ensure that doctors could carry out frequent administrative tasks without leaving the main workspace.

As for the **Patient Overview**, it aimed to serve as a central page for a doctor to start, recall or update a specific patient health data. This was iterated two times. In the first iteration, the design focused on centralizing and streamlining access to critical patient information. The page was horizontally divided into two primary sections: the left portion displayed key patient details such as demographics, chief complaints, and examination findings, while the right section allowed doctors to view, edit, or explore the patient's diagnosis history, prescriptions, referrals, and medical records. Although this structure achieved a clean and minimal interface, user validation sessions revealed that the simplification risked omitting important contextual cues and made frequent actions less intuitive. To avoid oversimplification and preserve usability in real clinical workflows, we revised the design based on user research insights and expert recommendations.

A second iteration followed up focused on grouping most of the features regarding the nature of the data they stored. This approach proposed a single view without the need of scrolling down divided in three main components (Figure 36):

1. A key stat component at the left upper side of the screen with all the details of the patient, including AI features.
2. A component at the right upper side of the screen including primary actions to connect with the patient such as “send a message” or “start a video call with the patient”. A complimentary set of shared actions followed below: “adding a document” or “requesting or creating” for Diagnose, Prescription and Referral features.
3. A Record Hub to serve as the main component to interact with other features such as: Treatments, Notes, Diagnosis, Prescription, Medical Records or request support through AI treatment. Given the nature of the data each contained, this section primarily aimed to reflect current information but additionally was thought to consider recalling past data history.

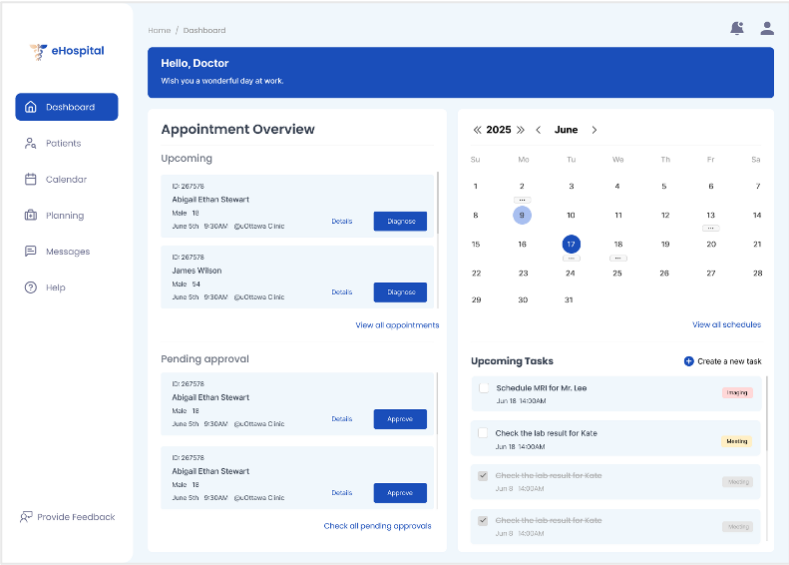


Figure 34. Redesigned - Doctor dashboard (MVP low-fidelity)

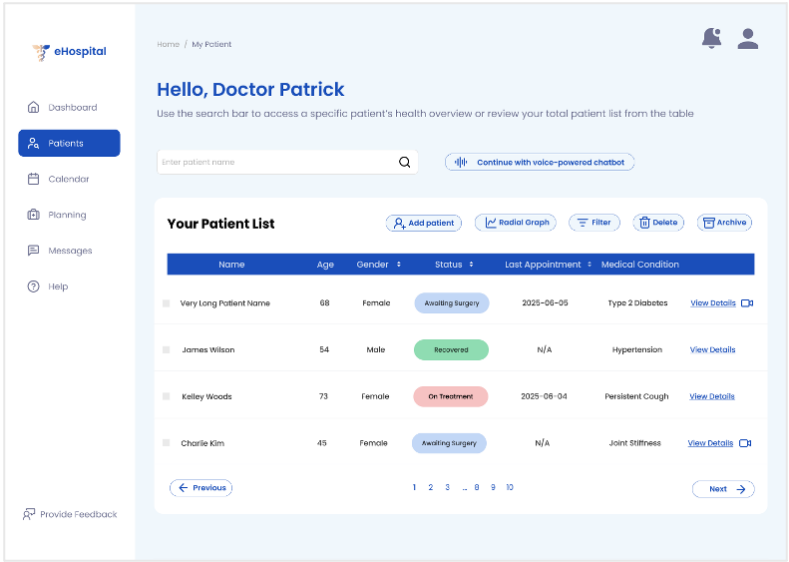


Figure 35. Redesigned - Doctor patient list (MVP low-fidelity)

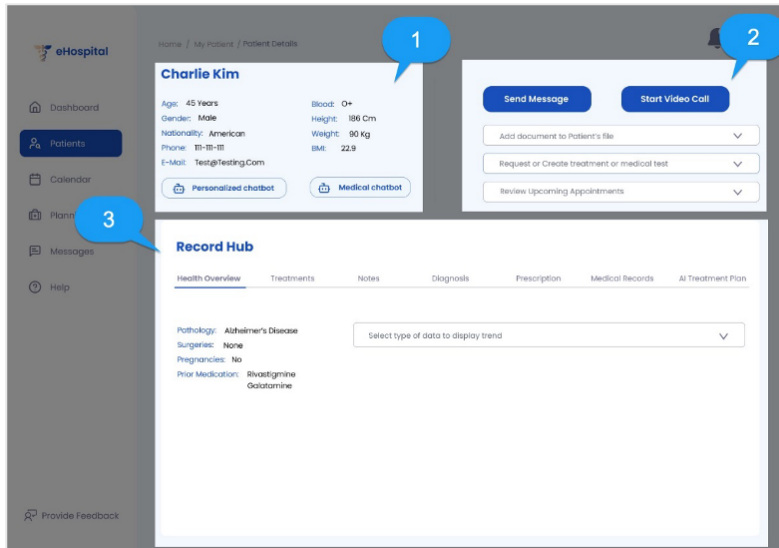


Figure 36. Patient Overview in the Doctor Portal (Beta Release – second iteration)

Functionalities

The MVP supported early interaction flows like accessing patient details, managing appointments, and tracking tasks, all from the dashboard. The purpose at this stage was to test early concepts with users and build a solid foundation for future iterations. The design decisions were guided by the principles of efficiency, visibility of system status, and minimizing cognitive overload, aligning with both UX best practices and clinical priorities.

Patient Portal

The MVP supported early interaction flows like accessing patient details, managing appointments, and tracking tasks, all from the dashboard. The purpose at this stage was to test early concepts with users and build a solid foundation for future iterations. The design decisions were guided by the principles of efficiency, visibility of system status, and minimizing cognitive overload, aligning with both UX best practices and clinical priorities.

Patient dashboard

We began the redesign by focusing on the first page patients see after logging into the system. The purpose of the dashboard is to provide a quick overview of key information, with access to detailed content available through corresponding pages. The goal was to display a sufficient amount of relevant information in a layout that remains clear and easy to understand.

To support this goal, we removed the quick access buttons in the top right corner, as they were either non-functional or duplicated with other functions. Since patients do not use the platform to manage daily schedules, we also removed the calendar and instead highlighted upcoming appointments. The layout was redesigned around the primary patient tasks identified in the IA section. By clearly separating the appointment overview and health summary, the new design allows patients to prioritize immediate scheduling information while still accessing personal health data. The layout follows a natural visual flow from left to right and top to bottom to support ease of use.

In addition to displaying information, the dashboard also provides quick entry points to key features such as appointment booking, appointment history, medication details, refill requests, and medical records.

From the dashboard, patients can access more information by clicking the “View Details” button. This action opens a pop-up modal that displays comprehensive appointment information, including the date, time, location, doctor’s name, appointment type, contact details, and current approval status. The call-to-action buttons within the modal are styled differently based on whether the appointment is in-person or virtual. For virtual appointments, the modal provides a direct entry point to the video call, minimizing the number of steps required to join the session.

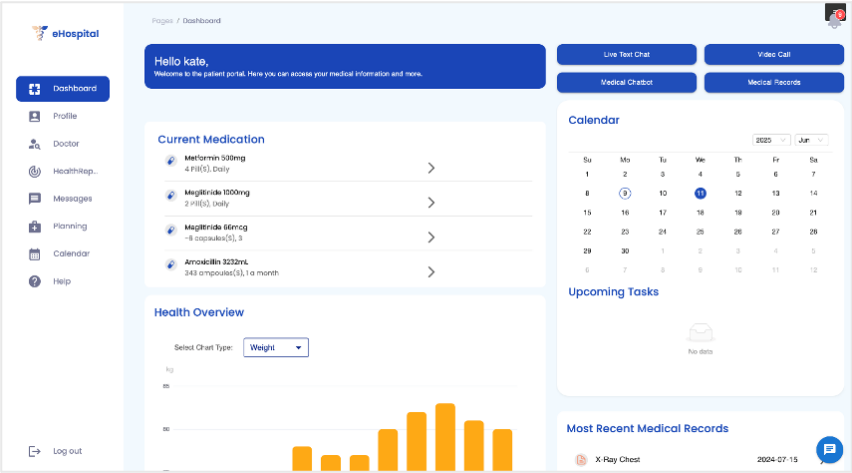


Figure 37. Current - Patient dashboard

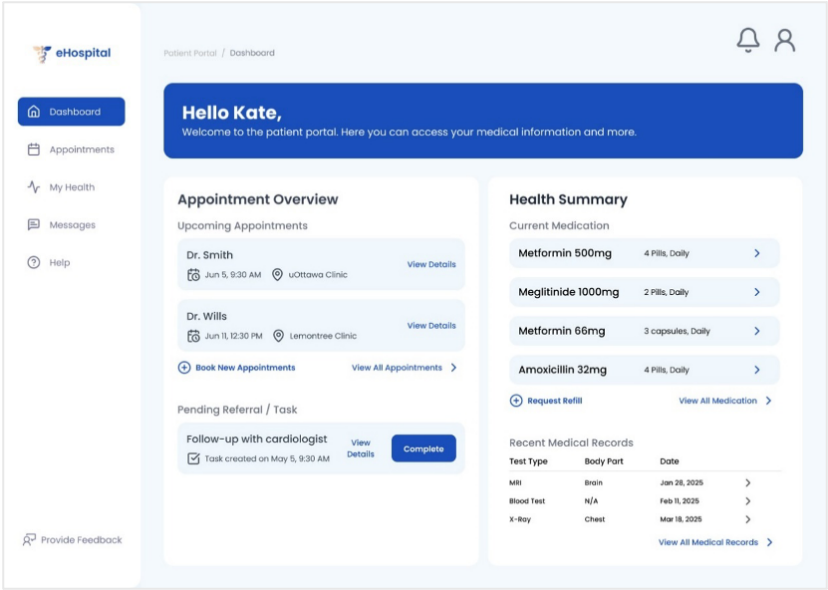


Figure 38. Redesign - High-fidelity Patient Dashboard

Booking new appointments workflow

From the patient dashboard, users can directly begin the appointment booking process through a prominently placed button. In the original design, this workflow was difficult to locate, requiring several steps to access. By surfacing this function and creating a dedicated page, we significantly improved accessibility and efficiency.

The new booking page presents a four-step process, all displayed within a single screen to avoid confusion from hidden stages. First, users select a medical specialty either by browsing a dropdown menu or entering keywords into a search bar. This dual-input method accommodates both users who have a specific need in mind and those who prefer to explore available options. Next, they choose the appointment type, such as general consultation, surgery, or lab testing, also through a drop-down selection.

Based on these choices, a list of available doctors appears on the left side of the interface. Each card displays the doctor's name, specialty, clinic location, and general availability. Users can scroll through the list to select a doctor and proceed to the next step, which involves selecting a time slot. Navigation arrows allow users to browse through date ranges, and a calendar icon offers direct access to specific dates.

Once all selections are completed, the "View Summary" button becomes active. Clicking it opens a modal where users can review the appointment details, confirm their choices, and leave a message for the doctor if needed. The appointment is booked upon clicking the "Book Appointment" button, though it remains pending approval from clinical staff.

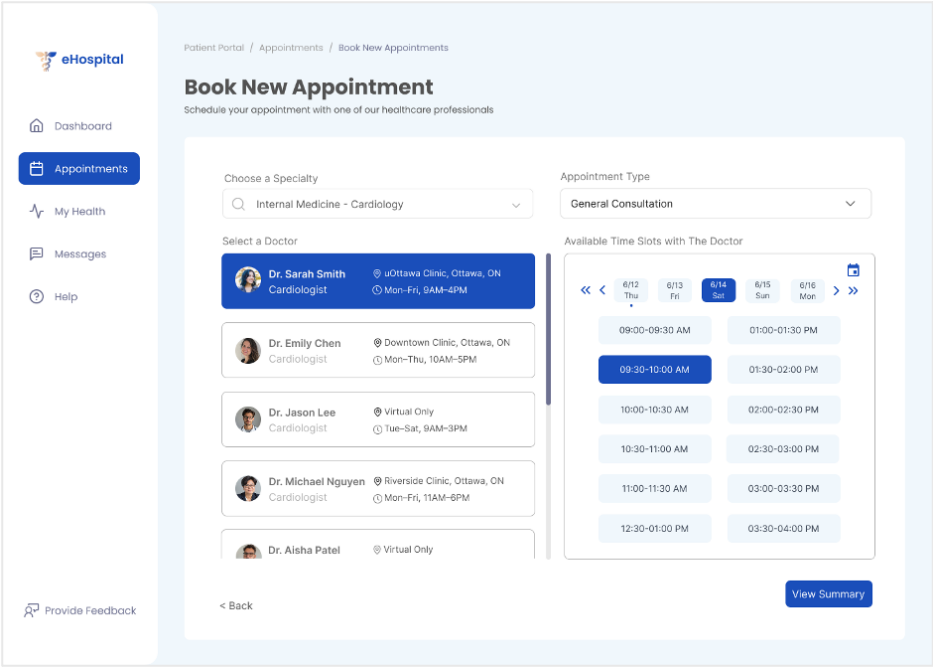
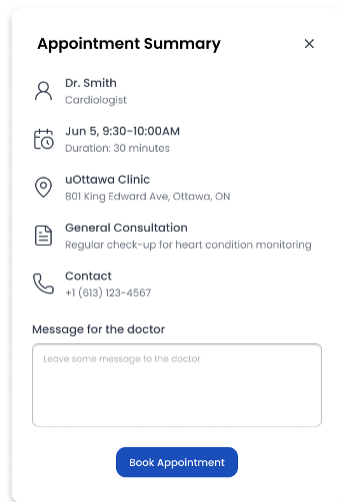


Figure 39. Redesign - Book New Appointment Page – Time Selected



Appointment Summary ✕

 **Dr. Smith**
 Cardiologist

 **Jun 5, 9:30-10:00AM**
 Duration: 30 minutes

 **uOttawa Clinic**
 801 King Edward Ave, Ottawa, ON

 **General Consultation**
 Regular check-up for heart condition monitoring

 **Contact**
 +1 (613) 123-4567

Message for the doctor

[Book Appointment](#)

Figure 40. Redesign - Book New Appointment Page – Appointment Summary Modal

My Appointments

From the dashboard, users can also navigate to the “Appointments” page by clicking the “View All Appointments” button. This page serves as a central hub for managing all upcoming, past, and canceled appointments, as well as surgery plans. Users can view appointments in either a list or calendar format, depending on their preference.

The page includes a tab navigation system and a search bar, helping users quickly locate relevant information. A call-to-action button in the top right corner allows quick access to the appointment booking workflow. In the upcoming appointments tab, each appointment card displays the date, time, location, and status. Depending on the appointment type, available actions may include editing, messaging the clinic, starting a video call for virtual visits, or viewing additional details. Surgery plan cards follow the same structure but exclude the video call option and display the doctor’s name instead of the appointment status.

In the “Past” tab, users can review previous appointments. Each past visit is marked as either completed or not completed. Completed appointments include detailed summaries with visit notes, prescriptions, and the doctor’s written feedback. In contrast, non-completed visits only show general appointment information. “Canceled appointments” appear in a separate tab. Each

card includes the cancellation reason and a “Rebook” button that links directly to the booking workflow, supporting users who may have canceled by mistake.

For users who prefer a visual approach, the “calendar view” displays appointment information by date. Selecting a specific day reveals a list of appointments in card format, helping users manage their schedule immediately.

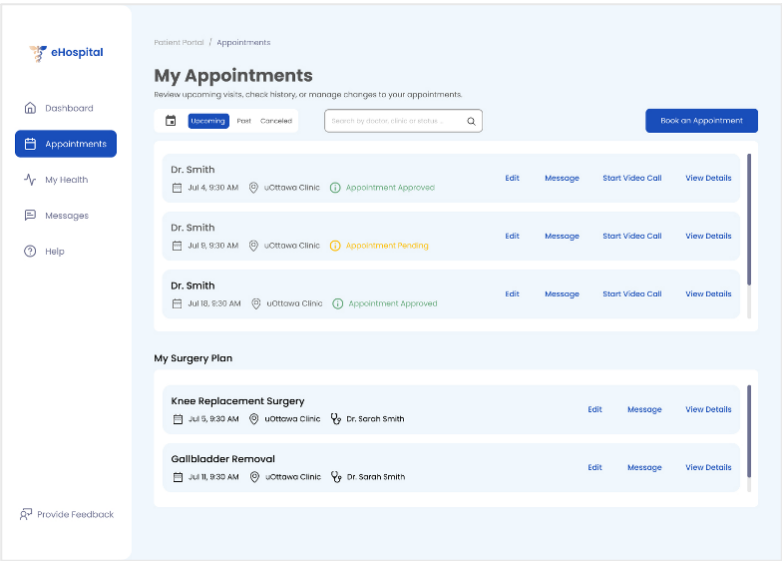


Figure 41.Redesign - My Appointments - Upcoming Appointments List

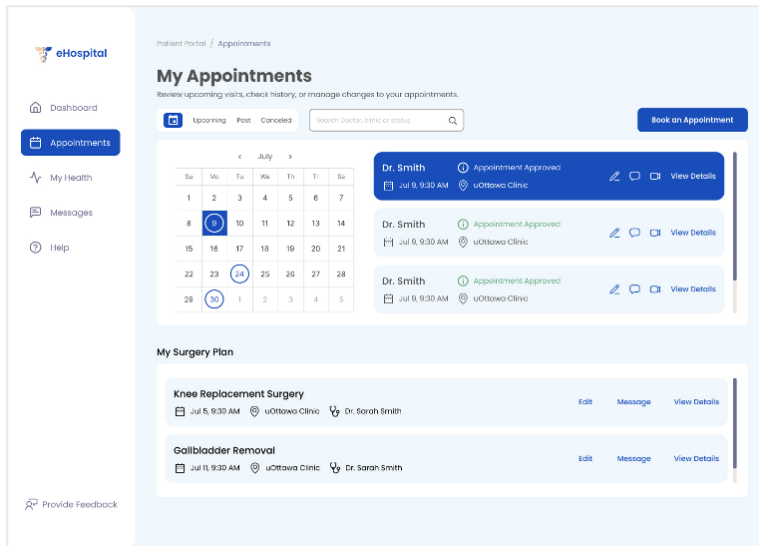


Figure 42. Redesign - My Appointments - Calendar View of Appointments

Underlying principles

Design system

A design system plays a crucial role in supporting user experience efforts by serving as a single source of truth. It provides a comprehensive set of standards for styles, patterns, and components used across the interface. This structured approach helps teams maintain visual consistency and ensure a user-centered experience. Without such a system, future design updates risk introducing inconsistencies, duplicating work, and increasing the complexity of maintenance due to the large number of interface elements such as pages and modals.

At the beginning of the project, our team was granted access to a Figma file containing an existing design system. Although maintaining or improving the design system was not originally within the scope of our work, the design team gradually made updates while redesigning both the doctor and patient interfaces. Due to time limitations and the significant effort required to manage a design system properly, this task was not a core focus during the project.

Nonetheless, a number of updates were made to enhance the system. Button, link, and action colors were extended to account for newly introduced components. Additional text and status

color styles were created to support consistency throughout the interface. A layout grid was implemented to help current and future designers align elements more precisely and improve readability. Button components were refined to define their appearance and behavior in various states. Outdated icons were replaced using the Lucid Icons plugin. Both the header and sidebar were adjusted to reflect the updated design and revised information architecture. Lastly, work on pop-up modals was initiated to promote consistency in the structure of cards and other interface components. (For the detailed design system, please refer to the link in the links and tools section of the report).

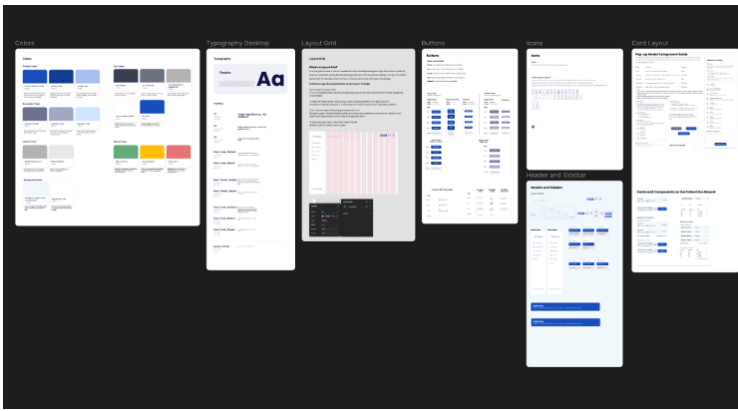


Figure 43.Redesign – Design system for E-Hospital

Implementation efforts

Patient Portal

On the patient side, our initial focus was on setting up the development environment and understanding the existing codebase. We cloned the original GitHub repository, resolved dependencies, and explored the routing and component structure to prepare for implementation.

Once familiar with the system, we began developing the Patient Dashboard, which serves as the entry point for patients. Following the new information architecture and Figma design, we implemented components to display the appointment overview, health summaries, and pop-up modals with appointment details. We used modal-based detail views and applied conditional rendering to differentiate between virtual and in-person appointments.

Doctor Portal

As for the doctor's side, once the initial setup process was done (Figure 44) and the cloning process was completed. The doctor's implementation began based on the wireframes provided. The new design for the doctor dashboard was always interchangeable during the project period as the feedback from doctors on the field was very valuable. The implementation was always changing based on updated wireframes, and we tried to keep up with it. The doctor dashboard and patient overview included new sections, better ways to view information, and most importantly a proper design.

```
D:\York\CS\Courses\UOttawa\25SUM\5902\Project\Frontend_Code\ottawa-ehospital-2025summer>git fetch teammate
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 1.89 KiB | 4.00 KiB/s, done.
From https://github.com/ahmadnozan/UIDesignTest
   acd1e87..f1b6222  main    -> teammate/main
D:\York\CS\Courses\UOttawa\25SUM\5902\Project\Frontend_Code\ottawa-ehospital-2025summer>git merge teammate/main
Auto-merging src/App.js
Merge made by the 'ort' strategy.
 package-lock.json | 2957 ++++++-----
 package.json      | 4 +-
 src/App.js        | 151 +-
 src/Testing/testingcode.txt | 1 +
 4 files changed, 1683 insertions(+), 1518 deletions(-)
 create mode 100644 src/Testing/testingcode.txt
```

Figure 44. Git merge code success

A key challenge was managing conflicting CSS styles from legacy components, especially when integrating new layouts. We also had to hardcode some data due to limited backend integration. Despite these hurdles, the dashboard closely matched our prototype and offered a more intuitive, streamlined experience for patients.

Second Prototyping and Testing

Overview

This iteration (Beta Release) focused on refining the interface and addressing user pain points based on structured feedback from both the professor and practicing physicians. The focus of this phase is refining the design of the doctor portal. The team visited Dr. Aziz, who shared his daily workflow using Oscar and TELUS, and Dr. Patrick, who took part in card sorting user testing to help the team restructure the information on patient overview design. After refining the prototypes based on the experience shared, the design was further validated through a

walkthrough session with Dr. Patrick to compare the results of the MVP and Beta Release prototypes.

Testing metrics

A key objective of the redesign process was to meet or exceed the usability targets defined in the MVP report. These benchmarks were initially set on a 1–5 scale. To ensure consistency in comparison, the original benchmarks set in the MVP report were recalibrated using a linear conversion formula to align with the 1–6 Likert scale used during testing. The ideal benchmark of 4.0 (on a 5-point scale) corresponds to approximately 4.8 on a 6-point scale, while the minimum acceptable level of 3.0 translates to 3.6. This adjustment allows for a more accurate and meaningful comparison between the original usability goals and the actual evaluation results.

The table below presents a comparison between the expected outcomes and the actual results observed during the user's research survey. The comparison between expected and actual results highlights the effectiveness of the redesign: in nearly every dimension of user experience, both patient and clinician profiles exceeded the original expectations. This indicates that the new prototype not only meets but, in many areas, surpasses the usability standards envisioned at the start of the project.

Area	Metric	Target (Ideal / Acceptable)	Doctor – Final Result	Patient – Final Result
Interface Clarity	Clarity of visual hierarchy, labels, and icons	$\geq 4.8 / \geq 3.6$	5.33	5.50
User Comfort	Emotional ease and sense of control during use	$\geq 4.8 / \geq 3.6$	5.00	5.00
Task Recognition	Ability to correctly identify intended actions from layout	Clear recognition / Some clarification allowed	5.00	5.89

User Satisfaction	Overall rating of the experience	$\geq 4.8 / \geq 3.6$	4.67	5.33
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Overall, the actual results validate the redesign approach and confirm that the new interface delivers significantly improved user experience.

These findings reinforce the value of iterative, user-centered design and provide a clear foundation for future development focused on personalization, trust in AI, and accessibility.

Translating clinical workflow into design: Insights from Dr. Aziz

As part of the exploratory phase of the redesign, an informal but highly valuable interview was conducted with Dr. Aziz, an experienced clinician and regular user of digital health platforms such as **TELUS** and **OSCAR**. During this session, Dr. Aziz navigated through both systems, offering a real-time, commentary-driven walkthrough of the features he finds most intuitive and clinically valuable. His feedback provided a rich understanding of what makes a digital health platform not only functional but efficient and clinician-friendly.

One of the first themes that emerged from his feedback was the importance of customizable workspace layouts. Dr. Aziz often works across multiple screens or windows simultaneously, allowing him to manage appointments, patient records, and prescriptions in parallel. He emphasized that systems which allow this kind of **multitasking flexibility** are crucial in fast-paced clinical settings, where switching back and forth between views can disrupt workflow. He also described how TELUS' role-based dashboard logic supports his day-to-day practice. Each clinic, he explained, is like a “bubble” within the system, isolating access to only the relevant data for a specific clinician or team. This “bubble” approach allows physicians to stay focused on their own patient caseload and tasks, without the distraction of unrelated information, ultimately reducing cognitive load and improving task efficiency.

A standout element in Dr. Aziz's demonstration was the search engine capability embedded in both TELUS and OSCAR. He praised how these systems allow him to find records, whether it be a medication, a physician, or a pharmacy, by typing just a few letters. Dr. Aziz praised the intelligent filtering of results: *“The search engine, that's what I like in the PS Suite. You type*

three letters and it gives you everything. Super good.”, said Dr Aziz. This kind of predictive search and intelligent filtering dramatically reduces the time spent searching through lists and is especially useful when handling a large patient volume. In discussing the prescription process, Dr. Aziz highlighted the value of built-in safety mechanisms, such as color-coded alerts for drug interactions. The system flags potential issues but gives physicians the option to override them with justification. This feature, he noted, respects physician autonomy while ensuring patient safety. He appreciated that the system does not lock the user out but rather prompts accountability through documentation. Another point of efficiency he emphasized was the use of predefined templates and shortcuts. In his day-to-day work, Dr. Aziz relies heavily on reusable clinical note templates for common conditions such as blood pressure follow-up, chest pain, or general consultations. These templates are accessible through shortcuts and dropdown menus, allowing him to complete documentation rapidly while maintaining consistency. Dr. Aziz also demonstrated how patient data, such as allergies, family history, immunizations, and risk factors can be entered through structured, layered forms. He particularly valued how the system guides him through relevant prompts (e.g., severity of allergy, confirmation status) without overwhelming him with information. This guided structure streamlines data entry and improved accuracy without compromising speed.

Lastly, Dr. Aziz noted the usefulness of embedded communication tools within the system. He showed how he can send and receive messages from clinic staff directly through the interface, often linked to a specific patient record. This kind of integration eliminates the need to switch platforms or rely on external messaging systems, enhancing workflow continuity.

Below is a filtered word cloud from Dr. Aziz’s interview. The most prominent terms now clearly reflect the core themes of his feedback, further reinforcing the interview’s clinical and usability focus.



Figure 45. Cloud Word from the Interview with Dr. Aziz

Overall, Dr. Aziz’s insights reinforced the need for a clinically intelligent, highly integrated, and customizable platform. His experience navigating TELUS and OSCAR helped the design team identify key features that clinicians expect from modern digital health systems, features that reduce friction, increase trust, and promote efficiency without removing clinical control. These insights directly informed several design decisions in the E-Hospital redesign, particularly around dashboard modularity, AI-supported features, and intelligent search and prescription workflows.

Design Refinements and Improvements

Doctor Dashboard and Patient list

We refined the overall design based on user feedback and professionals' suggestions. In beta release stage, the doctor dashboard information architecture was optimized to be more intuitive (Figure 19), the patient list was designed to be more efficient by utilizing the space (Figure 20). The detailed design refinements include:

Based on Professor and Client Feedback:

- Simplified visual hierarchy across patient dashboard and doctor interface for quicker orientation.

- Enhanced color coding for tasks with priority tags (e.g., "High Priority" in red, "Low Priority" in grey) to improve urgency recognition.
- Added functionality to hide or show completed tasks, helping users focus on daily priorities.

Based on Dr. Patrick's Feedback:

- Enabled case saving at any stage, with visual markers for unsigned and unbilled cases, addressing workflow continuity between doctor, clinical staff, and billing.
- Added clickable summary metrics to show total patient number, number of unfinished notes, etc.
- Patient list labels and table headers refined for clarity and alignment with medical documentation standards (e.g., "Medical Condition" was replaced by "Last Diagnosis").
- Simplified patient case status indicators (e.g., "Active", "Inactive") to reduce cognitive load.

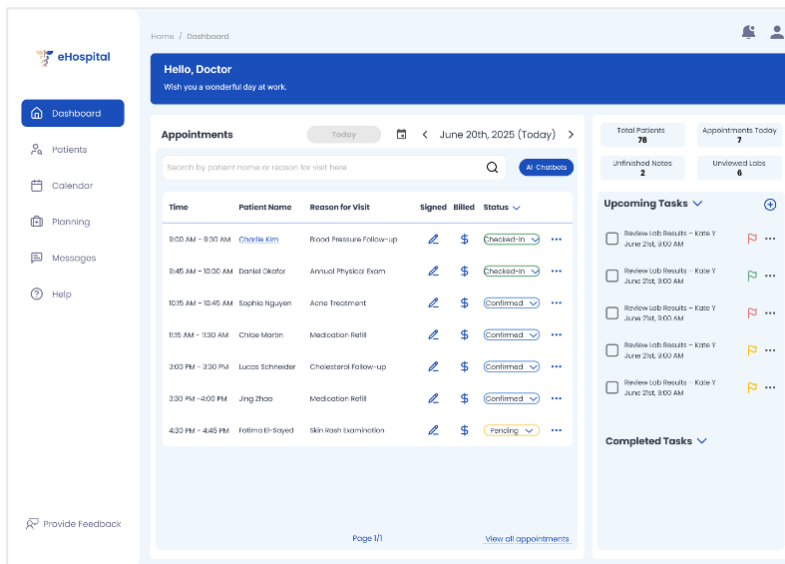


Figure 46. Dashboard in the Doctor Portal (Beta Release)

Home / Patients

Your Patient List

Enter patient name

Charts Radial Graph Filter Archive

Name	Age	Gender	Status	Last Appointment	Last Diagnosis
Charlie Gita	45	Male	Active	2025-06-05	Blood Pressure Follow-up
Very long Patient Name	45	Male	Active	2025-06-01	Seasonal Allergies
Chloe Barford	39	Female	Active	2025-06-03	Lower Back Pain
Sophia Nguyen	61	Female	Active	2025-06-03	Hypertension
Mason Clark	67	Male	Active	2025-06-02	Chronic Obstructive Pulmonary Disease (COPD)
Jam Patel	34	Male	Active	2025-06-01	Gastroesophageal Reflux Disease (GERD)
Jam Patel	34	Male	Active	2025-06-01	Gastroesophageal Reflux Disease (GERD)
Jam Patel	34	Male	Active	2025-06-01	Gastroesophageal Reflux Disease (GERD)
Jam Patel	34	Male	Active	2025-06-01	Gastroesophageal Reflux Disease (GERD)
Jam Patel	34	Male	Active	2025-06-01	Gastroesophageal Reflux Disease (GERD)
Jam Patel	34	Male	Active	2025-06-01	Gastroesophageal Reflux Disease (GERD)
Jam Patel	34	Male	Active	2025-06-01	Gastroesophageal Reflux Disease (GERD)
Jam Patel	34	Male	Active	2025-06-01	Gastroesophageal Reflux Disease (GERD)
Jam Patel	34	Male	Inactive	2025-06-01	Gastroesophageal Reflux Disease (GERD)
Jam Patel	34	Male	Inactive	2025-06-01	Gastroesophageal Reflux Disease (GERD)

Previous 1 2 3 4 5 6 7 8 9 10 Next

Figure 47. Patient List in the Doctor Portal (Beta Release)

Doctor's Patient Overview

The patient overview page functions as the main workspace for doctors during appointments. Informed by user interviews and a card sorting exercise, the team redesigned this key interface to better support clinical workflows (see Figures 48 and 49).

- Primary and secondary information was grouped and reorganized according to the natural visual flow of users.
- Clear sectioning was applied to help users locate and navigate information efficiently.
- Summaries of critical data were placed within easy reach, reducing unnecessary clicks and minimizing cognitive load.
- The “Encounter” section was given prominence, consolidating essential tools and actions frequently used during appointments.

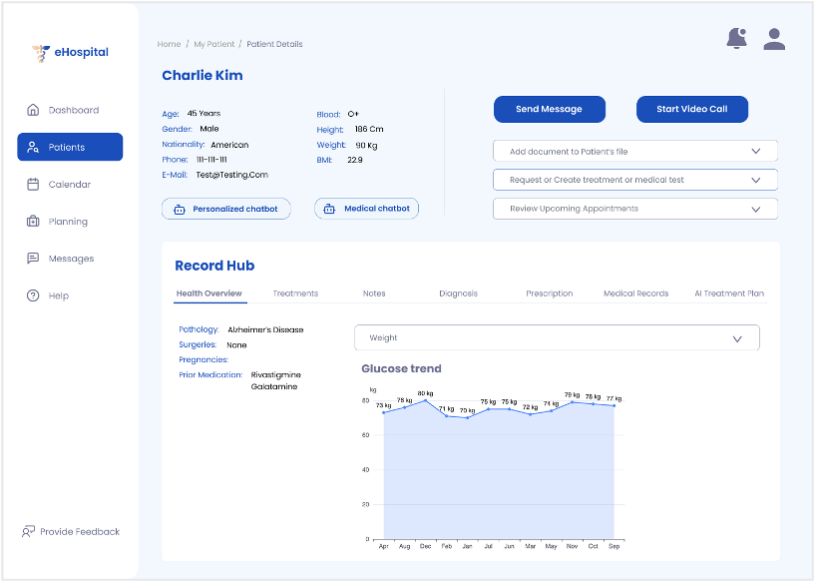


Figure 48.Patient Overview in the Doctor Portal (MVP)

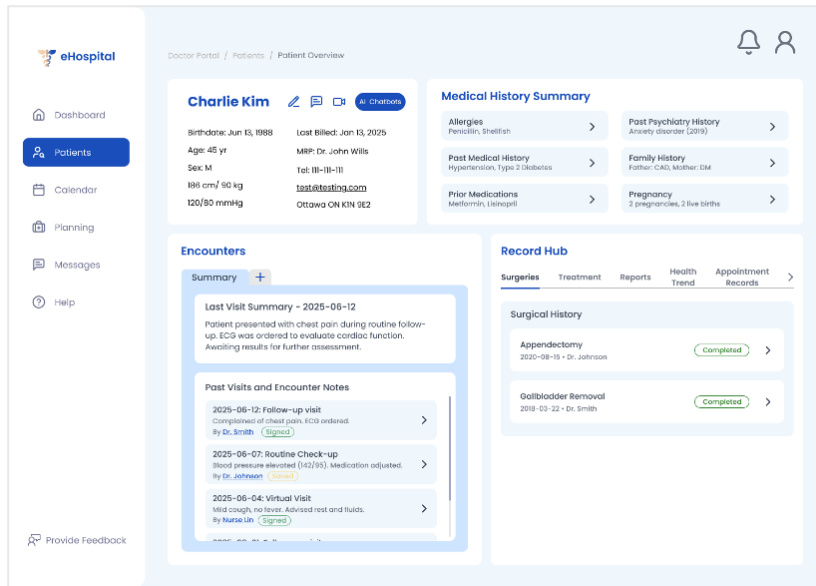


Figure 49. Redesign - Patient Overview in the Doctor Portal (Beta Release)

1. Patient profile/ demographics

- Presents a comprehensive view of the patient's basic information, including up-to-date medical data and contact details.
- Includes three action buttons that allow doctors to:
 - I. Edit the patient profile
 - II. Send a direct message
 - III. Initiate a video consultation
- Features an AI chatbot button that opens the assistant hub panel, which consolidates all AI tools available on the platform. This addresses the issue of dispersed AI functions, improving both visibility and accessibility of these tools.

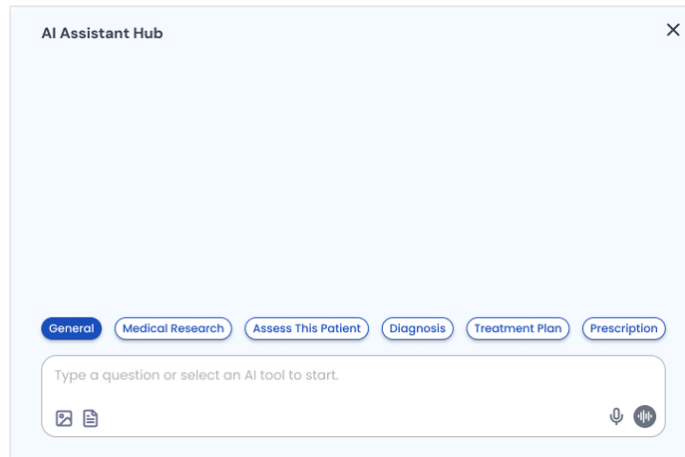


Figure 50. AI Assistant Hub Panel

2. Medical History Summary

- Provides a quick overview of a patient's medical history, with each card showing a one-sentence summary. This allows doctors to consider both current and past information during an appointment.
- In the previous version, this section was embedded within the larger "Record Hub." Based on analysis of daily workflows, the team recognized that some of this information should be immediately visible during appointments. As a result, the section was separated and placed directly on the overview page without requiring additional clicks.
- Contains six information cards commonly referenced by doctors:
 1. Allergies
 2. Past Medical History
 3. Prior Medications
 4. Past Psychiatry History
 5. Family History
 6. Pregnancy
- Each card allows users to quickly view, edit, or add information. For instance, the Allergies card supports these interactions with a single click, which is particularly useful during a patient's first visit.

Allergies	
Allergen: Penicillin Reaction: Anaphylaxis	Severity: Severe First Noted: 2010
Allergen: Shellfish Reaction: Hives, swelling	Severity: Moderate First Noted: 2018

+ Add Allergies

Figure 51.Redesign - Patient Overview - Medical History Summary - Allergies

3. Encounters

The encounter section allows doctors to take notes and complete key clinical actions during a patient visit. Based on input from Dr. Patrick, one encounter typically includes the following seven steps:

1. Examination
2. Impression/Plan
3. Diagnosis
4. Referrals (including specialists, bloodwork, radiology,...)
5. Prescription
6. Billing
7. Signature (can be saved and completed later if needed)

We developed a tab-based interface for this section, similar to browser tabs, enabling doctors to switch between previous encounter records and begin new ones during consultations.

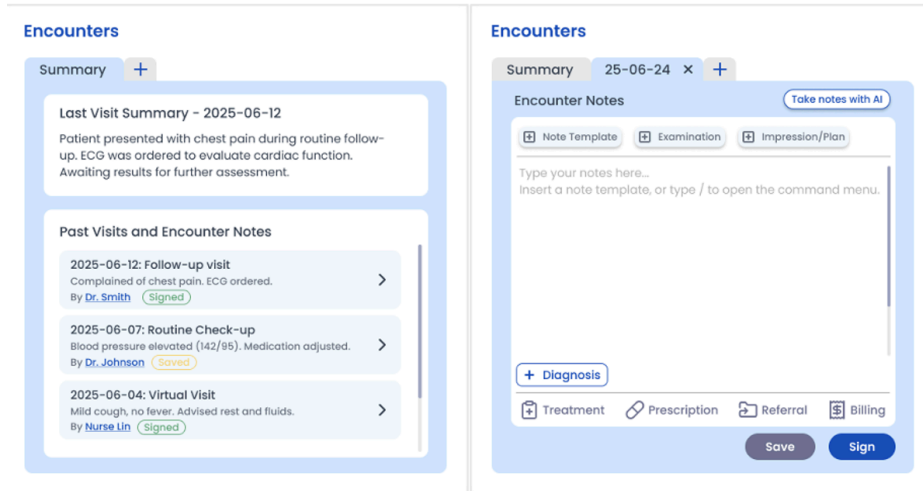


Figure 52.Redesign - Patient Overview - Encounter - Default and Add a New Encounter

- **Summary – default view**

The default display presents a summary of recent encounters. It includes the most recent encounter summary and a list of cards that provide entry points to each encounter.

- **Add a new encounter**

Users can start a new encounter note either by clicking the AI assistant icon in the top-right corner or by selecting from available templates such as note, examination, impression, or plan.

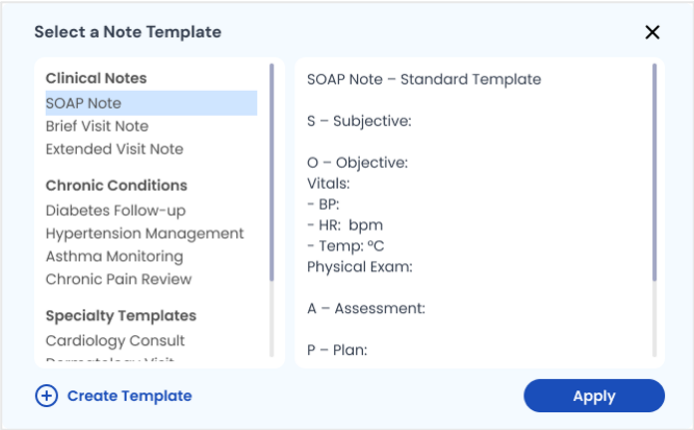


Figure 53.Redesign - Patient Overview - Encounter - Note Templates

➤ Once the note is completed, users proceed to select a diagnosis from a searchable list.

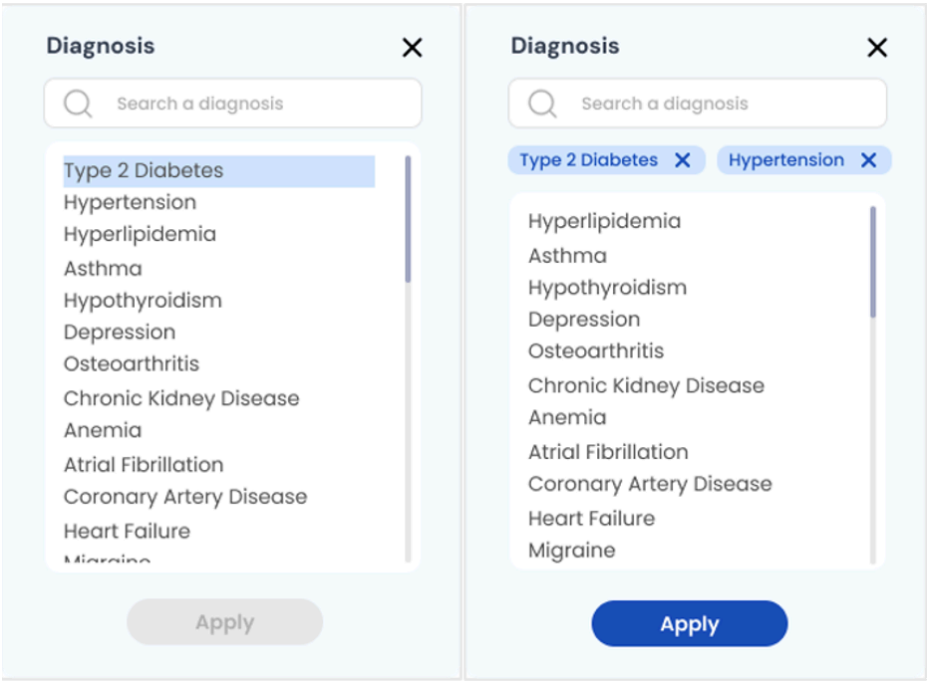


Figure 54.Redesign - Patient Overview - Encounter - Diagnosis List

- After the diagnosis is selected, the system allows the user to continue with prescribing medications.

The image shows a 'Prescriptions' form with a light blue header and a close button (X) in the top right corner. Below the header, there is a list of medications. The first entry is 'Metformin Capsules, 500mg, 4 Pills, Daily' with edit and delete icons. Below this is a section for adding new medications with the following fields:

- Medication Name***: A search bar with the placeholder 'Search a medication' and a dropdown arrow.
- Dosage Form***: A dropdown menu with the placeholder 'Select dosage form'.
- Strength***: A dropdown menu with the placeholder 'Select Strength'.
- Frequency***: A dropdown menu with the placeholder 'Select Frequency'.
- Quantity / Duration**: A field with a minus sign, a text input, and a plus sign, followed by an 'Add' button.

Below these fields is a '+ Add Medication' button. Underneath is a **Pharmacy*** section with a search bar and a dropdown arrow, containing the placeholder 'Search or select a pharmacy'. At the bottom of the form are two buttons: 'Save as a Draft' and 'eSign and Send'.

Figure 55.Redesign - Patient Overview - Encounter – Prescription

If needed, referrals can be created, edited, or deleted at this stage using a dedicated referral interface.

<

New Referral

Specialist

Bloodwork

Radiology

Specialist Type*

Q

Select a specialty

▼

Preferred Clinic

Q

Search a clinic or doctor

▼

Reason for Referral*

Send Referral

Referral Records

×

Specialist

Bloodwork

Radiology

Endocrinologist

Rideau Clinic

Neurologist

Civic Hospital

+

Figure 56.Redesign - Patient Overview - Encounter - Referral

- The final step in an encounter is billing. Patient and provider information is pre-filled to reduce the time and effort required to complete the billing form.

Billing Request

Location*

Q The Medical Center

Payment Issuer*

Ontario Health Insurance Plan

Billing Practitioner*

Q Dr. Patrick

Patient Name*

Charlie Kim

Health Card Number*

7878787870/TF

Service / Visit Date*

2025/06/26

MOH Group

1001234567

Facility

Q The Medical Center

Fee Item #1

Service Code*

Q 20104 Lacerations, Repair - Trunk

Diagnosis Code (ICD-9)*

Q Search a code or description

Base Amount*

CAS

54.4

Service Unit*

—

1

+

Fee Item #2

Service Code*

Q Search a code or description

Base Amount*

CAS

Service Unit*

—

1

+

Anesthesia Unit

—

+

Diagnosis Code (ICD-9)*

Q Search a code or description

Start Time*

08:00 AM

End Time*

08:30 AM

Fee Differential

Q Search or select a fee modifier

Holiday

Q Select a holiday (If applicable)

☐ Referring Practitioner
 ☐ Show Advanced Fields

Add

+ Add Another Fee Item

Internal Notes

Save as a Draft

Submit to MDBilling

Figure 57.Redesign - Patient Overview - Encounter - Billing

- Doctors can either save or sign the encounter note. Many choose to save their notes if they plan to revise or complete them later, since editing after signing is more restricted.
- **View past encounters**
Previously saved or signed notes are displayed alongside key details such as visit summaries, selected diagnoses, and the full content of the note. Action buttons are also available to help manage each record efficiently.

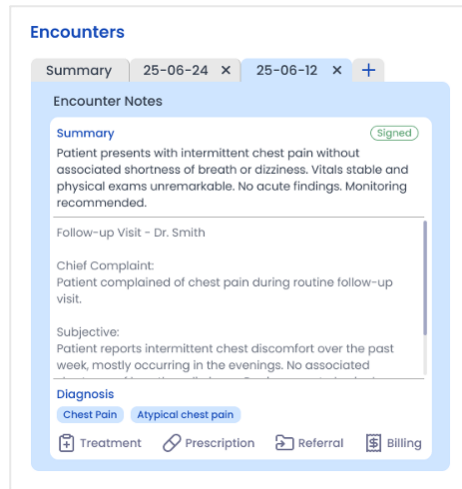


Figure 58.Redesign - Patient Overview - Encounter - View past encounters

3. Record hub

- This section contains patient information that is accessed less frequently but remains clinically relevant. Through the use of clear labels and a horizontal scroll layout, a large volume of data across various categories is presented in a well-organized and easily navigable format.
- There are 7 categories in the record hub, and it is expandable.
 - Surgeries
 - Treatment
 - Reports
 - Health Trend
 - Appointment Records
 - Referral
 - Documents



Figure 59.Redesign - Patient Overview - Record Hub - Health Trend

Validating design through iteration: Insights from Dr. Patrick

Dr. Patrick participated in two structured research sessions to evaluate the evolving design of the e-Hospital platform. These sessions were critical in understanding the platform's ability to support clinical workflows and enhance decision-making without introducing friction. His input reflected both deep clinical expertise and a strong awareness of the cognitive demands placed on healthcare providers in digital environments.

As mentioned earlier, the sessions were divided into two distinct phases:

- o Phase 1 emphasized evaluating static wireframes, navigation flows, and information organization.
- o Phase 2 involved a walkthrough of an interactive prototype with refined layouts, interface feedback, and early AI-enhanced functionality.

The feedback is best understood when mapped into the three main areas of physician interaction: the Dashboard, Patient List, and Patient Overview.

i. Dashboard – centralized access, prioritization of tasks

In Phase 1, Dr. Patrick recognized the intent behind the dashboard but noted challenges with visual balance and information hierarchy. The interface attempted to present many widgets and summaries but lacked guidance for the user's attention. Some sections were too prominent (for example messaging), while others, such as pending tasks, blended into the background. His main concern was the lack of prioritization: "Not everything should have the same weight. I want to see at a glance what's urgent, what's pending, and what can wait.", he said.

For Phase 2, the design introduced clearer layout spacing, bolder titles for actionable panels, and role-specific task summaries. These changes were positively received. Dr. Patrick acknowledged a more streamlined experience and reduced cognitive load, especially with the use of color-coded indicators and collapsible panels.

"The updated dashboard feels more purposeful; I don't have to scan everything to know what's critical. It naturally guides my attention.", declared Dr Patrick.

Still, he emphasized the importance of customizability, suggesting the dashboard should adapt to different specialties or usage patterns, such as for generalists vs. specialists. He also highlighted the potential for dashboards to support pre-visit preparation and post-visit follow-ups if designed with data recency and continuity in mind.

ii. **Patient List – fast filtering, visual cues, and efficiency**

The patient list was evaluated as a crucial component in navigating day-to-day care delivery. In Phase 1, the initial layout was found to be visually dense and lacking functional filtering. Search and filter options were either buried or ambiguous, making quick triage difficult.

He noted that row-level actions were not clearly differentiated from patient identifiers, leading to uncertainty about what clicking on a row would trigger. He also recommended status tags and emphasized the need to quickly surface patients needing immediate attention.

"I wasn't sure how the list was sorted or what those tags meant, it slowed me down more than it should have.", he said.

In Phase 2, filters were made persistent, the search experience was more predictable, and visual tags were introduced. The rows were reformatted to separate identity, status, and next action, improving scalability. Dr. Patrick confirmed that these changes improved his ability to locate and

prioritize patients more rapidly, aligning better with the fast-paced nature of clinical rounds and clinic workflows.

“Filtering is much quicker now. I can actually isolate what I need without clicking around too much.”, Dr Patrick said.

However, he still proposed the introduction of hover-over micro-actions (e.g., open file, view summary, add note), which would allow users to take direct action from the list without unnecessary page loads.

iii. Patient overview – clarity, context, and actionable summaries

The patient overview was the most deeply evaluated section and revealed the most transformative changes between phases.

In Phase 1, Dr. Patrick found the layout to be fragmented and too segmented, forcing him to scroll extensively and synthesize key information mentally. Important data points such as medications, allergies, lab results, and previous visit notes were located in separate sections that lacked consistent visual markers or prioritization.

“There’s a lot of jumping around, I have to scroll and mentally piece everything together.”, Dr Patrick said.

This feedback led to significant overhaul. By Phase 2, the overview was restructured into collapsible, thematically grouped modules. This not only reduced screen clutter but also allowed clinicians to focus on sections relevant to their immediate task.

“Now I get a clearer picture right away. Key details are grouped logically, and I don’t feel lost in the layout.”

Dr. Patrick appreciated the inclusion of a chronological timeline view of encounters and orders, which helped create a more coherent picture of the patient’s medical journey. He stressed the value of data density when meaningful, as long as the layout supported easy scanning and context retention.

Overall, Dr. Patrick found the improvements to be substantial, as he remarked: 'The improvements between the two versions are significant. I'd be comfortable using this in a live setting.'

The word cloud below visualizes the most frequently used and semantically meaningful terms from Dr. Patrick's feedback during the Phase 2 prototype walkthrough. The key terms suggest that the redesigned interface effectively enhanced **visual clarity**, **workflow coherence**, and **information prioritization**.



Figure 60. Cloud Word from the Interview with Dr. Patrick - Phase 2

This reflects a shift from scattered content in Phase 1 to a more structured and intuitive presentation in Phase 2, especially in the **dashboard and patient overview** sections. The emphasis on clarity and flow aligns with Dr. Patrick's overall positive reception and his increased confidence in the platform's readiness for real-world use.

i. AI Perception

Throughout both sessions, Dr. Patrick was attentive to how AI features were integrated into the workflow. He expressed enthusiasm for AI-assisted summaries, smart alerts, and proactive suggestions, provided these tools augmented decision-making rather than replace it.

This transparency and non-intrusiveness led him to rate the AI component highly in both sessions. However, he cautioned against overdependence or obfuscation, recommending that future iterations should include explanatory tooltips for how AI recommendations are generated, to build long-term clinician trust.

v. UX Ratings Comparison – Likert Scale: Phase 1 vs Phase 2 (Doctor profile)

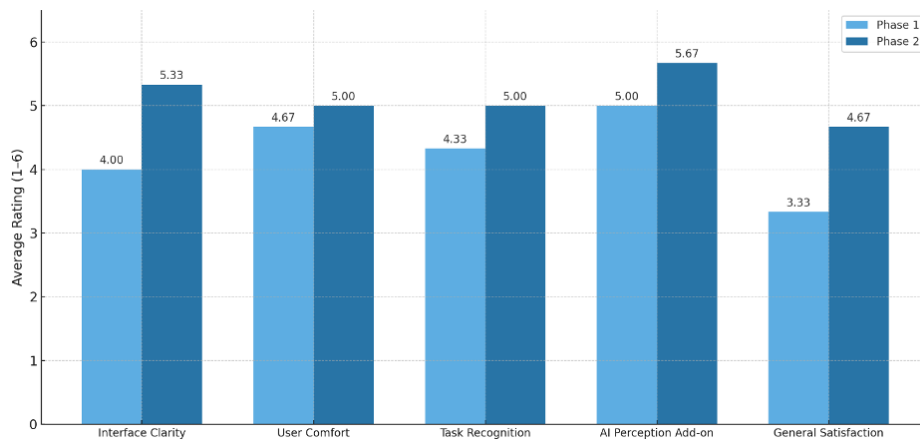


Figure 61. Dr. Patrick - UX Ratings Comparison (Phase 1 vs Phase 2)

The chart above presents Dr. Patrick's feedback across both iterations of the e-Hospital prototype. Ratings were collected using a Likert scale (1–6), where 6 indicates the highest level of agreement or satisfaction.

Interface Clarity: In Phase 1, interface clarity was moderately rated (4.0), reflecting some usability concerns. However, in Phase 2, a marked improvement to 5.33 was observed, suggesting that design changes contributed significantly to clearer visuals and better navigation. This improvement aligns with Dr. Patrick's qualitative feedback highlighting "a much clearer layout and more intuitive access to information."

User comfort remained high in both phases, with a slight increase from 4.67 to 5.0. This suggests that while the interface was already perceived as user-friendly, incremental enhancements in Phase 2, such as spacing, labeling, and reduced cognitive load, made interactions more pleasant.

Task recognition saw a steady score of 4.33 in Phase 1 and improved slightly to 5.0 in Phase 2. This indicates that the modifications improved Dr. Patrick’s ability to identify and perform key tasks with less ambiguity.

AI Perception Add-on: The AI feature was well-received across both phases. Scoring a consistent 5.0 in Phase 1 and increasing to 5.67 in Phase 2, this dimension shows that Dr. Patrick recognized the added value of AI support for decision-making and patient tracking.

General satisfaction showed the most substantial improvement, rising from 3.33 in Phase 1 to 4.67 in Phase 2. This increase is indicative of overall enhancements made between iterations, supporting a broader acceptance of the redesigned platform.

These results support the conclusion that iterative refinements based on user feedback, particularly in layout, labeling, and task flow, contributed significantly to the platform's usability and perceived value.

Enhancing accessibility and confidence: Insights from the patient users

Patient Voices: Qualitative reflections on the redesign

The most compelling evidence of improvement emerged from the voices of patients themselves. Their narratives highlighted not only the functionality of the platform but how it felt to use it, a critical aspect in health tech design, where emotional comfort and usability go hand in hand.

Phase 1: In the baseline version, participants frequently described the interface as “confusing,” “too crowded,” and “unclear.” Tasks like booking appointments or checking lab results were possible but left users feeling unsure about whether they had completed the task correctly. Some even mentioned hesitation or frustration in navigating menus, with certain features “hidden” or “not labeled clearly.” The interface lacked a coherent flow and demanded unnecessary mental effort from users who expected clarity and reassurance. One participant noted: *Some labels are not intuitive. I would never have imagined that 'referral' could be found under 'Planning'.*

Phase 2: In stark contrast, the redesigned prototype introduced intuitive layout groupings, clearer section headers, and logically sequenced actions. This led to a dramatic shift in user sentiment.

Participants reported a greater sense of confidence, ease, and control, as described in the following quotations about the redesign prototype:

“The steps were clear and straightforward, and I was able to book an appointment in just a few clicks without any confusion.”

“Key actions combined into the dashboard – Less click needed to complete key actions.”

Tasks that previously felt burdensome became more fluid. Users can scan content quickly, anticipate the next step, and complete actions with little to no uncertainty.

Participants also appreciated emotional design cues, such as friendly labeling and simplified forms, which helped them feel less overwhelmed.

Quantitative ratings – Likert scale

Alongside open feedback, participants rated their experience using a 1–6 Likert scale across the same five key UX dimensions (interface clarity, user comfort, task recognition, AI perception add-on, and overall satisfaction). The average scores of all participants show a notable improvement across every area.

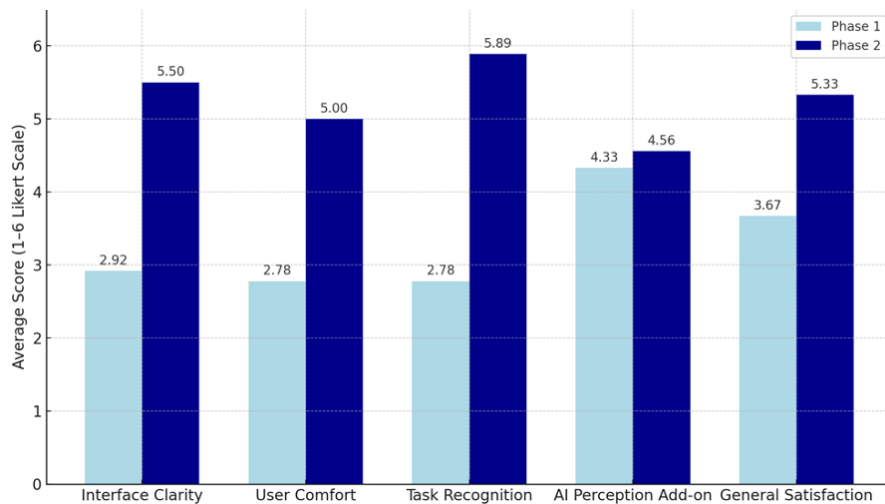


Figure 62. UX Ratings - Patient Profile (Current Stage of the Platform vs Redesigned Prototype)

The chart above compares average scores from **Phase 1 (current stage of the platform)** and **Phase 2 (redesigned prototype)** across these dimensions.

Interface Clarity showed the most dramatic improvement, rising from **2.92** in Phase 1 to **5.50** in Phase 2. This indicates that participants found the redesigned interface significantly more understandable and visually organized. This aligns with qualitative insights from the participants.

User Comfort also increased notably, from **2.78** to **5.00**, suggesting that the new layout provided greater emotional ease and confidence during task completion.

Task Recognition, initially rated at **2.78**, soared to **5.89** in the redesigned prototype. This suggests that participants were much better able to identify and understand what actions were expected of them.

AI Perception Add-on received moderate improvements (from **4.33** to **4.56**), suggesting that while the AI-supported features were appreciated, they weren't the primary source of user frustration in the initial design.

General Satisfaction increased from **3.67** to **5.33**, indicating a strong overall positive response to the redesigned interface. This reflects a higher level of trust, comfort, and usability in the updated prototype.

The quantitative ratings confirm what was observed in the qualitative walkthroughs: **the redesigned interface substantially improves the patient experience**. Particularly in the areas of clarity, comfort, and task flow, the prototype addresses major pain points identified during Phase 1. This strongly suggests that the redesign succeeded in enhancing usability for non-clinical users, a crucial target group for digital health platforms aiming to support patient self-management.

Together, the qualitative and quantitative findings point to a more accessible, less frustrating, and more empowering digital experience for patients navigating essential health tasks. These results affirm that the structural, visual, and flow adjustments made between iterations successfully improved both usability and emotional engagement, two pillars of a patient-friendly healthcare platform.

Conclusion

The testing results underscore the value of a user-centered design process, where iterative feedback and real-world context shape every stage of development. The redesigned prototype consistently outperformed the current platform across all measured UX dimensions for both patient and clinician profiles. Notably, average scores improved significantly in **interface clarity, user comfort, task recognition, AI perception, and general satisfaction**.

- **Doctors:** Overall satisfaction improved from **3.33** to **4.67**
- **Patients:** Overall satisfaction improved from **3.67** to **5.33**

By directly addressing workflow inefficiencies and emotional barriers, the new E-Hospital design emerges as a more accessible, intuitive, and trustworthy tool, empowering users to manage care more confidently and efficiently.

These findings not only validate the new design direction but also lay a solid foundation for future enhancements, including personalization, accessibility improvements, and transparent AI integration with user-centric control mechanisms.

The research approach was informed by best practices in UX methodology. As outlined by Pannafino and McNeil (2017) in *UX Methods: A Quick Guide to User Experience Research Methods*, selecting context-appropriate techniques is essential for capturing actionable user insights. Their framework guided our strategy for usability testing and feedback integration, ensuring that design refinements were not only responsive to user behavior but also methodologically sound.

Final Prototype

Overview

The final prototype integrates all improvements made since the Beta Release, incorporating feedback from users, usability testing, and expert input. While no changes were needed on the patient portal, the doctor portal underwent refinements, including new interaction patterns, updated information architecture, and added personalization through AI. These enhancements aim to improve usability and support a scalable, clinically aligned experience.

Key Design Improvements

Doctor Portal

- **AI Assistant Button Redesign and Relocation:**

The AI assistant entry point was redesigned to be more futuristic and moved to the persistent top banner across all pages, to ensure doctors can access it quickly from anywhere in the interface. This change supports smoother decision-making without needing to navigate away.

- **Improved Task Filtering:**

The “Upcoming Tasks” section now offers filter options. Doctors can choose to view Today’s tasks, completed tasks, or All tasks, depending on their planning needs. By default, completed tasks are hidden to reduce noise.

- **Color Coding Refinement:**

Appointment status colors were adjusted to reduce cognitive load during busy clinic hours. New status colors are aligned; different line styles were used to differentiate among statuses.

- **Clickable Bill and Sign Indicators in Appointment List:**

Visual indicators showing whether a case has been billed and signed are changed to clickable buttons, doctors can better manage which case needs to follow-up in the end of the day.

- **Enriched Summary Metrics**

New Summary metrics of “Unbilled Visits” and “Unsigned Notes” were added to align with the billed and signed indicators in the appointment list. Doctors can get an overview of unfinished cases.

For the final prototypes of Doctor Dashboard, please refer to the following Figure 63.

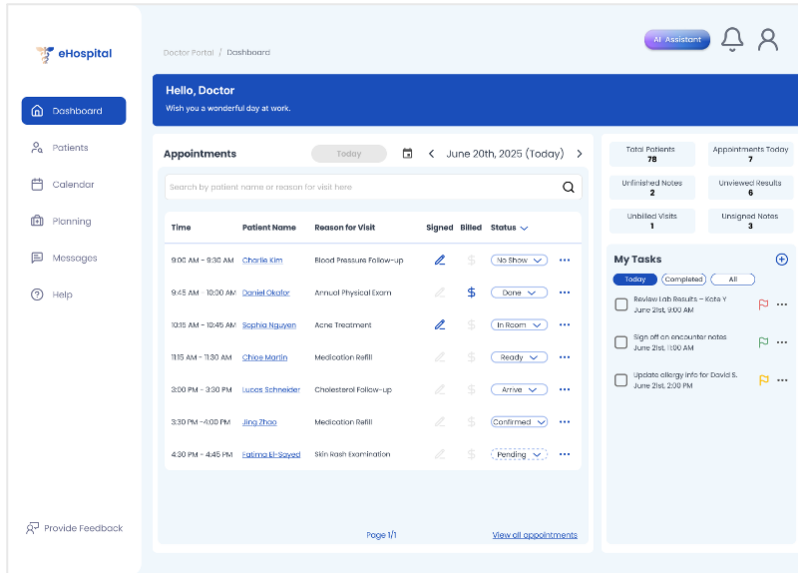


Figure 63.Redesign - Doctor Dashboard (Final Prototype)

Patient Overview

- Enlarged the encounter workspace:

Since it is the main working area for users, the team expands the section to make it clearer and more usable. For the visit history, we redesigned it to a two-column layout to allow users to choose from the list on the left and to see the summary on the right.

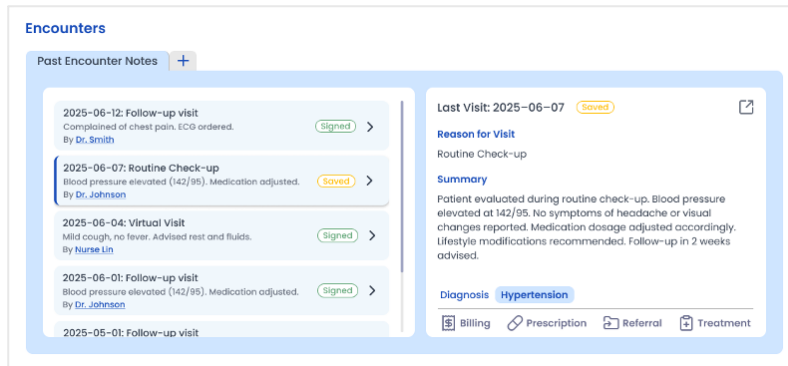


Figure 64.Redesign - Patient Overview - Past Encounter Notes

- **Rearrange the order of key action buttons in the encounter section:**

Based on the second testing with Dr. Patrick, we understood that billing is highly related to diagnosis. Thus, we rearranged the order of buttons to billing, prescription, referral, and treatment based on the frequency of use, making the workflow more natural for users.

The screenshot shows a user interface for adding a new encounter. At the top, there's a header 'Encounters' and a list of 'Past Encounter Notes' with dates '2025-06-12' and '2025-07-18'. Below this is a large text input area for notes. The input area has tabs for 'Note', 'Examination', and 'Impression/Plan'. A 'Take notes with AI' button is located in the top right corner of the input area. Below the input area is a '+ Diagnosis' button. At the bottom of the input area, there are four icons: a dollar sign for 'Billing', a pill for 'Prescription', a document for 'Referral', and a plus sign for 'Treatment'. 'Save' and 'Sign' buttons are at the bottom right of the input area.

Figure 65.Redesign - Patient Overview - Add a New Encounter

Include more information in the record hub:

Based on this change, the page layout has changed, making the record hub section larger as well. This allows us to include more information on the information cards, making it more efficient for users to access key information.



Figure 66.Redesign - Patient Overview - Record hub

About Implementation

Patient Portal

In the final prototype, our implementation efforts focused on the patient-side Session section, which includes three key subpages: Book New Appointment, View All Appointments, and Surgery Plan, as well as final adjustments to the Patient Dashboard. Our goal was to deliver a smooth, consistent, and user-friendly experience aligned with the design and usability goals outlined in the introduction.

Patient Dashboard

We completed the dashboard implementation by organizing upcoming appointment previews and providing shortcuts to medications, profiles, and the Session section. The layout followed our design system closely and ensured responsiveness and clarity across screen sizes.

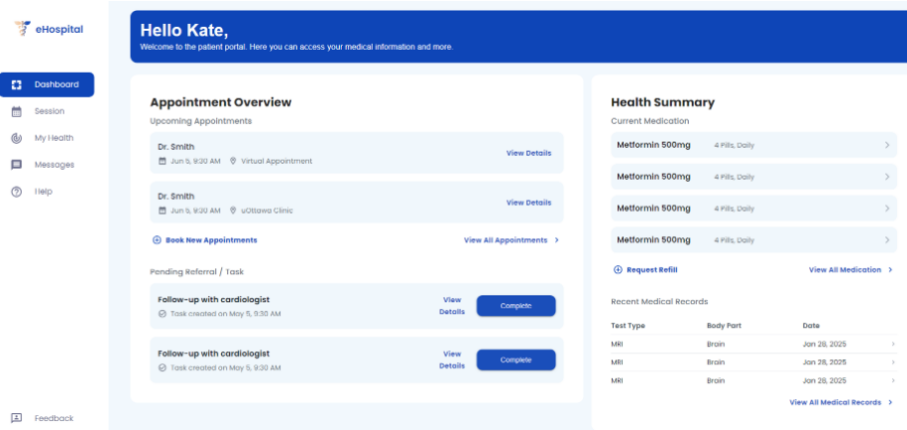


Figure 67. Patient Dashboard Implementation

Book New Appointment

This multi-step form allows patients to select a specialty, doctor, and available time slot. We implemented dropdowns with search functionality, a calendar view, and a confirmation modal. This flow was built to reduce confusion and cognitive load, especially for first-time users. Each step is clearly labeled and dynamically updated based on previous selections.

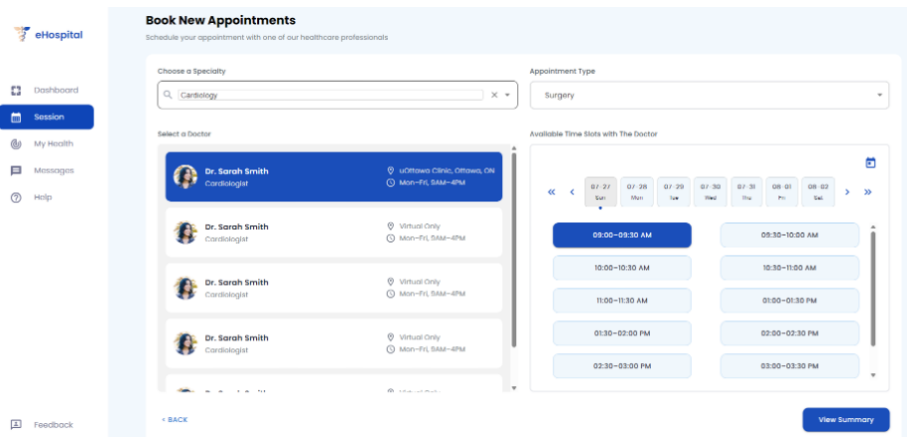


Figure 68. Book New Appointments Implementation

View All Appointments

This page enables users to review their upcoming, past, and cancelled appointments. We used tab-based navigation for filtering and integrated a scrollable list of cards with appointment details. Patients can click each entry to open a modal showing time, location, doctor, and appointment notes. This structure supports task recognition and makes it easy to access historical records.

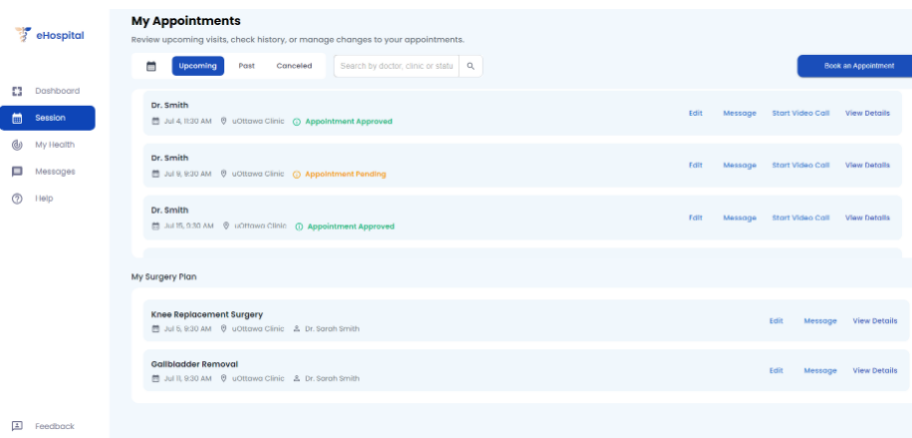


Figure 69. View All Appointments Implementation

Surgery Plan

We developed a simplified version of the Surgery Plan section, where patients can view upcoming or completed surgery arrangements. The information is displayed in a clean, table-like format with conditionally rendered status indicators, providing clarity without overwhelming the user.

Doctor Portal

In the final prototype the focus was on the patient overview. This involved converting high-fidelity wireframes into functional React components, ensuring that the interface supported real clinical workflows with minimal cognitive load. Our goal is to achieve consistency and better user experience. The doctor's dashboard was adjusted, a new section for tasks numbers and task list was added. In patient overview, the overall look was adjusted to fit the information and make it flow. This way it is easier to read and follow.

Doctor Dashboard

Integrated upcoming and pending appointment panels. The appointments list had buttons ready to change the status of the appointments. Summary metrics (e.g., total tasks, total patients) were also added ensuring a smooth view of the information. Lastly, upcoming tasks were added to ensure doctors have access to their daily clinical practices.

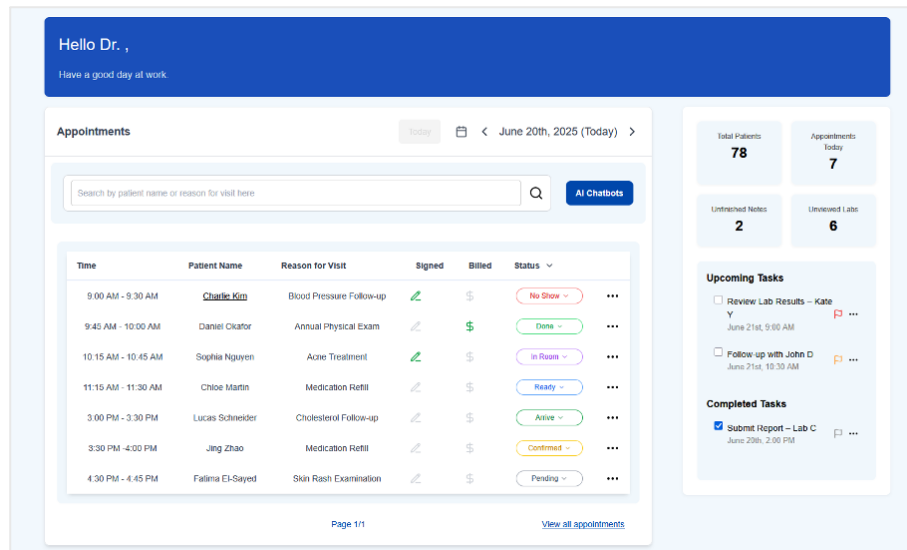


Figure 70. Doctor Dashboard Implementation

Patient Overview

The patient overview had a massive overhaul. First a new UI was used to showcase basic patient information (e.g., Age or Name). The medical record section was moved from the dashboard section to the patient overview and has a new look. The encounter section was enhanced for doctors to easily add their notes in. Lastly the records section at the bottom has all type of records for that patient.

Charlie Kim

Birthdate: Jun 13, 1988 | Last Billed: Jan 13, 2025
 Age: 37 yr | MRP: Dr. John Willis
 Sex / Gender: F / M | Tel: 111-111-111
 186 cm / 80 kg | test@testing.com
 120 / 80 mmHg | Ottawa ON K1N 9E2

Medical History Summary

Allergies Penicillin, Shellfish	Past Psychiatry History Anxiety disorder (2019)
Past Medical History Hypertension, Type 2 Diabetes	Family History Father: CAD, Mother: DM
Prior Medications Metformin, Lisinopril	Pregnancy 2 pregnancies, 2 live births

Encounters

Past Encounter Notes

- 2025-06-12: Follow-up visit
Complained of chest pain. ECG ordered.
By Dr. Smith Signed
- 2025-06-07: Routine Check-up
BP elevated (140/96). Medication adjusted.
By Dr. Johnson Saved
- 2025-06-04: Virtual Visit
Mild cough, no fever. Advised rest and fluids.
By Nurse Lee Signed
- 2025-05-01: Follow-up visit
Blood pressure elevated. Medication adjusted.
By Dr. Johnson Signed

Last Visit: 2025-06-12 Signed

Reason for Visit
Follow-up for chest pain

Summary
Patient presents with intermittent chest pain without associated shortness of breath or dizziness. Vitals stable and physical exams unremarkable. No acute findings. Monitoring recommended.

Diagnosis: Chest Pain | Appoint chest pain

Diets | Prescription | Referral | Treatment

Figure 71. Patient Overview Part 1

Record Hub

Surgeries | Treatments | Reports | Health Trends | Referrals | Documents

Surgical History

Cesarean Section 2021-11-10 Site: Lower abdomen Method: Surgical (C-section) Anesthesia: Spinal anesthesia Dr. Maria Chen	Recovery: Discharged after 3 days, recovered in 6 weeks Complications: Mild wound infection	Completed
Appendectomy 2020-08-10 Site: Lower abdomen Method: Laparoscopic Anesthesia: General anesthesia Dr. Johnson	Recovery: No follow-up required Complications: None	Completed
Gallbladder Removal 2018-03-22 Site: Lower abdomen Method: Laparoscopic Anesthesia: General anesthesia Dr. Smith	Recovery: Mild digestive symptoms for 1 week Complications: None	Completed

Figure 72. Patient Overview Part 2

Code examples

All the code snippets you see below are from the 2 repositories being pushed to the main branch.

```
const getStatusStyle = (status) => {
  switch (status) {
    case "No Show":
      return "text-red-600 bg-white-100 border border-red-400";
    case "Done":
      return "text-green-600 bg-white-100 border border-green-400";
    case "In Room":
      return "text-purple-600 bg-white-100 border border-purple-400";
    case "Ready":
      return "text-blue-600 bg-white-100 border border-blue-400";
    case "Arrive":
      return "text-green-700 bg-white-100 border border-green-600";
    case "Confirmed":
      return "text-yellow-600 bg-white-100 border border-yellow-400";
    case "Pending":
      return "text-gray-600 bg-white-100 border border-gray-400";
    default:
      return "text-gray-600 bg-white-100 border border-gray-300";
  }
};
```

Figure 73. Code sample from the status button of appointments table

```
/* Medical History Panel */
<div className="w-full lg:w-[60%]">
  <div className="bg-[#d8eafe] rounded-xl p-1 h-full">
    <div className="bg-white rounded-lg p-6 h-full shadow-sm flex flex-col items-start">
      <h2 className="text-2xl font-bold text-[#1f3c88] mb-4 text-left w-full">
        Medical History Summary
      </h2>
      <div className="grid grid-cols-2 gap-3 w-full">
        {
          (
            {
              label: "Allergies", value: "Penicillin, Shellfish" },
              {
                label: "Past Psychiatry History", value: "Anxiety disorder (2019)" },
              {
                label: "Past Medical History", value: "Hypertension, Type 2 Diabetes" },
              {
                label: "Family History", value: "Father: CAD, Mother: DM" },
              {
                label: "Prior Medications", value: "Metformin, Lisinopril" },
              {
                label: "Pregnancy", value: "2 pregnancies, 2 live births" },
            )
          ).map((item, i) => (
            <div
              key={i}
              className="bg-[#f1f8ff] hover:bg-[#d8eafe] transition rounded-md px-4 py-2 flex justify-between items-center w-full"
            >
              <div>
                <div className="font-semibold text-[#1f3c88] text-sm">{item.label}</div>
                <div className="text-[13px] text-gray-600">{item.value}</div>
              </div>
              <span className="text-gray-500 text-lg">🔍</span>
            </div>
          ))
        }
      </div>
    </div>
  </div>
```

Figure 74. Code sample from the medical records of Patient Overview

AI Assistant Hub

AI chatbot was added originally in the doctor dashboard but was moved in the final wireframe to the top right section of the dashboard to be used in every single page. This way at any point a doctor can access the AI chatbot.

Design Corrections and Validation

- Corrected UX text typos, such as label inconsistencies in headings (e.g., “Book New Appointment” title).
- Adjusted modal shadow styling for better visual consistency across components.
- Ensured alignment with spacing and font weight based on the Figma design audit.

Meeting Requirements

- Enhancing interface clarity through clean layouts and intuitive grouping.
- Improving task flow for appointments and health management.
- Ensuring usability for diverse patient types, including those less familiar with digital systems.
- Matching the visual and interaction patterns established in the introduction and design system.

Problem-Solution Alignment

Through the three stages of redesign (MVP, Beta Release, and Final Release), in the final handoff, we have addressed the problems discovered from the beginning of the project.

- The patient portal supports not only easy booking of an appointment but also sets the ground to continue empowering patients to have more control over their own data (medications, appointments, etc.).
- Clinical intuition is prioritized over system logic with a rigid, less intuitive data structure.
- Patient interface clarity improved hugely so as task recognition.
- New information architecture driven by user behavior, instead of backend logic.
- AI elements were perceived as useful and generate trust even if they couldn’t be tested (driven by the user feedback we built with doctors for example)

Ethical Diversity Considerations

Ethical issues or concerns

The ethical issue was related to our decision to use mock data while building the wireframes for both Patient and Doctor portals, due to the limited backend access we had and time constraints. This could lead to an ethical concern about the accuracy, including the risk of designing a

functional prototype that was not necessarily displaying real information from a Patient or a Doctor. To minimize this risk, we asked a real Doctor to walk through the prototype twice and flag anything that felt unintuitive from his daily experience and practice. This way, we were able to shape key sections for the Doctor Patient Overview and support the Common Good, ensuring that our redesign reflected the real the mental map and needs of healthcare professionals. At the same time, it allowed us to challenge our early design based on our own assumptions and available data and instead making users part of the design process.

The second ethical concern was regarding the inclusion of visible AI-related features, which we decided to group under a single button on key pages. By doing so, we recognized the risk that Doctors could misunderstand the role of the AI, thinking they could actively take part in their decision making without their consent. To reduce this perception, we grouped all AI features living in different components of the original Doctor Patient Overview page under a single button which requires the user to actively trigger it to select the type of support they need from the AI. During the usability testing, we asked open-ended questions to explore the Doctor's understanding and expectations around these features. His input confirmed that our design approach helped reduce confusion, and as a result it also aligns with the Right Lens by protecting the user's right to informed use and transparency while making a choice.

Diversity and Inclusion

The following are some principles that were considered into the design and development process of our project:

Equity: Some of our objectives for both portals embraced this principle by ensuring that the redesign simplified navigation paths. For example, we reduced the number of steps required to complete a task and used clear and friendly language to make portals more inclusive. This way, we incorporated equity in access for patients who may not be familiar with online healthcare applications.

Well-being: We respected this principle by considering how our redesign might affect users emotionally (comfort). For example, patients with special medical conditions could feel anxious navigating a portal that was not intuitive and difficult to understand. As a result, we created wireframes that were visually clean, grouping information logically to reduce fatigue.

Meanwhile, for Doctors our redesign prioritizes relevant clinical information so they could focus more on care without feeling mentally overloaded.

Democratic Participation: Our efforts were aligned to this principle by including real users' feedback for both portals to validate key pages. Their insights directly influenced how we shaped appointment flows, encounter notes and dashboards. Even though we did not reach a wide sample of Doctors, their needs were considered in every iteration including an interview with a different physician who showed us other healthcare systems he used. This approach helped us ensuring our early design decisions were not only based only on our assumptions.

Reflection and Lessons Learned

For design and research

Reflecting on the design and research journey throughout the E-Hospital project, several valuable lessons emerged, spanning user research, iterative prototyping, stakeholder collaboration, and ethical design decision-making. The experience reinforced the importance of grounding every design choice in real-world needs, maintaining agility, and embracing feedback as a catalyst for improvement.

Key challenges

One of the most persistent challenges was navigating design ambiguity in a complex domain like healthcare, where workflows are diverse, user expectations vary significantly, and data sensitivity is paramount. At the outset, our team had to interpret abstract usability concerns from the client and translate them into actionable design hypotheses, without access to full backend integration or analytics from the live platform.

Also, every design decision takes time and experience. We are aiming to provide a platform that looks reliable and professional to the users, which means its usability and aesthetics are equally important. To balance them, the design team spent much time referencing best practices, creating the best UI based on our context and understanding.

Additionally, conducting usability testing with limited participants required us to focus on depth over breadth. We had to carefully select tasks, craft relevant questions, and extract meaningful

insights from a small but diverse group, including doctors and patients with different levels of digital fluency.

Another challenge was designing AI readiness without having functioning AI components available to test. We had to rely on scenario-based testing and exploratory questions to anticipate how users might perceive and interact with future AI features.

Successes and achievements

Despite these challenges, the project produced several key successes:

- We implemented a user-centered research methodology, conducting pre/post usability studies that demonstrated measurable improvements in interface clarity, emotional comfort, and task recognition.
- Our iterative design process allows us to incorporate feedback at every phase, from client and TA reviews to real user insights. This helped refine both the Doctor Portal (modular dashboards and doctors' workspace) and the Patient Portal (streamlined booking and health data visibility).
- The redesign promoted inclusive design principles, reducing cognitive overload for doctors and enhancing digital accessibility for patients unfamiliar with healthcare platforms.
- Our application of benchmarking, combined with direct walkthroughs of platforms like TELUS and OSCAR, allowed us to align our design decisions with industry's best practices.

Key takeaways for future projects

1. **Designing with real users, early and often, matters more than designing for users.**
Our most impactful design improvements came from letting patients and doctors test, comment, and challenge our assumptions.
2. **Usability is not just about what works, it's about how people feel when using it.**
Emotional comfort, trust, and confidence are as important as task success, especially in health-related systems.

3. **Less is often more.**

Simplifying layouts, grouping content logically, and minimizing clicks resulted in both better feedback and better performance.

4. **Prepare for uncertainty but document the rationale.**

When backend data or AI features weren't available, we used assumptions, mock data, and clear documentation to maintain transparency and inform future developers or researchers who will continue the work.

Application for future endeavors

These reflections will carry forward into our future design and research efforts, particularly in complex, high-impact domains like healthcare. We now understand more deeply the value of designing with empathy, testing with purpose, and collaborating across disciplines to build solutions that are not only functional but meaningful.

For implementation

Patient Implementation

Working on the implementation of the patient-side features for the E-Hospital project was both challenging and rewarding. One of the biggest challenges was inheriting a partially functional codebase with legacy styling and inconsistent component structures. Setting up the environment and understanding how everything connected, from routing to state management, took significant time and patience. However, this process helped us learn how to navigate and contribute to an existing codebase effectively, which is a critical skill in real-world software development.

One major success was completing the entire Session section, including Book New Appointment, View All Appointments, and Surgery Plan, in a way that aligned with both the visual and functional expectations. We also became more confident in working with Material-UI v4, improving our skills in layout responsiveness, conditional rendering, and modal handling.

Doctor Implementation

Implementing the doctor side of the E-Hospital platform taught us the importance of aligning technical development with real-world clinical workflows. Translating complex Figma designs

into functional, responsive interfaces required not just front-end skills but also a deep understanding of how doctors navigate digital tools under pressure. We learned how to prioritize clarity and efficiency, especially when building components like the Encounter section, where doctors need to document notes, prescribe medications, and manage billing within tight time constraints. Working closely with the UX team helped me recognize how even small layout or interaction choices can affect usability in a healthcare setting, where speed and accuracy matter.

This experience also strengthened our ability to work within an existing codebase and resolve conflicts with legacy components. We faced challenges like inconsistent styling and fragmented component logic, which pushed me to adopt better practices in modular design and CSS management. Implementing AI assistant entry points and ensuring consistency with the design system taught me how to maintain scalability and visual cohesion across a large interface. Overall, this project reinforced the value of cross-functional collaboration, iterative refinement, and user empathy in building digital tools that genuinely support healthcare professionals.

Another challenge was managing design-to-code translation. While our Figma designs were detailed, some spacing, text, or shadow styles required adjustments during implementation. Through regular communication with the design team, especially Yu-Chen, we learned how to identify minor UX mismatches, validate changes with the designer, and apply corrections in code to maintain consistency across pages.

A key takeaway is the importance of close collaboration between design and development. Even small changes, such as adjusting a shadow or fixing a typo, can significantly impact the user experience. We also recognized the need to balance attention to detail with time constraints, learning when to polish and when to prioritize delivery.

Moving forward, we will carry these lessons into future projects, especially the ability to adapt to existing systems, communicate effectively with cross-functional teams, and focus on building user-friendly, accessible interfaces that address real-world needs.

For project management

Despite the fact our team exceeded the total amount of hours to complete the project, we maintained a positive dynamic, based in mutual respect, in order to deliver value to our Client,

Technical Advisor, Technical Support and future users. Foster camaraderie and nurture good communication were key for us. We created a Teams channel that served to store documentation, share daily status updates and as a space to have hybrid discussions, internal and retrospective meetings after each milestone. These tactics made us improve after we achieved each milestone for the course. In addition, they helped us to stay aligned while following an Agile approach. However, it is important to note that in some cases, some teammates with full or partial-time jobs struggled to report their daily status updates.

In addition, by reaching out weekly our Client and Technical Advisor and Technical Support, our team got more information to document decisions and get stakeholder feedback, approval and guidance. All were recorded followed by a minute meeting. These activities allowed us to get clarity on what was expected each week and organize our efforts according to our roles.

Some recommendations from a project management perspective involve improving the communication between designers and developers by sharing end-of-day summaries and explore how early integration of AI applications and tools can streamline repetitive tasks and boost productivity. For example, using ChatGPT to generate key takeaways based on an internal meeting or with the Client or Technical Advisor meeting recording.

Conclusion

The E-Hospital redesign project represents a successful application of user-centered design, collaborative iteration, and agile implementation in the context of digital health innovation. From the outset, our team was tasked with improving the usability, consistency, and AI-readiness of the existing Doctor and Patient portals. Through a structured design process and iterative validation with real users and clinical experts, we delivered on these objectives with measurable outcomes.

We began by conducting a thorough UX audit and benchmarking against industry leaders such as TELUS, Maple, and OSCAR, while also engaging directly with clinicians and patients to ground our design in real-world expectations. This research phase informed the redesign of critical user flows and page structures, specifically the Doctor Dashboard, Patient List, and Patient Overview, along with the Patient Dashboard, Appointments, and My Health sections.

Across all iterations, from the MVP to the final prototype, we prioritized clarity, task alignment, and cognitive ease. Our wireframes and interactive Figma prototypes incorporated modular layouts, simplified navigation, and clearly defined AI touchpoints, all of which were validated through usability testing with both user groups. Notably, all five UX metrics we tracked (interface clarity, user comfort, task recognition, AI perception, and overall satisfaction) showed significant improvement for both the Doctor and Patient portals.

On the implementation side, we successfully translated the design specifications into a functional front-end experience. React components were developed with attention to visual fidelity and layout integrity, and design system elements were standardized for future scalability. The result is a usable, intuitive, and scalable platform ready for real-world deployment and future integration of AI features.

In summary, this project achieved its core goals by:

- Delivering a high-fidelity, testable redesign aligned with user needs
- Producing measurable improvements in usability and satisfaction
- Implementing core components in code consistent with design prototypes
- Establishing a flexible design foundation for future feature growth.

These achievements reflect not only a redesign, but a transformation, positioning the E-Hospital platform as a modern, inclusive, and user-friendly tool capable of supporting both clinical workflows and patient empowerment in the digital age.

Potential impact and future applications of our solution

Our E-Hospital redesign for Patients and Doctors has the potential to enhance the digital healthcare experience for both users by simplifying routine tasks and streamlining workflows; and by enabling more meaningful interactions to reduce cognitive load. As Doctors benefit from faster access to key information, Patients receive a more personalized and responsive experience.

Furthermore, Doctors can intuitively explore AI tools as trusted companions to support, but not replace, their clinical judgement. In this sense, our prototype and implementation lay the foundation for integrating AI features in an intuitive approach, while promoting ethical inclusive design.

Even though this project focused on two roles only, with more iteration and proper UX research, the platform can grow to support other portals targeting Admin, Clinic Staff, Hospital Management, Pharmaceutical Offices, E-Pharmacy and Pharmacy-Admin to offer a consistent long-term value beyond and compete with other EHR systems available in the market.

Recommendations for further development or research

About design

AI Integration

While we designed entry points for an AI assistant across the platform, future development should focus on integrating explainable AI features. For clinical adoption, doctors need to understand why the AI makes a certain suggestion (e.g., differential diagnosis, treatment pathways). Clear rationales or evidence-based links should accompany recommendations. A recommendation would be to implement transparent AI feedback loops. Also, please consider patients' consents when doctors are accessing voice-empowered AI features.

Clinical Workflow Simulation

Although we tested the design with doctors like Dr. Patrick and reviewed competitor platforms, the next phase should include full in-clinic simulations of the prototype. This would assess real-time usability, efficiency during live consultations, and compatibility with EHR systems. Future development teams can consider partnering with a clinical setting or simulation lab to conduct contextual inquiries under time constraints.

Design with Consistency

A product that looks consistent makes users feel professional and reliable. For future UI redesign, please follow the design guidelines in the design system. With different goals or considerations, updating the design system is necessary, but please make sure you're changing them from the main components in the design system page of the Figma design file.

About implementation

To enhance the scalability and reliability of the platform, future development should focus on completing and fully integrating backend APIs. Some patient-side features, such as appointment

booking, medication refills, and surgery plans, relied on temporary data or partial mock-ups due to missing or unfinished API endpoints during our development timeline.

We recommend prioritizing the completion of these remaining endpoints and ensuring clear, consistent documentation to support seamless frontend-backend communication. Full API integration will enable the platform to transition from a working prototype to a deployable product.

For long-term success, we also recommend implementing unit and integration tests early in the development process. Automated testing would help reduce regression bugs, especially as new features are added or existing logic is refactored.

Lastly, continued collaboration between frontend and design teams will remain critical. Even after designs are finalized, unexpected issues can arise during implementation. Maintaining regular check-ins and validation reviews helps ensure that what's built remains aligned with the intended user experience.

By addressing these areas, future development can achieve a more complete, maintainable, and scalable E-Hospital platform.

Links and Tools

List of links to Figma files

Figjam - UX audit and information architecture planning:

<https://www.figma.com/board/G6aehGrZwhgzT7WVHhSlkg/5902-Team-workspace---Group-3?node-id=0-1&t=0NeoQkDs8WM2wFSG-1>

Figjam - UX research results, feedback and ideation (Doctor):

<https://www.figma.com/board/G6aehGrZwhgzT7WVHhSlkg/5902-Team-workspace---Group-3?node-id=539-6013&t=0NeoQkDs8WM2wFSG-1>

Figjam - User testing (Card sorting for doctor users):

<https://www.figma.com/board/G6aehGrZwhgzT7WVHhSlkg/5902-Team-workspace---Group-3?node-id=603-5790&t=0NeoQkDs8WM2wFSG-1>

Figjam - Implementation Validation:

<https://www.figma.com/board/G6aehGrZwhgzT7WVHhSIkg/5902-Team-workspace---Group-3?node-id=907-5954&t=0NeoQkDs8WM2wFSG-1>

Figma – eHospital redesign (Final Handoff):

https://www.figma.com/design/GsqGLPjV6r0gKhrgIPBuFo/eHospital-Redesign_5902_25Summer?node-id=3376-15227&t=s5z327Gm2Bwu5EfA-1

Figma – Design system: https://www.figma.com/design/GsqGLPjV6r0gKhrgIPBuFo/eHospital-Redesign_5902_25Summer?node-id=137-3990&t=s5z327Gm2Bwu5EfA-1

Figma – Doctor prototype:

https://www.figma.com/proto/GsqGLPjV6r0gKhrgIPBuFo/eHospital-Redesign_5902_25Summer?page-id=1131%3A11240&node-id=2224-27925&viewport=-5234%2C-78%2C0.85&t=lHyvF3rWGtajUaFG-1&scaling=scale-down&content-scaling=fixed&starting-point-node-id=1648%3A11526&show-proto-sidebar=1

Figma – Patient prototype:

https://www.figma.com/proto/GsqGLPjV6r0gKhrgIPBuFo/eHospital-Redesign_5902_25Summer?page-id=0%3A1&node-id=1732-35579&viewport=-2948%2C3634%2C0.64&t=9H6AeYfWAsgowAk9-1&scaling=min-zoom&content-scaling=fixed&starting-point-node-id=1732%3A35579

Links for GitHub Forks

Doctor Fork: <https://github.com/ahmadmozan/UIDesignTest.git>

Patient Fork: <https://github.com/Yang98-git/ottawa-ehospital-2025summer.git>

API List - Patient Dashboard APIs

- GET /api/appointments/upcoming
Returns a list of upcoming appointments with doctors and clinics.
- GET /api/tasks/pending
Retrieves follow-up tasks or referrals assigned to the patient.

- GET /api/medications/current
Returns the current medications list for the logged-in patient.
- GET /api/medical-records/recent
Fetches recent test results and medical record summaries.
- GET /api/medications/:medicationId
Gets detailed information about a specific medication (dose, doctor, start date, instructions).
- POST /api/refills/request
Submits a request to refill selected medications and sends it to the patient's preferred pharmacy.

Appointment Details & Session APIs

- GET /api/appointments/:appointmentId
Retrieves detailed information for a specific appointment (doctor, time, location, purpose, etc.).
- PUT /api/appointments/:appointmentId
Allows the patient to edit or reschedule an appointment.
- GET /api/appointments/history
Returns the full history of past and cancelled appointments.
- POST /api/appointments/book
Submits a new appointment request, including specialty, appointment type, doctor, and selected time slot.
- GET /api/surgeries/:patientId
Fetches the full list of scheduled and completed surgeries for the patient.
- GET /api/surgeries/detail/:surgeryId
Retrieves detailed information about a specific surgery, including doctor, clinic, date, duration, risks, and post-op plans.

API List (Doctor) - Dashboard APIs

- **GET /api/appointments/upcoming**
Returns a list of upcoming appointments for the logged-in doctor.
- **GET /api/tasks/today**
Retrieves the list of pending or scheduled tasks assigned for the current day
- **GET /api/appointments/summary-metrics**
Returns counts of unsigned notes, unbilled visits, and today's total patients.
- **GET /api/notifications**
Fetches unread alerts such as referral responses or lab results.
- **GET /api/calendar/events**
Returns scheduled events, appointments, and surgery blocks for the doctor.

Patient Overview APIs

- **GET /api/patients/:patientId/overview**
Retrieves general patient demographics, allergies, and health stats.
- **GET /api/patients/:patientId/history**
Returns structured medical history (e.g., medications, conditions, psychiatric background).
- **GET /api/patients/:patientId/encounters**
Lists past encounter summaries and metadata
- **POST /api/patients/:patientId/encounters**
Creates a new encounter entry (examination, notes, plan).
- **PUT /api/encounters/:encounterId**
Updates an existing encounter note.
- **GET /api/encounters/:encounterId**
Retrieves full details of a specific encounter.
- **POST /api/diagnoses/add**
Adds one or more diagnoses to a patient's active encounter.

- **POST /api/prescriptions/create**
Submits a new prescription for the patient.
 - **POST /api/referrals/create**
Creates a referral order for a specialist or lab.
 - **POST /api/billing/submit**
Submits billing information tied to a specific visit.
 - **GET /api/patients/:patientId/record-hub**
Returns patient documents: surgeries, treatments, reports, referrals, and trends.
- Data collection protocol and tools

User Research Guide for Patients – E-Hospital Platform Redesign (Before & After Testing)

Scope

This research targets **patients** as end-users navigating appointment booking, record management, and prescription functionalities in the E-Hospital platform.

Evaluation Strategy

- **Pre-implementation:** Evaluate the **current live version** of the E-Hospital platform.
- **Post-implementation:** Evaluate the **new redesigned prototype** of the platform.

Core dimensions:

- **Usability:** Clarity of interface elements, terminology, layout
- **User comfort:** Perceived ease, confidence, emotional response during interaction

Step 1. Platform Navigation (Before & After)

Patients will be asked to perform key tasks on the **redesigned prototype**, such as:

- Book an appointment
- View lab results
- Manage prescriptions
- Update profile

Observers will note:

- Points of hesitation or confusion

- Navigation flow
- Emotional response during tasks

The answers provided below are about:

☐ Current stage of the E-Hospital platform ☐ Redesigned wireframes

Step 2. General Questions (During or After Tasks)

- Are labels and icons easily understood?
- Are action buttons clearly visible?
- Were you able to accomplish your task without difficulty?
- Did you know what to do next at each step?

Step 3. Specific Questions (Semi-Structured Interview – Before & After)

- How easy was it to complete an appointment booking?
- What made you feel more/less in control of your information?
- Which parts of the interface felt confusing or unclear?
- Did the interface feel welcoming and easy to navigate?

Step 4. Post-Test Questionnaire (Quantitative)

Short surveys using Likert scales (1–6) will measure perceived usability and comfort.

<https://forms.office.com/r/qz43XJe7Zm>

(Phase 1): [Doctor] Wireframe-based user research guide for E-Hospital (module B) redesign

This user research guide outlines the structured methods to evaluate usability and user comfort of the redesigned wireframes for the E-Hospital platform. The guide emphasizes qualitative and cognitive assessment techniques.

Scope

The research targets the Doctors' user profiles. The evaluation strategy focuses on two core dimensions:

- **Usability:** Clarity of interface elements, terminology, and layout

- **User comfort:** Perceived ease, confidence, and emotional response during interaction

Date & Time: Jun 16, 2025, 9:00-9:45 am

Location: uOttawa CRX 5F

Participants: Carolina, Yu-Chen, Dr. Patrick

Purpose of the test:

The purpose of this session is to evaluate whether the current structure of the Patient Overview page and the redesigned workflows align with the doctors' mental models and daily tasks. The session consists of three parts:

1. An open card sorting activity to uncover how the doctor intuitively groups patient-related information.
2. A walkthrough of wireframes to gather feedback on layout clarity, visual hierarchy, and task alignment.
3. A short post-test questionnaire to collect quantitative feedback on usability and perceived effectiveness of the design.

Step 1. Card sorting test (20mins)

Scripts:

Now we'd like your help organizing the kinds of information that might appear on a patient overview page. Imagine you're opening the screen where you typically review a patient's personal and clinical information. It's the kind of page you'd use to take action during an appointment.

We've prepared a set of cards, each representing one type of patient-related data or content. Please group these into categories that make sense to you, based on how you'd prefer to see this information structured in a real clinical setting.

If you think some groups should be nested or have subcategories, that's totally fine. Feel free to add layers as you see fit. You're welcome to rename the group titles however you like, and you can move cards around at any time. If you feel that some information is missing, please feel free to add more information.

If anything is unclear, feel free to ask, but there's no right or wrong way to do this.

Questions:

- What kind of information would you want to see right away when opening a patient record?
- Can you walk us through why you grouped these together?
- Are there any items you think should be combined or separated further?

Step 2. Wireframe walkthroughs (20mins)

Participants will be exposed to select wireframes reflecting common tasks. The goal is to assess layout clarity, visual hierarchy, and task recognition.

Scripts:

Hi, thank you so much for taking the time to join us today.

We're currently redesigning part of the E-Hospital platform, and we'd love to hear your thoughts on some wireframes we've prepared.

There's no right or wrong answer. We're here to understand what makes sense to you and what doesn't.

We will show you a few pages of a medical platform design. Please say out loud "what you see" and "what you think" when you look at those pages.

As they go, ask:

- How do you feel about this page?
- Does anything here feel unexpected or unclear?

If they're stuck, ask:

- What are you looking for that you can't find?
- What would you expect to see on a page like this?

Short interview

Participants will be asked these follow-up open-ended questions to gather further qualitative insights.

- Which screens felt most confusing?
- Are there any features missing for your daily tasks?
- How does this design support your daily clinical flow?
- Are there any features redundant or unnecessary to you?
- Overall, any feedback on this platform?

Step 3. Post-Test Questionnaire (quantitative insights) (5-10 mins)

<https://forms.office.com/r/6aqXdRMXgM>

Additional questions:

- Are labels and icons easily understood?
- Are call-to-action buttons clearly visible?
- Are expected functions easily discoverable?
- Which visual elements helped you find tasks faster?
- Would this design reduce your documentation or navigation effort?

Phase 2 - [Doctor] Wireframe-based user research guide for E-Hospital (module B) redesign

This user research guide outlines the structured methods to evaluate usability and user comfort of the updated interactive prototype for the E-Hospital platform (Module B), mainly the Doctor profile (dashboard and patient list). Building on insights from the first round of testing, this phase focuses on validating recent design improvements and identifying remaining gaps.

Scope

The research targets the Doctors' user profiles. The evaluation strategy focuses on two core dimensions:

- **Usability:** Clarity of interface elements, terminology, and layout
- **User comfort:** Perceived ease, confidence, and emotional response during interaction

Date & Time: Jun 22, 2025, 9:00-10:30 am

Location: uOttawa CRX 5F

Participants: Carolina, Yu-Chen, Veldony, Dr. Patrick

Purpose of the test:

The purpose of this session is to evaluate whether the updated prototype of the Doctor Dashboard and Patient's List and Overview pages align more effectively with doctors' mental models and daily tasks, and to validate improvements made since the first round of feedback.

Structure of the Session

The session consists of three parts:

1. Warm-up & Contextual Expectations (10 min)
2. Interactive Prototype Walkthrough (65-70 min)
3. Post-test Questionnaire (10 min)

Step 1. Warm-up & contextual expectations (~10 min)

Scripts:

Since our last session, we've made adjustments based on your feedback. Today we'll explore the updated interactive prototype. We're interested to hear whether this better supports your clinical workflow and if it addresses any of the concerns you raised previously.

Preliminary questions (before navigating the new version):

- Do you recall any friction points in the last version that you are curious to recheck today?
- Are there specific issues you are hoping to address?
- Is there anything you are particularly curious to check today?

Step 2. Interactive prototype walkthrough (~65-70 min)

Participants will interact with the updated prototype, covering typical clinical workflows. The goal is to assess layout clarity, visual hierarchy, task recognition, and usability improvements.

Scripts:

You can now explore the updated prototype. Please tell us what you see, what you think, and whether this feels clearer or easier to use than before.

Prompts during walkthrough:

- Does this now match your expectations better than the previous version?
- If you had to replace the word “**Appointment**” on the left sidebar, what alternative term would you suggest? (Ideally, use a word with 12 characters or fewer.)
- What is working well for your daily tasks?
- Are there areas still causing confusion?
- Is anything important still missing?
- Is navigation smooth and intuitive?
- Do terminology and labels feel clear?
- Any visual elements that feel helpful or distracting?

Short interview

Participants will be asked these follow-up open-ended questions to gather further qualitative insights.

- Are there any features missing for your daily clinical tasks?
- How does this new version support your daily workflow?
- Overall, do you feel this version has improved from what you saw previously?
- In your opinion, how can AI tools better support your diagnostic or treatment tasks?
- Any final feedback on the platform?

Step 3. Post-Test Questionnaire (quantitative insights) (10 mins)

<https://forms.office.com/r/6aqXdRMXgM>

The questions below can be asked again.

Additional questions:

- Are labels and icons easily understood?
- Are call-to-action buttons clearly visible?

- Are expected functions easily discoverable?
- Which visual elements helped you find tasks faster?
- Would this design reduce your documentation or navigation effort?

REVAMPING THE UX/UI AND FUNCTIONALITY OF INTELLIGENT DIGITAL SOLUTIONS (IDS)

The frontend of the E-Health Portal was redesigned to create a smoother, more intuitive experience for both patients and healthcare providers. The previous interface had issues such as inconsistent buttons, unclear menus, missing labels, and a layout that made it difficult for users to quickly find important medical information. The redesign focused on improving clarity, accessibility, and overall workflow. The navigation structure was reorganized to clearly separate key features like appointments, prescriptions, medical history, feedback, provider tools, and patient dashboards. Visual consistency was established through standardized color themes, readable fonts, and well-structured cards and tables for displaying clinical data. Search functionality was improved with smarter filtering and clearer result formatting, making it easier for clinicians to locate patient files, past visits, and test results. Interactive components such as dynamic forms, alerts, and confirmation prompts were added to help prevent errors and guide users through the system. The design was also made fully responsive so it works smoothly on desktops, tablets, and phones, which is essential for modern digital health systems. Overall, the updated frontend transforms the portal into a more professional, organized, and user-friendly platform that supports faster decision-making, reduces confusion, and prepares the system for future integrations like predictive analytics, real-time alerts, and AI-assisted workflows.



Figure 75. IDS Homepage structure before enhancements

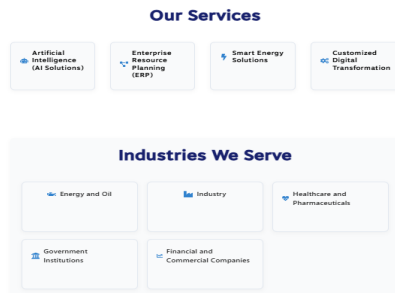


Figure 76. Newly implemented “Services” and “Industries We Serve” sections

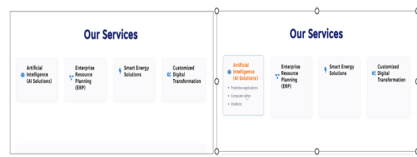


Figure 77. Newly implemented sections featuring hover interactivity for improved user engagement

Commented [FJ1]: edit figure titles



Figure 78. Header navigation before (left) and after (right) fixing anchor link functionality



Figure 79. Responsive behavior of the IDS webpage across mobile devices

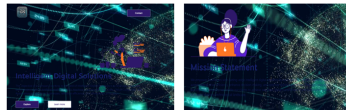


Figure 80. Background video testing: dark-on-dark readability issue

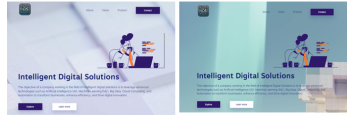


Figure 81. Background video testing: washed-out effect from light video

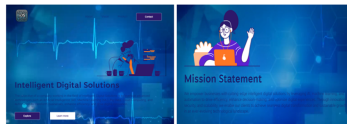


Figure 82. Background video testing: blue-themed video mismatch

3.11 Frontend System Design and User Interface Implementation [6]

The frontend of the e-health system was designed with a strong focus on usability, clarity, and efficiency, ensuring that users can interact with the platform smoothly in real clinical environments. The interface follows a clean and structured layout, using consistent design elements such as dashboards, navigation bars, data tables, forms, and alerts to help users quickly understand where to find information and how to complete tasks. Special attention was given to minimizing cognitive load by grouping related features together and using clear labels and straightforward workflows. This allows clinicians to access patient records, appointments, and system features without unnecessary complexity or confusion.

The frontend was also developed to support responsive and adaptive design, enabling the system to function effectively across different screen sizes and devices, including desktops, tablets, and mobile devices. This flexibility ensures accessibility in various healthcare settings, such as clinics, hospitals, or remote work environments. Interactive elements such as buttons, dropdowns, search fields, and confirmation messages provide immediate feedback to users, reducing the risk of input errors and improving overall reliability. Input validation is handled at the frontend level to ensure that required fields are completed correctly before data is sent to the backend, improving data quality and system stability.

In terms of system integration, the frontend communicates securely with backend services through RESTful APIs to retrieve and submit data in real time. This enables dynamic

updates of patient information, appointment schedules, and system responses without requiring page reloads. Sensitive data is handled carefully through secure communication channels, ensuring alignment with healthcare privacy requirements. Visual indicators such as notifications, status badges, and alerts are used to highlight important updates, system actions, or required user attention. Overall, the frontend serves as the main interaction layer of the platform, transforming complex backend logic and database operations into an accessible, user-friendly experience that supports efficient workflows, accurate data entry, and high-quality healthcare delivery



Figure 83. Inconsistent card layouts and information grouping observed in the initial system.

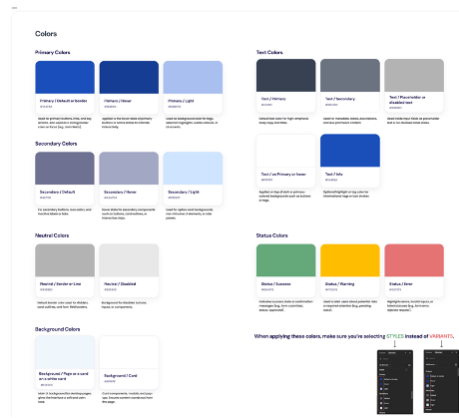


Figure 84. Standardized color palette and semantic color definitions used in the design system.

Typography



Headings

H1 Poppins Bold 36px / 40px	Usage: Page titles (e.g., "My patients")
H2 Poppins Semibold 24px / 32px	Usage: Section titles (e.g., "Upcoming Appointments")
H3 Poppins Semibold 20px / 28px	Usage: Card titles or primary data labels

Figure 85. Typography hierarchy and usage guidelines defined for eHospital interfaces.

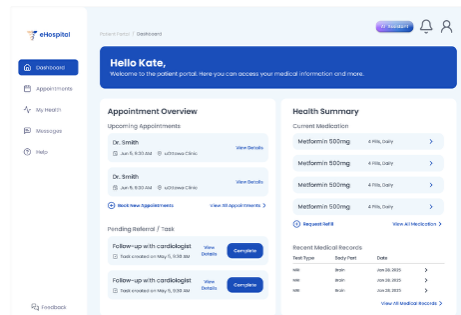


Figure 86. Spacing rules and layout consistency applied across clinical screens.

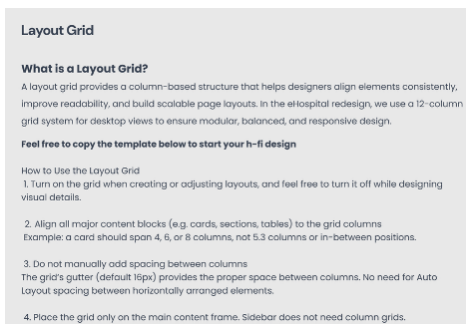


Figure 87. Layout grid structure used to align and scale interface components.



Figure 88. Button variants and interaction states defined in the design system.

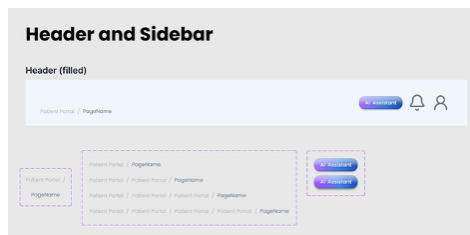
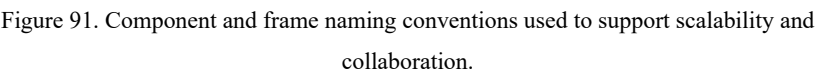
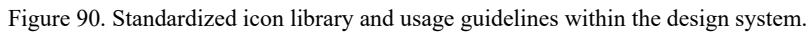


Figure 89. Standardized header and sidebar navigation components for patient and clinician interfaces.



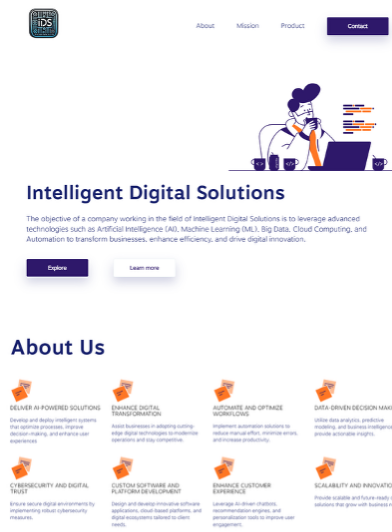


Figure 92. Component Initial IDS website homepage showing original content structure and navigation.

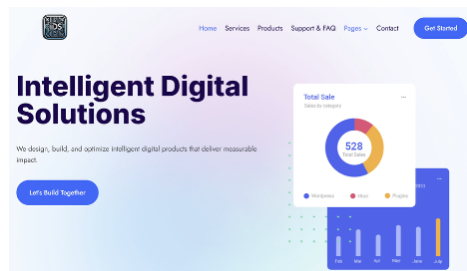


Figure 93. Proposed site map and navigation architecture for the redesigned IDS website.

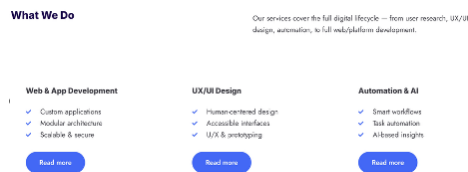


Figure 94. Service detail component layout showing icon integration and consistent spacing.

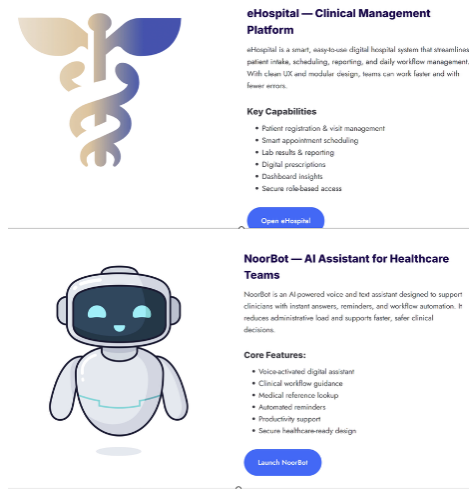


Figure 95. Call-to-Action sections and contact prompts designed to increase conversion.

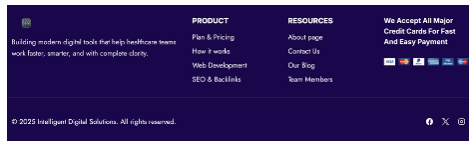


Figure 96. Footer and global site components (links, contact info, and legal sections).

Feel free to contact us for any inquiries or support needs

We usually respond within **1–2 working days**.

Whether you're looking for platform details, project support, or integration guidance, we're here to help.

Socials: [Facebook](#) [Twitter](#) [LinkedIn](#)

First Name

Last Name

Email *

Subject

Message *

Figure 97. Contact page to increase client reach

4. Backend Documentation

The backend is the server-side portion of the e-Hospital website. This is the section of the e-Hospital that is not visible, but it is what ensures that everything on the website functions correctly. It is essential to note that Python is the coding language used to create backend code that is compatible with the rest of the site.

The backend is the middleware between the user interface and the database, which handles data transactions and processes requests. Within the backend, the data is being retrieved and stored using either API endpoints (NoSQL) or structured queries (SQL). When it comes to AI-driven decision support engines specifically, the backend processes patient data, runs machine learning models, and generates personalized treatment recommendations. It integrates electronic health records (EHRs), medical literature, and patient preferences, thereby handling API calls, processing data, and implementing system logic.

The majority, if not all, of the information included in this section is taken from past reports. This is the section of the work where the work itself cannot be observed, and therefore, logging into any of the portals was not particularly informative. The backend documentation section includes a lot of necessary information, including the Application Programming Interface (API) documentation and reference-specific documentation.

4.1 API Documentation [7]

The main document provided regarding the APIs is referenced and lists 18 different APIs used in the e-Hospital. The various APIs are listed below, along with their respective uses.

1. Record Image Upload
 - Disease diagnostic, record updating, and record reviewing
2. Retrieve Image Record by Patient's Phone Number
 - Disease diagnostic, record updating, and record reviewing
3. Retrieve Image Record by Record ID

- Disease diagnostic, record updating, and record reviewing
- 4. Fake ML API for Image Record
 - Disease diagnostic, record updating, and record reviewing
- 5. Retrieve Basic Information by Patient's Phone Number
 - Disease diagnostic, record updating, and record reviewing
- 6. Retrieve Raw Data Record by Patient's Phone Number
 - Disease diagnostic, record updating, and record reviewing
- 7. Disease Prediction Update
 - Disease diagnostic, record updating, and record reviewing
- 8. Check Authorized Patient List
 - Privacy
- 9. Authorize or Unauthorize Record to Doctor
 - Privacy
- 10. Get Available Labs
 - Lab appointment
- 11. Get Schedule of a specific Lab
 - Lab appointment
- 12. Schedule Lab Appointment
 - Lab appointment
- 13. Check the Scheduled Appointment for the Patient
 - Lab appointment
- 14. Check the Scheduled Appointment for the Lab

- Lab appointment

15. Get The Number of Available Doctors in Each Specialization

- Number of available doctors in the system for each specialization

16. Get a List of Current Available Dentists

- Available dentist nearby

17. Update Doctor Availability

- Allows doctors to update their availability

18. Retrieve Diseases' symptoms

- Returns a list of diseases and their symptoms

4.2 Reference-specific Documentation

This section provides a summary of the backend application's functionality across different sections of the e-Hospital, based on specific papers from previous projects.

4.2.1 e-Hospital: Intelligent Clinical Decision Support [2]

The Intelligent Clinical Decision Support system intelligently automates the reception and processing of requests for AI-generated treatment plans. In Figure 75, a diagram appears to describe the functioning of the components in relation to one another. The multi-layered network functionalities stem from the backend receiving orders from the Web user interface. Following this, the orders are allocated to the relevant diagnosis module, which retrieves information from the knowledge base, comprising patient data, treatment protocols, and other medical information, as well as a patient record and treatment history database. These two subsystems are data sources that provide sufficient information for information processing.

The backend also features an applied machine learning model for a multitude of patients, which, after receiving data from the diagnosis module, prepares individual treatment plans. This model, along with the others, has been trained on large datasets to achieve optimal output. The

sophisticated architecture of the backend enables further development of the system through supplementary data input or improving the algorithms of its constituent parts.

Figure 75 illustrates this architectural design, together with the user interface, diagnosis module, knowledge base, databases, and machine learning model. It highlights the ways in which these components work together to deliver innovative, real-time clinical decision support services.

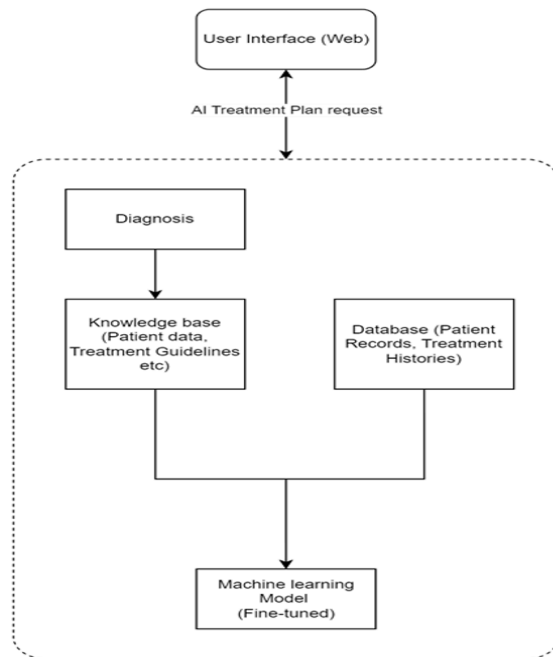


Figure 75. Architecture Diagram of the Intelligent Clinical Decision Support

4.2.2 E-Health Medical Chatbot [8]

The backend of the E-Health Medical Chatbot is implemented using a microservices architecture model, which ensures scalability, modularity, and efficient management of data processes. The main components of the backend are an API gateway which controls interactions between the chatbot interface and backend microservices, a proprietary database (NoSql/Sql)

which captures, stores, and manages patient interactions with the chatbot and the NLP engine that processes user queries using artificial intelligence healthcare models (e.g. BERT, MedGPT). For secure user access, login and user access verification are provided by an integrated OAuth 2.0, which protects healthcare privacy regulations, such as HIPAA, and the authorization processes. Additionally, there are Prometheus- and Grafana-enabled systems that monitor real-time API performance and chatbot analytics. The backend communicates with external Electronic Health Records (EHRs) using HL7 FHIR-compliant APIs, enabling easier retrieval and updating of patient information. Figure 76 illustrates the service interaction, database structure, and security features design.

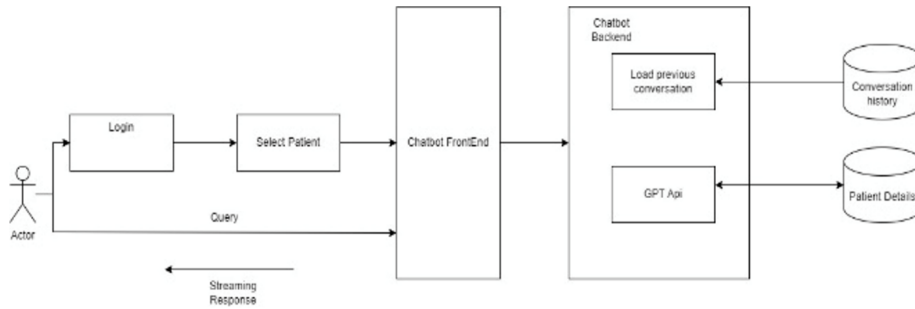


Figure 76. Architecture Diagram of the E-Health Medical Chatbot

4.2.3 e-Hospital AI Treatment Plan Enhancement and Optimization [10]

The server-side architecture, which enables AI-driven decision-making and patient-centered treatment plan generation, is the most sophisticated component of E-Hospital's AI Treatment Plan Enhancement and Optimization infrastructure. The backend architecture processes patient information in a secure environment, integrates AI models for personalized treatment recommendations, and optimizes real-time clinical inputs to inform treatment plans. The primary functionalities encapsulated in the backend are data storage and retrieval, integration of patient health records, and deployment of AI models to optimize treatment predictions and other processes within the system. The backend implements a microservices architecture for better separation of concerns, enabling scalable and modular services such as patient data retrieval and storage, treatment recommendation creation, and AI model servicing.

The database design is built to accommodate medical histories, laboratory tests, images, and other data that are considered either structured or unstructured, providing great flexibility. APIs are provided for the frontend components to communicate smoothly with the backend and interact with other e-health platforms, providing integration and instance update features. The application is fortified with multi-layer security, including the encryption of patient data, token-based authentication, and role-based access control, to ensure patient privacy and data integrity. Moreover, there is an integration of backend monitoring and logging tools to guarantee system traceability, reliability, and performance, thus allowing the AI algorithms for treatment planning to be continuously improved.

4.2.4 ML Model Evaluation and Interpretation [11]

The API is built on Flask, a Python web microframework, which serves endpoints to accept machine learning models along with the datasets to be analyzed. A POST request directed towards the endpoint /evaluate triggers the backend process, where the model file and dataset (model.pkl or h5 and dataset.csv, respectively) are uploaded. Then, the system checks the file for format compliance. The data is then passed on to be preprocessed - including but not limited to: normalization, feature extraction, and data sanitization. After the data is prepared, the machine learning model is loaded, and the accompanying function to evaluate the model is called. This function fetches the relevant metrics like accuracy, F1 score, cross-validation scores, and confusion matrices, cumulatively referred to as 'distinct metrics'. If the model allows network-based probability forecasts, the predicted probabilities are computed. In the wake of implementing mechanisms for explainability, the backend utilizes SHAP summary plots to provide feature importance, while the model's classification performance is visualized through ROC curves. The findings, including the summary SHAP plots, the ROC curve, and all results, are stored in the system's storage and can be retrieved from relevant API endpoints. These visuals are further saved as images that can later be retrieved through the API. This functionality is integrated into the e-Hospital platform and accessed through a user-friendly interface for medical professionals, enabling easy interpretation and assessment of the ML model's performance.

The backend architecture effectively deals with the upload, evaluation, and visualization of the model. Through the API, users can upload a model and a dataset. Before evaluation, the system verifies that the data format is correct. The system calculates performance metrics and generates visualizations, which can be accessed through API endpoints. The modular architecture, comprising data processing, model evaluation, and visualization, is well-suited for integration into the e-Hospital platform as well as other machine learning models.

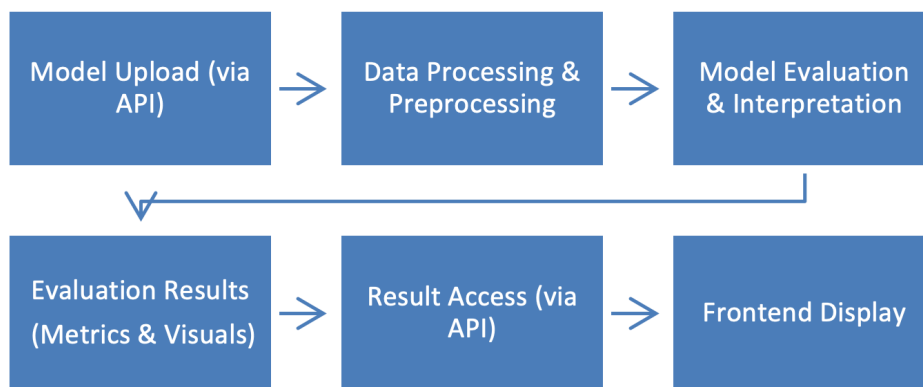


Figure 77. Architecture Diagram of the ML Model Evaluation and Interpretation

4.2.5 AI-Powered OHIP Billing Backend System [12]

The AI-Powered OHIP Billing Assistant's backend follows MERN stack framework, using Express.js and Node.js for a modular, scalable, and API-centric architecture. At its core, a RESTful API layer takes care of user login and billing record retrieval, AI-suggested code insertion, and other essential tasks. The backend consists of 2 primary data models: User and Bill. They are represented as documents in a MongoDB Atlas cloud database selected by the Mongoose ODM. User credentials along with personal information are protected through password hashing with secure login endpoints, while billing records are managed in terms of CRUD functions and associated with user accounts. AI code extraction is facilitated via a specific endpoint /bill-code, accessing Open AI's GPT-4 through Retrieval Augmented Generation (RAG) for improved billing code recommendation efficiency. The architecture incorporates clean, modular APIs which allow effortless integration with prospective governmental hospital software, and layers focused on validating the data guarantee precision

and consistency of all records. With the given backend structure, optimized speed, minimal upkeep, and effortless integration to pre-existing e-health systems is ensured.

4.2.6 Backend Architecture of the Chatbot for eHospital Diagnosis Module [13]

We have used python and Fast API to develop the backend of eHospital Diagnosis Module for Chatbot, aiming to provide an integrated intelligent system to afford real-time medical support in the form of chatbots. The system is capable of distinguishing between two sub-chatbots: Diagnostic Model selection and Diagnostic Support. Each diagnostic unit has been given a set of RESTful API interfaces that receive text requests from the user, control their conversation, and call sophisticated models for AI computations. Whatever the user provides as input, be it in form of text, speech or image, is captured and forwarded to the central backend. For voice input, OpenAI's Whisper-1 speech-to-text model transforms speech to text (multimodal interaction). This transformed text, along with any medical images submitted, is sent to appropriate API endpoints. In diagnosing chatbots, the backend uses GPT-4 which cross relates symptom descriptions to diagnostic models and tells users the body organs they could perform tests on while providing hyperlink or direct link to respective model's page in the e-hospital system. For support, the Diagnostic Support Chatbot augments GPT-4o, a text and image understanding multimodal AI, so that they produce informative, context specific and medically precise replies. Each chat message—user input, AI response, and optionally transmitted images—is marshaled into structured message objects and transferred between frontend and backend as FormData. In order to ensure session persistence and data consistency, the backend keeps both transient and long-term chat histories. Long-term storage is implemented using AWS RDS MySQL for structured text data and AWS S3 for image uploads. Bespoke Python tools written with PyMySQL and boto3 handle all database and cloud storage interactions, such as message fetching, chat loading, and session removal. All messages in storage include metadata such as user ID, patient ID, session timestamp, role (user/AI), and optional image keys, facilitating granular session tracking. The backend also supports chat continuity with endpoints for session history storage and retrieval, enabling doctors to resume where they left off on complicated diagnostic work flows with context preserved. This advanced backend infrastructure supports a dynamic, secure, and insightful user experience, rendering the chatbot an attractive digital assistant for contemporary clinical workflows.

4.2.7 Chatbot for Doctors [14]

The e-hospital chatbot system has a thin, efficient backend design, integrated with the frontend React interface in a tightly-coupled fashion for a responsive and reliable experience. While most of the core logic of model selection is in the frontend, the backend is mainly concerned with file uploading, multimodal data processing (including some pre-processing), and routing the request to the appropriate services. In total, the processes built on Node.js and Express collect and securely transmit patient data uploads, such as MRI scans, X-ray images, and diagnostic reports, on to the LLaVA model. With regards to file uploads, their role is pre-processing the file(s), temporarily holding the data, and then sending appropriate data to the model for processing without introducing any waiting period. The backend also controls all interaction with the model repository by enabling the e-hospital to provide access to an indexed representation of all diagnostic models that are available on the platform. Overall, this was an appropriate solution to allow a thin backend logic engine, keep the overall system architecture modular and scalable, and make a case for rapidly deployed, maintainable, and safe data flow systems; and in clinical settings, they are essential.

4.2.8 AI-Integrated Backend Services [15]

The e-hospital management system's backend was built entirely using Flask, which served as the center point for user request processing, server-side logic management, and integration between the frontend, AI chatbots, and external databases. The Flask backend was critical in processing all operations involving user authentication (authentication required for even the simplest operations), authorization based on roles, and API routing for Noor Bot RAG-based chatbot and all other peer-integrated models. The backend was responsible for retrieving, preprocessing, and forwarding multimodal data (e.g., text input and uploaded medical images) to AI models responsible for prediction. Flask's minimalistic, modular framework allowed efficient organization of RESTful API endpoints and easy management of different use cases, from booking appointments to answering clinical questions. The ability to integrate third-party storage with AWS S3 and deploy a system using Heroku provided excellent scalability and reliability. The middleware functionality was useful for managing the different request validations, session management, and error logging, greatly improving back-end reliability and maintainability. In

summary, the backend was an essential component of the e-hospital management system in facilitating real-time intelligent interaction within a secure, extendable healthcare environment.

4.2.9 AI-Orchestrated Backend Pipeline for Medical Document Digitization [16]

The backend of the Document Digitizer System is the processing engine, providing all server-side processing, from file incoming file reception, to intelligent data extraction. The backend is created with FastAPI and designed for speed and scale, performing crucial pre-processing steps involving image compressions, resizing, erasing suspicious data, and validation. They need to normalize and optimize the documents that are uploaded - many of which are photographs taken in questionable light conditions at questionable resolutions - in preparation for the downstream use of AI. Once validated, the documents are uploaded to Cloudinary, an image hosting provider that offers effective media management and access to images in the cloud, allowing for needed security when storing images and for high retrieval convenience for downstream processor execution, if needed. Based on the nature of each document, the backend dynamically pushes each file to the corresponding vision based AI model. For handwritten prescriptions, the Qwen 2.5 model is used and it is very good at reading out medication names, dosages and doctor orders. For more complicated documents, such as Ontario lab requisition forms, Gemini Flash 2.0 is deployed. Gemini Flash 2.0 is a high-context model that learns the context of highly formatted documents with the capability of parsing structured data. The backend captures the AI model output in a form that is formatted JSON readily for integration into the e-Hospital platform or elsewhere for any automated process such as filling out forms on any computer. The modular backend has a great capacity for reliability, throughput, extensions to support new document types, and new AI models, and, as mentioned before, it's still a concerted effort to ensure data privacy, efficiency with low client-side load, and minimal processing cost by filtering out what could potentially be a very large amount of input prior to triggering the vacant computational resource of an AI model. In general, the backend is a robust and intelligent character that embodies the user facing frontend, and computational heavy lifting from the AI units that comprise the document digitization pipeline.

4.2.10 Voice-Assisted Past Visits Record System [17]

The backend of the Voice-Assisted Past Visits Record System in the e-Hospital platform was implemented using FastAPI, a contemporary Python web framework known for high performance and automatic validation features. This backend creates a bridge between ReactJS on the front end and a MySQL database as well as connects to OpenAI's GPT-based Speech-to-Text API to automatically document useful medical visit records. During a visit, when the doctor speaks, the audio feed is sent to the backend which processes it securely and asynchronously and transforms it into a structured text document using OpenAI's API. Once the transcription has been completed, it is reviewed and stored. The RESTful endpoints facilitate several functions, such as submission of audio, verification of patient/doctor details, validation of data records, and ultimately the storing of complex records. Several security considerations and precautions were cleared: HTTPS requests, JWT tokens, and role based access for sensitive patient data. The backend also connects to a Redis cache, which provides a slightly faster response time, and manages scheduled background tasks (e.g. backups and cleanups). All transcriptions and manual visit records are stored relationally in a MySQL database with foreign keys to doctors and patients. The backend logic takes care of error handling, and error reporting, provides real-time front-end user feedback, and keeps data format consistent. It used Postman, Navicat, and GitHub to validate APIs, test and debug, and track overall development progress. Overall, the backend supports fast, accurate, and secure voice-to-record processing and reduces administrative workload while enhancing the quality of clinical documentation for doctors using the e-Hospital platform.

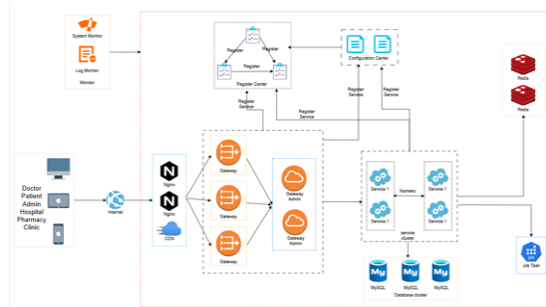


Figure 78. System Deployment Architecture

4.2.11 Feedback Management System [18]

The backend of the Feedback Management System is designed using FastAPI, a modern web framework for Python that allows fast, scalable development of RESTful APIs. The backend will serve as the core logic layer between the frontend chatbot and the system's database, AI services, and cloud storage, all of which produce rich feedback (assessment of failures) for the healthcare doctor. The backend has been conceived using modular routing that means there are dedicated separate endpoints that will handle different aspects of the feedback, including patient-side (chatbot) feedback collection, doctor-side chat interactions, and file-based feedback searches. Once a patient submits feedback through the chatbot, the data is validated using Pydantic models (Python developer IRL) to ensure the accuracy and general consistency of the data. Following validation the Pydantic model validated text is passed through the OpenAI API to classify the feedback based on what it is about (for e.g. is it service related or treatment related) and severity (low, medium or high). As the responses are based on more than just individual messages, once the system checks the patient's feedback history from the database (MySQL) to see if there is a trend pattern with the agenda that indicates if the issue has gotten worse. If an issue has gotten worse over subsequent feedback entries, the feedback is flagged as urgent, potentially prompting AI generated suggestions for the doctor (the document AI point of view (POV)). Metadata can be separately/simultaneously stored or appended to each feedback entry in your database in a structured format like severity, classification and time stamps. Once the doctors document the shared feedback whether it is classified as urgent or not, it will proceed through the feedback management workflow process, consisting of feedback triggers, and trends. Additionally, daily summaries of patient feedback are compiled into PDF reports and uploaded to AWS S3, allowing for scalable, secure access to historical records, with the added benefit of a powerful and scalable cloud-based solution. The backend supports intelligent file searching by incorporating OpenAI's Assistant API and vector database for semantic search capabilities. This allows the doctor to ask natural language questions such as "Did any patients report dizziness this week?" and receive grounded answers based on the indexed feedback file. Security is a very strong emphasis for this system; all API credentials and database configurations are managed through environment variables and logs are kept for all system actions including every AI prompt call, and each alert generation to provide accountability and auditability to the system. This backend architecture supports responsive AI-based clinical decision support while also

processing real-time patient feedback and being modular and extensible in order to scale across the healthcare environment.

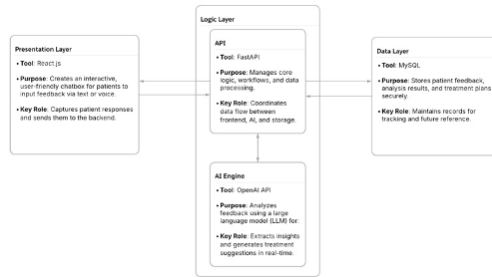


Figure 79. System Architecture

4.2.12 Backend Infrastructure of the Intelligent Medical Research Assistant [19]

The Intelligent Medical Research Assistant (IMRA) backend is developed in Python using the Flask framework and divided into three components: `main.py`, `app.py`, and `create_db.py`. The modular structure reduces the complexity of controlling user requests from uploading a PDF to AI-generated responses. The `main.py` file initializes the Flask app, controls user session activity and UUIDs, makes sure old data sessions are cleaned automatically, and includes REST endpoints for `/api/upload` & `/api/stream`. When the user uploads a PDF, the server sends a request to the document-processing pipeline, which performs the following tasks: parse the PDF, create a vector index using Chroma, extract metadata (title, abstract, keywords), and discover similar articles using PubMed & Semantic Scholar APIs.

The `app.py` module can use both PyPDF2 and pdfplumber to parse PDF documents. It handles bad PDF formats through fall-back techniques that rely on GPT-based keyword inference, or OCR for scanned documents. It extracts the title, abstract, and keywords from the papers to then look up relevant academic literature. Those articles will be sorted by keyword overlap (between the original paper and papers that were returned) and text similarity to improve relevance.

The third module, `create_db.py`, manages vector storage through Chroma DB and LangChain. It chunks and embeds uploaded documents through Chroma vector embeddings,

which enable retrieval-augmented generation (RAG). During interaction with the user through the chat interface, the system queries the Chroma vector store to get the most relevant chunks of text from the uploaded PDF, which are then turned into a prompt with the user's prompt, to send to OpenAI's GPT-4 model. The goal is to ensure that it grounds answers in the uploaded PDF document, which should help minimize hallucination and increase accuracy.

At every layer, security and efficiency are in place at the temporary session, streaming with server-sent events (SSE) to give a real-time response and modular APIs to make it easier to scale and/or swap AI providers. The technology stack is set up for fast document analysis and in-context response generation for a fast user experience.

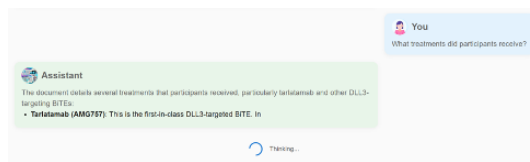


Figure 803.5 Streaming response from the assistant with partial content displayed as it arrives

4.2.13 Comprehensive Backend Architecture of the Voice-Enabled RAG-Based Medical Chatbot [20]

The backend of the RAG-based medical chatbot system is built using Flask. The backend is the heart of the system and is responsible for performing all critical functions such as document ingestion and vector embedding, retrieving documents based on similarity, and returning user voice input/appropriate voice output, and generating a final response. One of the main innovations we utilized is creating an automated document ingestion pipeline. The backend will not rely on the front end where a user can upload medical PDFs. Instead, the backend is programmed to curate and collect medical PDFs from approved sourcing. Each document will be processed (parsed) with PyPDF2 or PyMuPDF, cleaned, and pre-processed prior to embedding it. Once the medical text has been cleaned, it will be embedded using OpenAI's text-embedding-ada-002 model, which turns the text report into high-dimensional semantic vectors. The vector database developed with Pinecone is a vector database, optimized for fast and scalable similarity search, which allows for quick retrieval of the appropriate information presented in real-time.

When a user submits a query (either typed or spoken), the backend will address it through multiple stages. If the input is an audio file it will be processed by the Whisper API or Google Speech-to-Text to create text from the speech. Next, the backend will implement a vector similarity search inside Pinecone using the user input as context. The top retrieved relevant documents from the medical domain are aggregated into the user's query--now termed as an augmented prompt--and passed onto OpenAI's GPT model. This allows the chatbot to produce text-based responses from a combination of pre-trained knowledge as well as recent information based on documents.

After a response is produced, the backend has the option to send it through a Text-to-Speech (TTS) engine (gTTS library or pyttsx3) to produce audio of the reply. The text and audio will be returned to the front end to display the response and for playback--demonstrating a multimodal interaction paradigm.

This backend architecture supports an integrated end-to-end medical chatbot, while preserving accuracy, timeliness, and context-rich responses. The architecture is constructed in a modular and scalable way to integrate into larger platforms like the University of Ottawa's e-Hospital system. Future extensions could involve using patient-specific data to build personalized responses, including patient outcome-tracking modules, and making the chatbot multilingual.

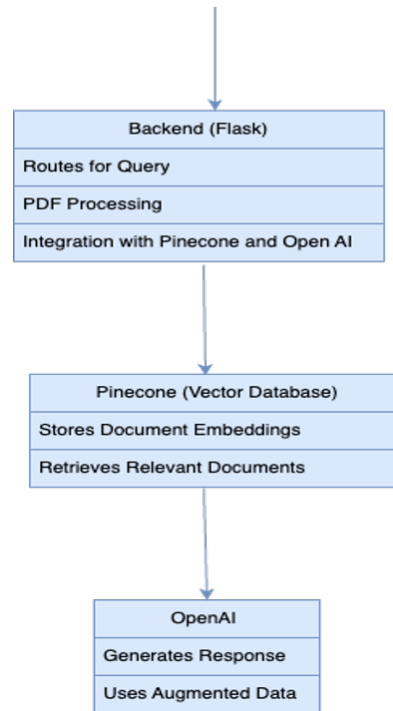


Figure 814. System architecture flow from frontend input to backend processing, document retrieval in Pinecone, and response generation by OpenAI

4.2.14 Pharmacy Inventory Backend Architecture [21]

In a pharmaceutical inventory system, the backend was developed in Python, as its many libraries made it easy to act on parts of the essential logic, for example tracking medicines, linking prescriptions and automating stocks. At first a lightweight file-based system was adopted locally, using SQLite3 for all the development and testing. This was suitable for simulating local small pharmacy scenarios. Then, as the project was going into production, MySQL was adopted as it has much more suitable properties to run simultaneously with many other users - safety, concurrency, and ability to manage extensive, large-scale patterns of transactional data.

The backend code is mainly responsible for connecting the users interface to the inventory database and managing the medicine stock levels real time. In other words, it will

automate the deduction process after you fill a prescription and update the stock levels. When entering or filling a prescription in the pharmacy software the backend accesses the Prescriptions Table, verifies the stock levels from the Medicines Table and inputs the stock transactions into the Inventory Updates Table through structured SQL transactions. The backend is able to automate these transactions helping to minimize errors associated with manual data entry. Similarly, the backend exposes secure API endpoints with the use of JSON for the user interface, allowing the pharmacists to add new medications, delete expired drugs, and view or search the current inventory.

The group overcame many challenges during the developing cycle, particularly when it came to syncing regards to the data between sessions, the delayed real-time updates and correcting the POST request errors when submitting a request from the user interface to the backend APIs when hosted on Heroku to name a few. Nevertheless, by adjusting the API logic into smaller usable bundles as well as optimizing the database queries the backend ultimately became more useful regarding data integrity and responsiveness.

Prescription integration was an impactful improvement: the system not automatically calculates out stock levels when a prescription is entered, keeping current stock levels viewable without manual intervention. The back end was proactively developed with future scalability in mind to accommodate future additions, such as automated re-order reminders, role-based permissions, and audit logging to meet regulatory requirements. In sum, this back-end layer is critical to enabling automation, accuracy, and efficiency of medicine management in e-Hospital environments.

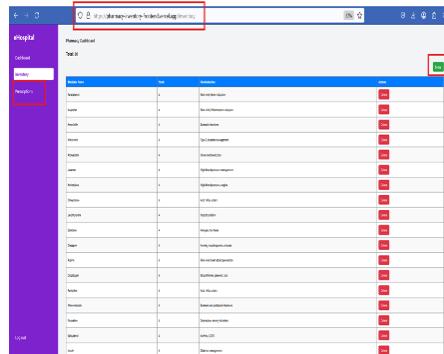


Figure 825.Final Prototype

4.2.15 Voice-Enabled Medical Chatbot [22]

The Healthcare Voice Chatbot backend is developed with the Flask web framework, acting as the processing engine in between the React frontend to the AI services (OpenAI Whisper and GPT-4). In the beginning, they were able to accept real-time audio files from doctors via an `/analyse_audio` route. Upon receiving the audio, the backend pre-processes the audio for resampling, normalizing, and formatting the input to 16kHz WAV file for transcribing as result. The OpenAI Whisper model is utilized directly on the CPU for availability, and transcribes the audio into clear, natural language text. The text transcript is then processed for analysis by GPT-4 in two stages. The first in which GPT-4 runs on the content to extract medical keywords, which are rendered a timestamped .txt log so the text is traceable. Secondly, GPT-4 is asked to produce a medically structured summary that includes symptoms, potential diagnoses, follow-up questions, and initial threaded treatment options. The message is returned by the backend in a JSON object form that is ready to be rendered directly. Finally, the temporary media files are removed after processing to further ensure system performance. This multi-layered architecture supports responsiveness in real-time, reliable transcriptions, and smart summarizations which make it usable during live clinical consultations. Built-in, error-handling functions identify and alert users to missing input or incorrect requests and, with the inclusion of fallback logic, we have guaranteed service continuations, of great value to the user across the service layers. The back-end enables secure and always-on, doctor-patient voice interactions

made possible with AI, significantly reduces documentation manually, and can improve clinical judgement and decision-making while working within the e-Hospital platform.

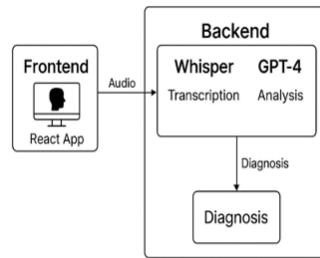


Figure 836. Architectural Diagram

4.2.16 Enhancing Decision-Making with a Voice-Enabled, AI-Powered Medical Assistant [23]

The backend architecture of the voice-enabled, AI-powered medical assistant was designed as a modular, secure, and high-performance framework capable of integrating multiple intelligent components into a unified clinical workflow. Developed using FastAPI, the backend serves as the central orchestrator between the frontend user interface, Dialogflow NLP engine, speech-to-text and text-to-speech services, and the AI reasoning layer. Its primary role is to handle real-time requests, manage active user sessions, and facilitate communication between APIs, ensuring low latency and reliable performance. Each incoming audio query from the clinician is first routed to the Whisper or Google Cloud Speech-to-Text API for transcription, after which the transcribed text is parsed and analyzed by the Dialogflow CX engine to detect intent and extract key medical entities. The backend then invokes domain-specific clinical decision support logic—including rule-based algorithms and LLM-based inference modules—to generate structured diagnostic prompts, test recommendations, or guideline references.

All backend processes were implemented with strict adherence to healthcare data privacy standards such as PIPEDA and HIPAA, ensuring that no patient-identifiable data is permanently stored. Temporary session data are managed in memory using asynchronous event loops, and encryption protocols are applied during data transmission between APIs. To enhance scalability

and modularity, the backend supports containerized deployment through Docker, enabling independent scaling of subsystems such as the NLP engine, inference model, and data access layer. Logging and monitoring were incorporated via a custom middleware layer that records API latency, response accuracy, and exception handling for debugging and system audits.

In addition, the backend exposes RESTful API endpoints that can be extended for EHR integration, allowing the assistant to pull or push structured patient data securely in future iterations. These endpoints are version-controlled and documented through Swagger UI, ensuring maintainability and transparency for future development. The architecture also supports integration with Hugging Face Transformers for fallback NLP and question-answering tasks, enabling the system to respond intelligently even when Dialogflow returns uncertain intents. The backend's event-driven design minimizes redundant API calls, improving efficiency while maintaining real-time responsiveness during dialogue.

Overall, the backend functions as the intelligent control hub of the system—coordinating multimodal inputs, processing clinical context, and delivering actionable insights in under two seconds on average. Its modular design not only supports current functionalities like voice-to-text, LLM inference, and clinical decision support but also provides a scalable foundation for future enhancements such as EHR interoperability, predictive analytics, and automated case documentation. This backend layer, therefore, represents the technical backbone of the entire assistant—bridging conversational AI, clinical reasoning, and data integrity within a robust and compliant architecture.

4.2.17 Intelligent Health Informatics System for Patient's Treatment Plan [24]

The backend of the Intelligent Health Informatics System serves as the computational core that bridges user interactions from the GUI and voice interface to the AI reasoning and data management layers. Developed primarily in Python, the backend was designed with modularity, security, and scalability in mind to support real-time clinical decision-making. Its architecture integrates multiple processing pipelines that handle patient data ingestion, NLP-based inference, and treatment recommendation generation. Once patient information is received—either through manual input or voice recognition—the backend validates, structures, and stores the data

temporarily using pandas data frames for manipulation and dynamic updates. Through the transformers library, the backend leverages Bio_ClinicalBERT, a transformer-based language model fine-tuned on biomedical corpora, to process natural language inputs and infer treatment suggestions using masked language modeling. To ensure contextual accuracy, the backend employs a carefully engineered prompt construction layer that dynamically inserts patient demographics, vital signs, and diagnoses into predefined clinical templates, enhancing model interpretability and response consistency.

The backend also coordinates the AI suggestion module, which interacts with the inference engine to generate medically appropriate outputs in real-time. Each treatment recommendation request follows an event-driven sequence: patient data retrieval, prompt formulation, Bio_ClinicalBERT inference, output parsing, and GUI return response. This modular event chain enables asynchronous task execution and prevents interface lag during intensive processing. Furthermore, the backend incorporates an error-handling subsystem that detects incomplete or ambiguous patient inputs, prompting clinicians to refine entries for improved inference accuracy. Real-time voice inputs are processed via the speech_recognition library, converted into structured text, and fed into the backend's input manager for further analysis, ensuring seamless multimodal integration between audio and textual interfaces.

From a systems perspective, the backend emphasizes data integrity and clinical compliance. Although current storage uses CSV as a lightweight local database, the backend is designed with a clear migration pathway toward SQL or NoSQL databases for scalable deployment in hospital networks. Role-based permission control, session encryption, and access logging mechanisms are planned for future integration to meet healthcare data standards such as PIPEDA and HIPAA. Additionally, backend APIs can be extended to synchronize with Electronic Medical Record (EMR) systems, allowing bidirectional data flow between clinical databases and the AI recommendation module. The modular backend structure supports future enhancements such as the inclusion of GPT-4-based reasoning or federated learning models for continuous model updates without centralized data sharing.

Performance testing of the backend confirmed its ability to handle concurrent operations efficiently, processing treatment suggestions in under one second per request on standard

hardware. Latency measurements across different modules showed stable response times and minimal computational overhead. Overall, the backend serves as the system's intelligent control hub, managing data flow, coordinating AI-driven inference, and ensuring robust, secure, and context-aware recommendations for clinical use. Its flexible, modular design not only supports current functionalities but also lays the groundwork for a scalable, cloud-integrated clinical decision support infrastructure.

4.2.18 Secure Prescription Parsing API [25]

The backend of the Medical Prescription Parsing API forms the core computational framework that powers the entire audio-to-prescription workflow, managing everything from data ingestion and transcription to structured data extraction and secure document generation. Built using FastAPI, the backend operates as a lightweight yet highly scalable RESTful microservice capable of processing multiple simultaneous audio submissions with minimal latency. Its architecture is composed of several coordinated modules—Audio Handling, Speech Recognition, Text Parsing, Data Structuring, and PDF Rendering—each designed to perform a specific, optimized function in the processing pipeline. When an audio file is submitted through the `/submit_audio` endpoint, it is temporarily stored in a secure environment under ephemeral storage policies that automatically delete files after processing to preserve patient confidentiality. The backend then invokes OpenAI's Whisper model, which performs high-fidelity transcription with multilingual support and noise resilience, effectively handling complex medical vocabulary, drug names, and dosage instructions under real-world acoustic conditions.

Once transcription is complete, the backend transitions into its Natural Language Processing (NLP) and data extraction phase. This stage uses a hybrid rule-based and regular expression parsing engine that identifies and structures critical prescription components such as patient identifiers, medication names, dosages, administration frequencies, prescriber credentials, and treatment notes. These extracted entities are serialized into a structured JSON object, ensuring interoperability with electronic health record (EHR) systems and downstream analytics tools. To maintain accuracy and adaptability, the backend employs a modular parser configuration, enabling developers to easily update extraction rules or integrate advanced NLP modules like Named Entity Recognition (NER) models in future releases.

Following data extraction, the backend activates the PDF rendering module, which utilizes ReportLab to populate a standardized prescription template with the parsed data. This module ensures consistent formatting, legibility, and compliance with clinical documentation standards, while embedding digital signatures and timestamps for traceability. The generated PDF is returned to the client along with the structured JSON data, enabling both human-readable and machine-readable outputs for seamless workflow integration. Throughout this process, the backend enforces strict security and compliance protocols, including Transport Layer Security (TLS) for encrypted communications, CORS restrictions and IP whitelisting for controlled access, and audit logging mechanisms that track operations without storing personal health information (PHI).

Performance evaluation revealed that the backend achieves exceptional responsiveness, completing end-to-end processing—audio upload, transcription, parsing, and PDF generation—within an average of 4.7 seconds on standard hardware. Its asynchronous I/O model enables concurrent processing of multiple requests while maintaining consistent response times, making it highly suitable for clinical environments with high data throughput. Furthermore, the backend’s modular microservice architecture supports horizontal scalability, allowing deployment across distributed cloud infrastructures while maintaining centralized access control. Overall, the backend functions as the intelligent and secure foundation of the Medical Prescription Parsing API—transforming complex, unstructured voice data into standardized, actionable, and legally compliant medical documentation while ensuring privacy, reliability, and scalability for real-world healthcare deployment.

4.2.19 AI-Powered Multi-Scenario Medical Billing Assistant For Doctors [26]

The backend of the AI-Powered Multi-Scenario Medical Billing Assistant is implemented using a dual-service architecture combining FastAPI and Express to manage request routing, audio processing, and AI-driven billing logic. FastAPI handles core billing operations such as input validation, OHIP/non-OHIP scenario classification, and integration with the Retrieval-Augmented Generation (RAG) engine. Express is used to support real-time streaming features, particularly for the speech-to-speech workflow powered by the OpenAI Realtime API. When a billing request is received—whether from manual form data or Whisper-generated transcription—the backend preprocesses the input, retrieves relevant billing entries from the

Pinecone vector store, and sends the structured context to GPT-4 for code extraction and reasoning. All generated billing codes, fees, timestamps, and policy-based multipliers are stored in a relational SQL database through a set of RESTful endpoints. The backend design ensures modularity, low latency during speech interactions, and reliable persistence of billing records, enabling seamless communication between the user interface, AI engine, and database layer.

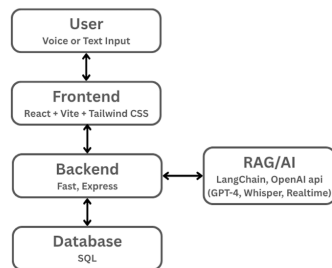


Figure 107. System Architecture

4.2.20 Health Informatics AI Chatbot: An Intelligent Patient Data Analysis System [27]

The backend of the Health Informatics AI Chatbot is built as a high-performance service layer designed to handle natural language queries, orchestrate AI-driven reasoning workflows, and execute database operations with clinical reliability. The team implemented the backend using FastAPI, chosen for its asynchronous architecture and low-latency request handling, which is critical for real-time medical data retrieval. At the core of the backend is an intelligent SQL agent pipeline powered by GPT-4o and LangGraph, enabling multi-step reasoning, schema-aware query generation, and automated routing of user intent. The backend converts clinicians' natural language questions into validated SQL queries, executes them through an optimized MySQL connection pooling framework, and returns structured responses in multiple formats—text summaries, tables, key insights, and error-aware feedback. To ensure stability under clinical workloads, the backend integrates a singleton connection manager, reducing connection overhead by more than 80%, along with a layered logging system that records query latency, agent steps, memory usage, and model inference times. Additionally, the backend provides dynamic patient-based routing, secure API endpoints for cross-component communication, real-time audio synthesis for multimodal interaction, and robust fallback mechanisms that guarantee meaningful responses even when queries fail. Through these combined components, the backend

acts as the system’s computational engine—delivering reliability, scalability, transparency, and medical-grade performance in handling complex patient data analysis.

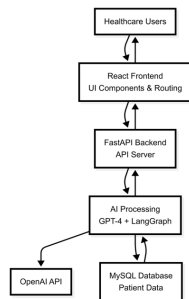


Figure 108. System Architecture

4.2.21 Automation on conversion of Paper-based records to digital text for Data Classification [28]

The backend of the automated paper-to-digital medical record conversion system functions as the core computational engine responsible for handling OCR processing, multimodal data extraction, structured text generation, and file management. Built in Python, it orchestrates the full processing pipeline by receiving document-processing requests from the React frontend, validating incoming files, and routing them through a multimodal OCR workflow powered by Google’s Gemini 1.5 Flash model. Once a file is uploaded, the backend performs preprocessing on PDFs and images by reading file bytes, detecting MIME types, and preparing properly formatted prompts to send to the Gemini API. Through the model’s advanced vision-language capabilities, the backend retrieves both raw OCR outputs and structured interpretations of medical content, enabling extraction of clinical information that traditional OCR tools cannot process—especially mixed handwritten notes, nonlinear text, and forms containing both typed and free-form entries. In addition to interacting with the Gemini API, the backend applies domain-specific regular expression patterns, text-normalization rules, and Python-based cleansing utilities to refine the extracted text and ensure consistency in medical terms, dates, patient identifiers, and diagnostic information. Once processed, the backend automatically generates two parallel outputs: unstructured text files in .docx format, preserving the full transcription for detailed review, and semi-structured .csv files that group information into predefined medical categories for seamless ingestion into future EHR systems or healthcare

databases. The backend also manages local file storage, organizes output directories, and returns a structured response to the frontend containing processing summaries, file names, and completion indicators. This architecture ensures modularity, reliability, and extensibility—allowing additional capabilities such as cloud storage integration, real-time database connectivity, and continuous processing pipelines to be incorporated in future development. Overall, the backend acts as the intelligent processing layer of the system, coordinating OCR tasks, semantic data extraction, text structuring, and output generation with a design optimized for accuracy, scalability, and clinical relevance.

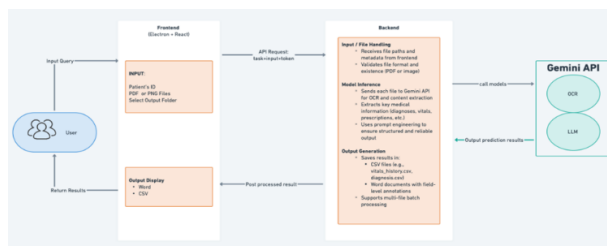


Figure 109. OCR Conversion system layout

4.2.22 Enhancing Clinical Decision-Making with a Voice-Enabled, AI-Powered Medical Assistant [29]

The backend of the voice-enabled, AI-powered medical assistant serves as the system's core computational engine, orchestrating all logic, data processing, and external service communication required for real-time clinical decision support. Built using FastAPI, the backend provides a lightweight, asynchronous framework that enables low-latency interactions between the user interface, NLP components, and third-party APIs. It manages session states, authenticates requests, and routes clinician inputs to the appropriate processing pipeline, ensuring that each user query is handled securely and efficiently. Incoming speech data is first transcribed through the integrated Whisper API, after which the backend performs input validation and forwards the structured text to the LLM-driven reasoning module responsible for clinical inference. The backend also communicates with Dialogflow CX for intent classification, fallback handling, and retrieval of pre-defined conversational flows. A dedicated clinical decision layer, implemented as a set of rule-based modules and medical logic functions, supports symptom interpretation, differential diagnosis generation, and guideline lookup. Additionally,

the backend integrates Google Cloud Speech-to-Text and gTTS for seamless bidirectional voice interaction, while optional HuggingFace Transformer models enhance the system’s ability to interpret complex medical queries. All components are designed to operate in a modular, container-ready architecture, allowing easy scaling, model replacement, and incorporation of future EHR connectivity. Through robust error handling, logging, and test coverage—including unit, integration, and conversational workflow testing—the backend ensures reliability, clinical safety, and consistent performance in high-pressure healthcare environments.

Appendix A – Sample Code Snippets

```
def detect_intent(text, session_id="1234", language_code="en"):
    """Detect intent using Dialogflow"""
    session_client = dialogflow.SessionsClient()
    session = session_client.session_path(DIALOGFLOW_PROJECT_ID, session_id)
    text_input = dialogflow.TextInput(text=text, language_code=language_code)
    query_input = dialogflow.QueryInput(text=text_input)
    request = session_client.detect_intent(request=dialogflow.DetectIntentRequest(session=session, query_input=query_input))
    return request.query_result

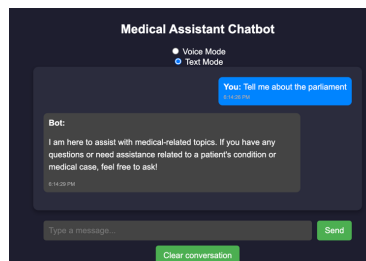
except LongRPSError as e:
    logging.warning("Dialogflow call failed: %s", str(e))
    return None

def update_state(session_id, session_state, new_state):
    """Update the session state using Firestore"""
    prev_state = session_state.get(session_id, {}).get("current", {}).get("state")
    history = session_state.get(session_id, {}).get("history", []) + [prev_state]
    session_state["id"] = session_id
    session_state["prev_state"] = prev_state
    session_state["current_state"] = new_state
    session_state["history"] = history
    logging.info("Current state: %s, Previous state: %s", new_state, prev_state)
    logging.info("New history: %s", history)
```

Appendix B – Sample Dialogflow Intents



Appendix C – Screenshots



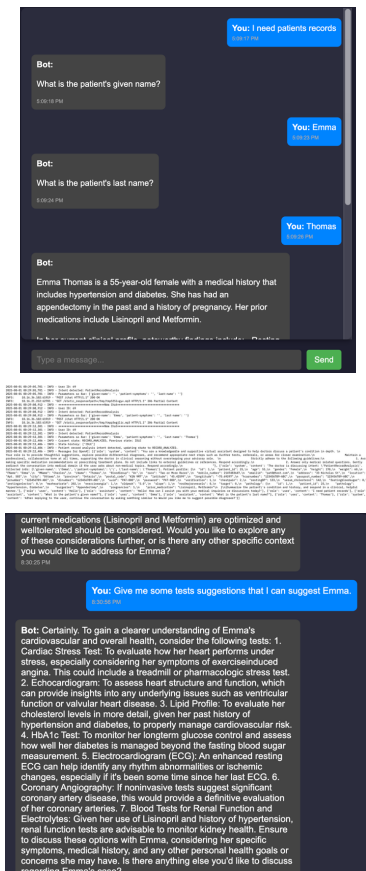


Figure 110. Sample code

4.2.23 A Conversational AI Framework for Medical Education: Integrating Literature, Clinical Data, and Real-Time Research [30]

The backend of the “Conversational AI Framework for Medical Education” forms the core intelligence and operational layer of the system, responsible for orchestrating data retrieval, AI reasoning, workflow automation, and integration with external services. At the heart of the backend is the N8N workflow automation platform, which coordinates multiple interconnected components, including the RAG (Retrieval-Augmented Generation) AI agent, document processing pipelines, and communication systems. This backend infrastructure manages complex

tasks such as routing user queries, selecting the appropriate knowledge source, performing semantic searches, and generating real-time responses while maintaining context through buffered memory management. The backend is strongly supported by multiple database systems, which provide both structured and unstructured data access. The local eHospital MySQL database offers comprehensive clinical records and administrative data through secure RESTful API endpoints, allowing the AI agent to retrieve patient demographics, test results, and other sensitive medical information. Simultaneously, the system employs a Pinecone vector database to store private and public knowledge bases, enabling semantic search over large collections of medical literature and personal documents. Additional storage support is provided through AWS S3 for document management, and SQLite is used for authentication and user account management. These database systems collectively allow the backend to efficiently access, organize, and process diverse data sources, ensuring that the AI agent can deliver accurate, structured, and contextually relevant information to users. By integrating workflow automation, AI reasoning, and robust database support, the backend forms a scalable, secure, and intelligent foundation for the medical conversational agent, capable of supporting both clinical decision-making and medical education.

4.2.24 AI-Assisted Chest X-Ray Diagnostic System Overview [31]

This project presents a complete AI-assisted diagnostic support system designed to help clinicians interpret chest X-ray images by combining image-based disease prediction with patient history and large language model reasoning. The system allows users to upload a chest X-ray image along with a patient history document through a web-based interface, after which both inputs are processed in a cloud-hosted environment on AWS. A deep learning classification model, implemented using TensorFlow and Keras, analyzes the X-ray image to detect the likelihood of 14 thoracic disease categories and outputs confidence scores for each condition. When one or more disease predictions exceed a defined confidence threshold, those diagnoses are selected; otherwise, the system returns the highest-confidence prediction to ensure a clinically meaningful output. The confirmed diagnosis is then combined with the patient's medical history and passed to a large language model, which generates a structured list of recommended follow-up tests to support further clinical decision-making. The entire system is

containerized using Docker and consists of a clearly separated frontend and backend architecture, enabling automated deployment, scalability, and centralized logging. The backend is responsible for model execution, preprocessing, inference, and communication with the LLM, while the frontend manages file uploads, diagnosis requests, and user interaction through a Gradio-based interface. Model training was conducted using the ChestX-ray8 open-source dataset, with approximately 30,000 images used for training and 10,000 for validation, requiring careful handling of class imbalance and preprocessing standardization. Multiple model versions were evaluated, and a TensorFlow/Keras implementation was ultimately selected due to its superior ROC performance and training efficiency compared to alternative DenseNet-based approaches. Overall, the system demonstrates an end-to-end workflow that integrates deep learning, cloud deployment, and LLM-based reasoning to reduce diagnostic uncertainty, support clinicians—especially less experienced practitioners—and establish a foundation for future extensions such as explainability tools, ensemble learning, and broader clinical integration.

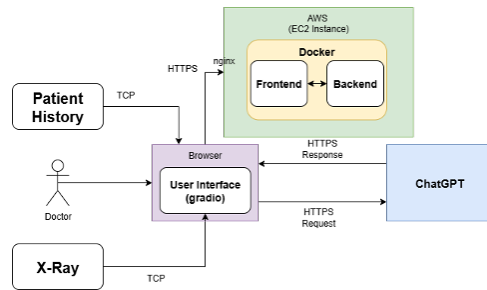


Figure 111. Block diagram depicting solution workflow

4.2.25 Architecture and System Orchestration for Autonomous OR Turnover Simulation [32]

The backend of the Autonomous Operating Room Turnover Simulation system is responsible for orchestrating simulation control, autonomous navigation, perception processing, data fusion, and automated reporting within a unified and reproducible robotics framework. It is built around ROS 2 as the core middleware, enabling asynchronous, message-driven communication between navigation, sensing, anomaly detection, coverage mapping, and reporting modules. At the foundation, NVIDIA Isaac Sim operates as the simulation backend,

providing a physics-accurate digital twin of the operating room, realistic sensor emulation, and deterministic time control. Through the Isaac ROS 2 Bridge, simulated sensor outputs—including LiDAR scans, depth images, point clouds, odometry, and transform frames—are published directly to ROS 2 topics, while velocity commands generated by Nav2 are consumed by the simulator to drive robot motion in real time.

Within ROS 2, the backend is structured as a collection of loosely coupled nodes, each with a clearly defined responsibility. The navigation backend leverages the Nav2 stack to handle localization, costmap generation, global planning, local control, and Behavior Tree execution. A custom coverage controller node generates a boustrophedon waypoint sequence that is fed into Nav2 as a series of navigation goals, allowing deterministic coverage execution while preserving reactive obstacle avoidance and recovery behaviors. In parallel, the perception backend subscribes to depth camera topics and converts raw depth frames into 3D point clouds, performs RANSAC-based plane fitting to estimate the OR floor, computes geometric residuals, and assigns continuous anomaly likelihood scores using a Gaussian model. These scores are discretized into a grid-based anomaly map that is temporally fused across frames to suppress noise and emphasize persistent irregularities.

A dedicated coverage mapping backend node accumulates robot pose updates into a high-resolution spatial grid, tracking visit frequency for every cell in the OR workspace. This enables quantitative evaluation of inspection completeness, revisit density, and coverage uniformity. The anomaly and coverage grids are aligned in a shared world coordinate frame, allowing backend fusion into composite risk maps that highlight both inspected and anomalous regions. All intermediate data products—poses, sensor streams, anomaly scores, heatmaps, and metrics—are logged using ROS 2 bagging and internal data structures to support deterministic replay and offline analysis.

At the final stage, the backend includes an automated reporting module that aggregates navigation statistics, coverage metrics, anomaly localization results, and visual artifacts into a structured one-page OR turnover QA report. This module programmatically generates heatmaps, overlays anomaly contours, computes performance metrics such as recall, precision, and localization error, and exports results in machine-readable formats (JSON, NumPy arrays) as

well as human-readable summaries (Markdown or PDF). The backend is designed to be simulation-agnostic at the interface level, meaning the same ROS 2 nodes and data flow can be deployed on physical robots with minimal modification. Overall, this backend architecture provides a scalable, modular, and clinically oriented foundation for validating autonomous OR quality-assurance robots through high-fidelity simulation prior to real-world deployment.

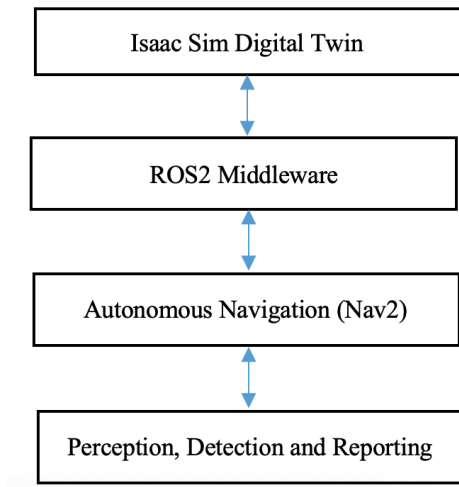


Figure 112. System Overview

4.2.26 AI-Based Pre-Operative Psychological Screening System [33]

This report presents the design, implementation, and evaluation of an AI-assisted pre-operative psychological screening and decision-support system aimed at improving the assessment of patients' psychological readiness for surgery, an area that is often under-addressed in clinical workflows due to time constraints, reliance on static questionnaires, and the absence of integrated decision-support tools. The system is grounded in established psychological theory and validated clinical instruments, integrating constructs from the Generalized Anxiety Disorder scale (GAD-7), Patient Health Questionnaire (PHQ-9), Amsterdam Preoperative Anxiety and Information Scale (APAIS), and Cognitive Appraisal Theory to assess four key domains: anxiety, mood, surgery-specific fear, and coping readiness. Unlike traditional static screening

approaches, the system employs a Dynamic Question Generator that selects a balanced set of 18 questions per session from curated, theory-driven domain pools, introducing controlled variability while preserving clinical validity and scoring consistency. User responses are captured using a Likert-scale format and processed by a backend architecture implemented with FastAPI, where domain-level scores are computed, severity thresholds are applied, and features are assembled for machine learning inference. A Random Forest classifier, trained on a synthetic dataset of approximately 250–300 patient profiles due to the lack of real clinical data, predicts an overall psychological risk level, selected for its robustness, interpretability, and suitability for small, structured datasets. Although the explicit risk category is not visually displayed in the user interface, the model's output directly influences the interpretation logic, ensuring that AI-driven risk estimation remains central to the system's conclusions. To address the critical need for transparency in healthcare AI, the system incorporates a deterministic explainable interpretation engine that transforms numerical scores and machine learning outputs into structured, clinically readable narratives, identifying primary psychological contributors, secondary influencing factors, and protective elements such as strong coping readiness, while avoiding opaque generative language models. The frontend, implemented in React, presents results in a clear and structured format, combining quantitative domain summaries with qualitative narrative insights to support efficient clinical review without manual scoring. End-to-end testing demonstrated reliable dynamic question generation, accurate domain scoring, stable machine learning inference, and consistent interpretation generation, with the Random Forest model achieving approximately 85–90% accuracy and showing particularly strong performance in identifying moderate-to-high risk cases, which are most clinically relevant in pre-operative contexts. The discussion highlights the system's advantages over conventional tools, including reduced subjectivity, improved efficiency, integrated multi-domain assessment, and explainable AI support, while acknowledging key limitations such as reliance on synthetic data, simplified threshold-based interpretations, lack of true adaptive questioning, and absence of formal clinical validation. The report concludes that the proposed system successfully demonstrates the feasibility of combining psychological theory, dynamic assessment, machine learning, and explainable decision-support into a unified pre-operative screening workflow, and outlines future directions including validation with real patient data, prospective clinical studies, advanced model comparison and tuning, enhanced explainability using techniques such as SHAP, deeper

integration with electronic health record systems, and expansion toward adaptive testing and longitudinal psychological monitoring.

4.2.27 AI-Assisted Real-Time Polyp Detection and Risk Labelling [34]

This project presents a lightweight deep learning-based system designed to support real-time colorectal polyp detection and basic risk classification during colonoscopy. The motivation is to reduce missed lesions, which remain common even with modern endoscopic equipment, and to provide assistive visual feedback without disrupting clinical workflows. Using the POLAR dataset of annotated colonoscopy images with histopathology labels, two small YOLOv8n models were trained: a one-class detector to localize polyps and a two-class detector to distinguish low-risk (hyperplastic) from high-risk (adenomatous or sessile serrated) lesions. The system was intentionally designed for low latency and easy integration into e-hospital or robotic endoscopy environments. After training and validation on POLAR, the models were applied frame by frame to external colonoscopy videos from the HyperKvasir dataset to evaluate generalization in a realistic setting. Detected polyps were displayed with color-coded bounding boxes and confidence-based risk labels, demonstrating stable real-time overlays at clinical frame rates. Results show that the pipeline is feasible as an assistive “second observer” that highlights suspicious regions and provides coarse risk information while keeping the endoscopist in control. Although performance is limited by dataset size, class imbalance, and domain shift between datasets, the work demonstrates a practical proof of concept for integrating compact YOLO-based models into real-time colonoscopy systems, with clear directions for future improvements using larger datasets, unified architectures, and fully annotated video data.

4.2.28 Medical Prescription Parsing API: A Secure and Intelligent Audio-to-Prescription Processing System [35]

The backend of the Medical Prescription Parsing API forms the core computational framework that powers the entire audio-to-prescription workflow, managing everything from data ingestion and transcription to structured data extraction and secure document generation. Built using FastAPI, the backend operates as a lightweight yet highly scalable RESTful microservice capable of processing multiple simultaneous audio submissions with minimal

latency. Its architecture is composed of several coordinated modules—Audio Handling, Speech Recognition, Text Parsing, Data Structuring, and PDF Rendering—each designed to perform a specific, optimized function in the processing pipeline. When an audio file is submitted through the `/submit_audio` endpoint, it is temporarily stored in a secure environment under ephemeral storage policies that automatically delete files after processing to preserve patient confidentiality. The backend then invokes OpenAI’s Whisper model, which performs high-fidelity transcription with multilingual support and noise resilience, effectively handling complex medical vocabulary, drug names, and dosage instructions under real-world acoustic conditions.

Once transcription is complete, the backend transitions into its Natural Language Processing (NLP) and data extraction phase. This stage uses a hybrid rule-based and regular expression parsing engine that identifies and structures critical prescription components such as patient identifiers, medication names, dosages, administration frequencies, prescriber credentials, and treatment notes. These extracted entities are serialized into a structured JSON object, ensuring interoperability with electronic health record (EHR) systems and downstream analytics tools. To maintain accuracy and adaptability, the backend employs a modular parser configuration, enabling developers to easily update extraction rules or integrate advanced NLP modules like Named Entity Recognition (NER) models in future releases.

Following data extraction, the backend activates the PDF rendering module, which utilizes ReportLab to populate a standardized prescription template with the parsed data. This module ensures consistent formatting, legibility, and compliance with clinical documentation standards, while embedding digital signatures and timestamps for traceability. The generated PDF is returned to the client along with the structured JSON data, enabling both human-readable and machine-readable outputs for seamless workflow integration. Throughout this process, the backend enforces strict security and compliance protocols, including Transport Layer Security (TLS) for encrypted communications, CORS restrictions and IP whitelisting for controlled access, and audit logging mechanisms that track operations without storing personal health information (PHI).

Performance evaluation revealed that the backend achieves exceptional responsiveness, completing end-to-end processing—audio upload, transcription, parsing, and PDF generation—

within an average of 4.7 seconds on standard hardware. Its asynchronous I/O model enables concurrent processing of multiple requests while maintaining consistent response times, making it highly suitable for clinical environments with high data throughput. Furthermore, the backend's modular microservice architecture supports horizontal scalability, allowing deployment across distributed cloud infrastructures while maintaining centralized access control. Overall, the backend functions as the intelligent and secure foundation of the Medical Prescription Parsing API—transforming complex, unstructured voice data into standardized, actionable, and legally compliant medical documentation while ensuring privacy, reliability, and scalability for real-world healthcare deployment.

5. Database Documentation

The database is the organized data portion of the e-Hospital website. This section of the e-Hospital is also not visible, but it is where all necessary information is stored, including patient records, test results, and other relevant data. The database is where all of the information is stored and retrieved from.

Similar to the backend, the majority, if not all, of the information included in this section is taken from past reports. The database documentation section of the report will be divided into the following subsections: MongoDB documentation, MySQL documentation, and OpenAI.

5.1 MongoDB Documentation [36]

This type of database is mainly used to record such as images. Upon examining the e-Hospital website, it becomes apparent that many of its features require image input and storage, including chatbots, patient records, and other applications. These images include CT scans, MRIs, X-rays, and other diagnostic tests. According to the referenced paper, there are two databases on the MongoDB server: htdata and mldata. The htdata got its name from the health tests data. It is used to store patients' health tests. At the time when the paper was written, the only data that gets collected includes:

- CT-Scan_Abdomen
- CT-Scan_Breast

- Cell-Images
- Endoscopic
- MRI_Brain
- X-Ray_Lung
- Kidney
- Template

```

_id: ObjectId('640b53f86d51186048452aa3')
patient_id: 25
RecordDate: "2023-03-01 19:00:01"
  file: Object
    filename: "image"
    originalname: "Image_20230215111208.jpeg"
    encoding: "7bit"
    mimetype: "image/jpeg"
    buffer: BinData(0, '/9j/4AAQSkZJRgABAQAAASABIAAD/7QA4UGhvdG9zaG9wIDMuMAA4Qk1NBQAQAAAAAAAAAQk1NBQAQAAAAABDUHYzZjwCyB0mACZjs...' )
    size: 109975

```

Figure 113. Example of Patients' Health Test

The mldata is created to store data for machine learning training. Since we are only beginning to utilize machine learning in our projects, there is currently no data in this database. The following images and brief descriptions provide step-by-step instructions on how to use MongoDB. You first start with the interaction with the web using APIs. All of the APIs in the e-Hospital were listed in the backend documentation section.

POST http://localhost:5000/imageUpload

Params Authorization Headers (9) Body Pre-request Script Tests Settings

none form-data x-www-form-urlencoded raw binary GraphQL

Key	Value
image	C378P2_thinF_IMG_20150620_133111a_cell_86.png
phoneNumber	6131230000
recordType	Cell-Images
recordDate	2023-04-01 12:00:00

Body Cookies Headers (7) Test Results

Pretty Raw Preview Visualize JSON

```

1
2  {"success": "New Image created.",
3   "id": "6470f05e91a6f20ac34b0807"}
4

```

Figure 114. Visual Representation of an Image Upload

Once that step is complete, we need to ensure that both data points and the Python script are inputted into the same directory. Before executing the script, which will ensure that the database is updated.

```

1 # pip install pymongo
2 from pymongo import MongoClient
3 import glob
4
5 # Connect to the server with the hostname and portNumber.
6 connection = MongoClient("mongodb+srv://ehuser:ehuser@hospital.mgo2.mongodb.net/test")
7
8 # Connect to the Database where the images will be stored.
9 database = connection["testdata"]
10 collection = database["X-Ray"] # update the collection name to the record type like "X-Ray"
11
12 # Scan all .jpeg files in the current directory
13 fileName = glob.glob("*.jpeg") # change the ext that matches your files
14 # file = "Image_20230215111200.jpeg"
15
16 id = 132 # change the id of the patient if needed
17 for file in fileName:
18     # Open the image in read-only format.
19     with open(file, "rb") as f:
20         contents = f.read()
21         length = len(contents)
22         record = {
23             "patient_id": id,
24             "RecordDate": "2023-03-01 19:00:00", # change the record date if needed
25             "file": {
26                 "originalname": file,
27                 "encoding": "utf-8",
28                 "mimetype": "image/jpeg", # change the type based on the file ext
29                 "buffer": contents,
30                 "size": length
31             }
32         }
33     # Now store/put the image.
34     x = collection.insert_one(record)
35     print(x.inserted_id)
36
37 # switch to next id
38 id += 1

```

Figure 115. Python Script for the MongoDB

Lastly, we need to verify the test details by locating the datasets and clicking on them to view the added data.

As mentioned at the start of this section, MongoDB can be used to train machine learning datasets. The reference paper explains the process of adding data to the database for machine learning purposes, and that starts with the file type, which can be a CSV file or a group of .jpeg images. You start by uploading the file and ensuring the data is in the same directory as the database, then you download the files, and lastly check the files. The images below show the script for each of the three steps.

```

9 # Connect to the Database where the images will be stored.
10 database = connection['mldata']
11
12 # Create an object of GridFS for the above database.
13 fs = gridfs.GridFS(database, "template") # update the collection name to the record type like "BloodTest_Dataset"
14
15 # Scan all .jpeg files in the current directory
16 fileName = glob.glob("*.jpeg") # change the ext that matches your files
17 # file = "image_20230215111208.jpeg"
18
19 for file in fileName:
20     # Open the image in read-only format.
21     with open(file, 'rb') as f:
22         contents = f.read()
23
24     # Now store/put the image via GridFS object.
25     fs.put(contents, filename="image.jpeg") # update the name to some identifiable like "healthy.jpeg" or "diseased.jpeg"
26

```

Figure 84. Uploading the Files for MongoDB - ML training

```

8 # Connect to the Database where the images will be stored.
9 database = connection['mldata']
10
11 # Create an object of GridFS for the above database.
12 fs = gridfs.GridFS(database, "template") # change the name to your collection
13
14 i = 0
15 # Retrieve all file under the collection
16 for grid_out in fs.find({"filename": "image.jpeg"}): # change the name to your identifiable file name
17     print(f"Retrieve file in index {i}")
18     i += 1
19     file = grid_out.read()
20
21 # Write data to the current directory
22 f = open(f'image{i}.jpeg', 'wb') # change to your correct file type.
23 f.write(file)
24 f.close()
25

```

Figure 85. Downloading the Files for MongoDB - ML training

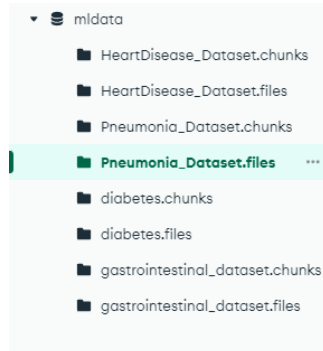


Figure 86. Checking the Files for MongoDB - ML training

5.2 MySQL Documentation [37]

This type of database is used to record raw data. It is essential to note that some of the created databases do not currently have a frontend feature, but they were created for use when the frontend feature is implemented. All of this can be clearly seen in the references paper. This section discusses five different categories in the MySQL database.

Starting with the Account Info, which contains information such as clinic_admin, doctors_registration, hospital_admin, lab_admin, master_admin, nurses_registration, and patient_registration.

The second database is the Account Related Info, which contains the following list of databases: clinic_recordauthorized, clinic_servicehistory, doctor_recordauthorized, hospital_recordauthorized, lab_appointment, lab_recordauthorized, nurse_recordauthorized, psychology_adhd_questionnaire, psychology_depression_questionnaire, psychology_dr, and psychology_patients.

The third one is the Health Test, which contains the bloodtests, bloodtests_liver, ecg, eye_test, mri_alzheimer, physical_test_cad, physical_test_ck, physical_test_hd, physical_test_ms, realtimetest_heart_disease, and tumor.

The fourth database is the Diseases database, which has a list of diseases, including alzheimer, arrhythmia, brain_tumor, breast_cancer, breast_disease, cardiovascular, chronic_kidney, coronary_artery_disease, diabetes, gastrointestinal_disease, heart_disease, juvenile_myopia, kidney_stone, liver_diseases, malaria, multiple_sclerosis, and pneumonia.

Lastly, there is a section for uncategorized databases, categorized under the "Others" database, which includes active_record_disease_list, contact_us, lims_patient, lims_test_fields, and symptoms_checker.

The paper also guides us through the steps to add a new database for a new type of health test. We can simply create a new table for the new health test as long as it is the original health test, as can be seen in the image below.

ecg	
Column Name	Data Type
id	PK, int
patient_id	FK, int
(record data)	...
...	...
RecordDate	datetime

Figure 8719. MySQL Documentation - Health Test Schema

However, if the disease type is not already in the database, then we need to use the script shown below to add it.

```
CREATE TABLE cardiovascular (
  id INT NOT NULL AUTO_INCREMENT,
  patient_id INT NOT NULL,
  prediction_date DATETIME NOT NULL,
  prediction VARCHAR(50) NOT NULL,
  accuracy VARCHAR(50),
  record_type VARCHAR(50),
  record_id VARCHAR(50),
  PRIMARY KEY (id),
  FOREIGN KEY (patient_id) REFERENCES patients_registration(id),
  UNIQUE (patient_id)
)
```

Figure 88. MySQL Documentation - Creating a New Disease Type

5.3 General Database Information

This section is a continuation of the one presented in the backend documentation section, where it was mentioned that each of the following subsections is based on final reports submitted in the previous semester of work related to the e-Hospital and the work presented in this technical paper.

5.3.1 Intelligent Clinical Decision Support and Medical Chatbot Enhancements [1]

The database design for Intelligent Clinical Decision Support and Medical Chatbot Enhancements outlines the storage, retrieval, and management of patient information, clinical guidelines, chatbot communication, and decision-making algorithms. The system utilizes a relational database (PostgreSQL) to capture demographic data, medical history, and

consultations in real time. A NoSQL database, specifically MongoDB, is used to store, retrieve, and analyze unstructured logs of bot interactions and conversations. The FHIR (Fast Healthcare Interoperability Resources) standard is used to integrate the chatbot and the decision-support system with Electronic Health Records (EHRs) using the appropriate application programming interfaces (APIs). The schema includes tables for patient profiles, clinical rules, response logs, and outputs from AI models, all of which are integrated with foreign key constraints to ensure referential integrity. Search indexes enhance the speed of query execution, and this is particularly beneficial in decision-making scenarios where the system requires symptom inputs. The system is built on a microservices architecture. Thus, individual services handle user login, chatbot NLP, clinical data interpretation, and data retrieval independently. An API gateway on the backend fulfills the requests, and a message queue (such as RabbitMQ or Kafka) provides decoupled communication between components. The architecture incorporates a comprehensive RESTful API layer that enables seamless access to front-end applications without compromising interaction with backend databases, while ensuring regulatory compliance with HIPAA standards.

5.3.2 e-Hospital: Intelligent Clinical Decision Support [2]

Intelligent Clinical Decision Support (ICDS) System integrates the database to serve the purpose of management, accessibility, and security of patient data. MySQL offers a powerful and scalable framework to store a diverse collection of patient information including, but not limited to, electronic health records (EHRs), treatment plans, medical histories, and diagnostic data. It allows real-time synchronization between the patient data and AI generated system insights, therefore enabling timely decision-making for healthcare professionals. Another important aspect of the database technology's infrastructure is the merging with a Python backend because it allows seamless transitions from the frontend UI to the backend processes. In order to protect patient confidentiality, the database uses role-based authentication systems, which according to the data security laws, permits access only to authorized personnel such as doctors. In addition, the database's architecture is designed to allow for retrieval and querying of patients' information which can then be used for making timely decisions about diagnosis, treatment, and subsequent care. Regular validation procedures and testing ensure the database's integrity and ensure that all stored data remains accurate and compliant with healthcare regulations.

The construction of the AI model for Intelligent Clinical Decision Support (ICDS) is done methodically to ensure effective and efficient development of the decision support system. In the architecture diagram **Error! Reference source not found.**, the process initiated with data collection commences with gathering preliminary datasets such as electronic health records (EHRs) from various healthcare entities to enable the AI model construction. This is followed by data preprocessing in which the data is verified and cleaned for use with the GPT model. The model is most suitable for this task due to its excellent capabilities for natural language processing and generating deeper-relevant medical answers. Fine-tuning the model improves its understanding of the clinical language along with the recommendations generated. The model is now loaded into the ICDS dashboard which is the interface for clinicians to input patient data and receive generated outputs from AI. Integrated systems are rigorously tested and evaluated in order to ensure that they meet clinical performance standards prior to deployment for operational use in the healthcare environment. Performance and compliance on the system is managed and maintained through perpetual monitoring and updating. Healthcare professionals need to provide their comments to continually improve every phase, confirming that the AI model and dashboard serve the users' needs and augment the patient care results. This design principle fosters collaboration between data scientists, healthcare professionals, and information technology experts to ensure effective deployment of the ICDS system.

5.3.3 e-Hospital Receptionist Chatbot [8]

The database for the E-Hospital Receptionist Chatbot has been structured to manage and store patient and appointment information to enhance smooth interactions between the patients and the healthcare providers. The database architecture allows for different entities such as patients, healthcare providers (doctors), appointments, messages, and tasks. All entities are designed to allow easy management and retrieval of information whilst ensuring integrity.

- Patients: The patient table contains records relevant to the demographic attributes such as name, contact information, and the medical history of the patient. It also monitors the dealings the patients have with the chatbot, including but not limited to appointment request and communication with doctors.

- Healthcare Providers (Doctors): The provider table includes information pertaining to the doctor's other qualifications as well as his /her credentials, the areas of specialization, and free time slots. This table is linked with the appointment table so that scheduling as well as tracking of the appointments with particular patients can be done efficiently.
- Appointments: The appointments table holds data for anticipated appointments which has patient id, doctor id, date and time of the appointment, and status of the appointment. This enables and facilitates easy modification, cancellation, as well as confirmation of appointments through the interface of the chatbot.
- Messages: The messages table saves every interaction that occurs between a patient and a doctor. Each message is linked to the specific patient's and doctor's details along with date and time to keep a comprehensive record of all conversations.
- Tasks: The management of routine tasks such as follow-up visits, reminders for medication, and even some diagnostic procedures is made easier with the help of a separate table. This aids both the patients as well as the doctors in managing healthcare processes in an organized manner.

These entities are integrated in the database system and related information is stored in other tables. For consistency across the system, primary and foreign keys serve as relational integrity constraints. For security, sensitive information like medical records and personal details are protected through encryption using standard protocols. Additional regular data backups are set in place to maintain problem-free operation of the system and avoid data loss.

Designed with regard to user convenience, the architecture of the Appointment Management Chatbot system ensures effective communication with patients and healthcare providers. The system has several components that form the backbone of efficient communication and provide a seamless user experience.

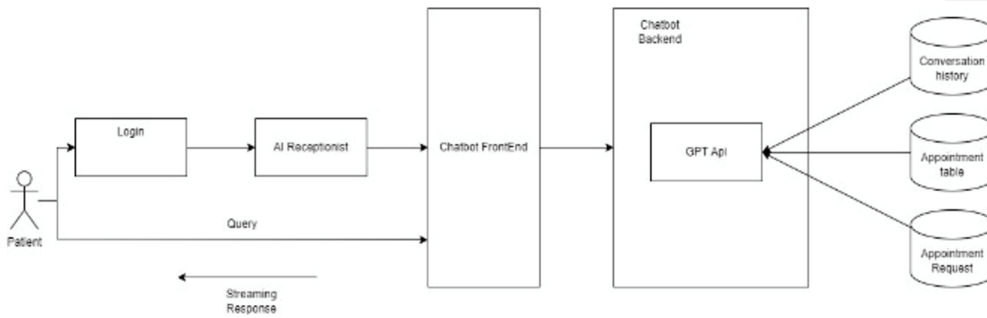


Figure 89. Architecture Diagram of the e-Hospital Receptionist Chatbot

5.3.4 E-Health Medical Chatbot [9]

The database section of the e-hospital system plays a crucial role in supporting the E-Health Medical Chatbot in several key ways, including user authentication and role-based access. The chatbot requires secure authentication to ensure that only authorized users (e.g., doctors, nurses, or patients) can access specific medical information. The e-Hospital database stores user credentials, access levels, and medical roles, enabling the chatbot to determine the appropriate level of response. Medical Knowledge Base Integration, where the chatbot relies on a structured medical database that includes disease symptoms, treatments, drug interactions, and clinical guidelines. The e-Hospital database can store and regularly update this information to ensure the chatbot provides accurate and evidence-based responses. Query handling and data retrieval, where the chatbot interacts with the e-hospital database to retrieve relevant medical information, such as appointment schedules, doctor availability, or lab test results. Structured queries allow the chatbot to fetch necessary details dynamically based on user requests. Logging and monitoring interactions focus on conversations between users and the chatbot, which can be stored in a database to track system performance, identify frequent medical inquiries, and ensure compliance with healthcare regulations. This data can be analyzed to enhance chatbot responses and identify areas for improvement in medical guidance. This section addresses the implementation of privacy and data security measures, where the database section utilizes encryption and de-identification techniques to safeguard sensitive user data. The chatbot filters responses based on stored privacy settings, ensuring it does not disclose confidential patient information.

5.3.5 e-Hospital AI Treatment Plan Enhancement and Optimization [10]

To facilitate effortless integration, retrieval, and management of patient treatment data for the E-Hospital AI Treatment Plan Enhancement and Optimization project, a comprehensive documentation of the database is necessary. The database design should have separate structured tables for the patients' demographic details, their medical history, AI treatment plans, physician treatment alterations, and treatment outcome evaluations. It is necessary that each table determines primary keys (for instance, Patient_ID, Treatment_ID) and establishes foreign keys to maintain relationships between tables in addition to defining indexing strategies for faster access to data. Moreover, the documentation needs to include data types, restrictions, and structural relationships in order to avoid discrepancies. There needs to be an adequate mechanism that enables revisions made to treatment plans to be tracked to ensure AI provided action recommendations and physician approved modifications can be audited. Comprehensive security measures including role-based access control (RBAC), data encryption, and HIPAA and PIPEDA compliance need to be established for the system. In addition, the system needs to allow for real time updates and integration with current hospital management systems through the use of API endpoints and HL7/FHIR protocols. Having clear documentation on backup, recovery, and maintenance routines is necessary to maintain integrity and accessibility of the data.

The AI treatment plan e-hospital system uses a chatbot framework to simplify communication between the doctors and the AI model. A physician must log into the platform, select a patient from the list, and the system will check if any chat history exists. If there are prior conversations the system picks records from MySQL server and sends to the chatbot interface to facilitate continuity of recommendations. In the absence of prior conversations, the backend retrieves the patient's basic information which includes but is not limited to case history, measurements, and other treatments they have been subjected to through an API call. This guarantees that the AI treatment plan is tailored to the specifics of the patient. The back end of the chatbot does conversation management and generation with the help of tuned GPT models allowing for flexible understanding of user intentions. To ensure trust and compliance to medical standards, the system grants or withholds access based on role permissions. This allows the system to secure sensitive information. These features allow better automation of interaction and

interrelated decision-making processes within the e-hospital chatbot interface which makes use of AI technologies.

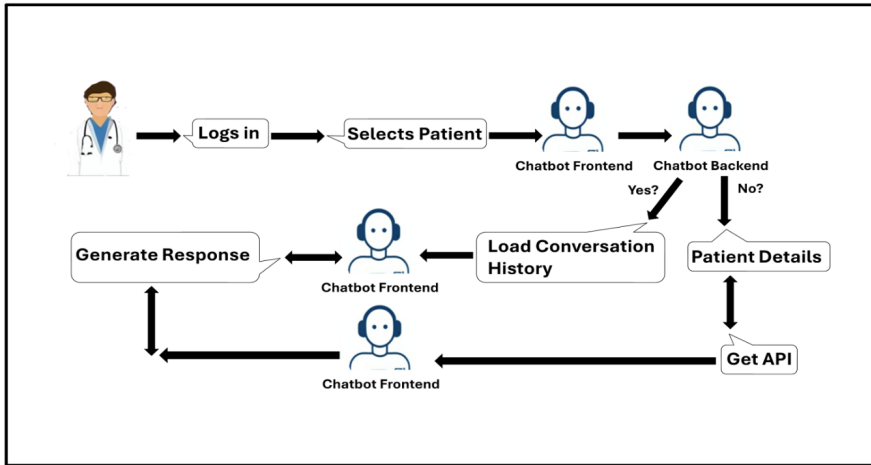


Figure 90. Architecture Diagram of the AI Treatment Plan

5.3.6 ML Model Evaluation and Interpretation [11]

The ML Model Evaluation and Interpretation system uses a relational database as the central repository for machine learning models, datasets, and evaluation results. This database will streamline data and file integration with the e-Hospital platform since model files and patient data will automatically be uploaded and downloaded. Some notable tables include Models containing model metadata such as the model name, version, format, and date uploaded, Datasets which keeps records of a dataset ID, source, feature count, and whether or not the dataset has been preprocessed, and Evaluation_Results containing accuracy, F1 score, ROC curve values, and SHAP importance scores. The backend uses structured API endpoints to interact with the database enabling an effective model evaluation; model evaluation does not require direct file upload because the retrieval and query process is automated. Ingestion of real patient records as datasets will be automated in future versions for higher model evaluation and adaptability efficiency in the e-Hospital environment.

5.3.7 E-hospital Recognize Database Structure [26]

In an ongoing effort to improve the e-Hospital platform, the database underwent an extensive reorganization catering to issues of scalability, reusability, and intended future integration with AI-based diagnostic and decision-support tools. The previous database model had suffered from fragmented, ad hoc tables which could neither support the emerging models of AI, nor provide a common structure for patient information. The new design has taken up a life cycle-based approach to health care management borrowed from the best practices in TELUS CHR and EPIC systems to ensure that everything is well-connected from clinical interviews to patient outcomes post-treatment. A key central feature is the integration of structured and unstructured data, where structured would include lab tests and patient demographics, and unstructured would include medical images. The reorganization introduces new core tables of organs, diagnosis codes, and patients' histories that include family histories, medical histories, surgical histories, social histories, vital histories, and allergy histories, each linked in a relation to the main patient registration table. This not only improves the speed of reception of data and accuracy of the information but also facilitates the implementation of additional features such as AI diagnosis prediction based on background and clinical data. A major innovation is the proposal to include standardized outcome assessments such as PROMIS, which can be stored in JSON format as a flexible, expandable questionnaire linked to treatment plans. This architecture facilitates outcome measurement in mental, physical, and social domains employing systems like CAT (Computer Adaptive Testing). By and large, the novel database is structured to offer a solid spine for clinical inference, decision support, and longitudinal digital health research, citing real-world datasets like MIMIC for direction in future development and evaluation. The final ERD in the appendix reveals this enhanced architecture, forming a scalable basis for future AI model integration and system enhancement.

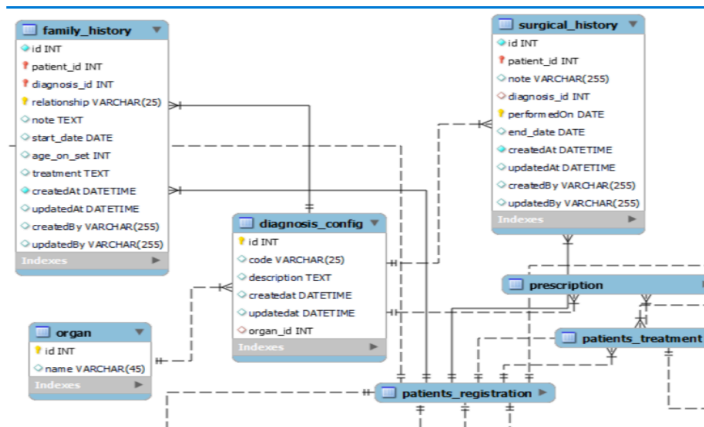


Figure 91. Diagnosis Code Integration for Patient Management

5.3.8 E-hospital Data Governs Policy [27]

The e-hospital platform runs on a mixed data setup that uses tidy SQL tables for structured records and MongoDB for free-form documents, giving the system room to grow as digital healthcare changes. A thick, layered policy watches over the data so that each bit stays accurate, private, reachable, and in step with Ontario rules such as PHIPA, PIPEDA, and OHIP billing. Within that guardrail the platform juggles patient files, clinician profiles, appointment logs, test results, x-rays, prescriptions, treatment notes, machine-learning alerts, invoices, drug supplies, and even real-time chat threads.

Every data domain in the system is maintained through modularized tables, well-structured according to strict schema standards. Tables are to be named in snake_case with every table granted a primary key, for example, patient_id, doctor_id, etc., as well as foreign keys with referential integrity conditions that cover cascading updates or deletes. All schema objects must maintain fields for metadata, including created_at, updated_at, created_by, and updated_by to help traceability and accountability for auditing purposes. Tables are normalized up to Third Normal Form (3NF) to prevent redundancy, though justified denormalization is permitted for performance-related reasons. Any database structure change must be followed by a formal Change Request (CR) and must receive approval from the Database Governance Committee, including with tagging for version control and documentation update of the schema.

To maintain interoperability and data consistency, the platform maintains a centralized Data Dictionary and Entity Relationship Diagram (ERD) which are refreshed at every release. The data dictionary comprises metadata for every table and column: data types, default values, constraint rules, business logic if derived, and real-world examples. This metadata repository aids in traceability, helps speed up the onboarding of new developers and analysts, and is a single source of truth for the evolution of the system-wide schema.

Access control in the system is enforced using RBAC and ABAC mechanisms. Some of the roles considered are: Patient, Doctor, Specialist, Clinical Staff, Analyst, and Admin, with each role having certain access privileges. The database records the accesses, including records of such information as user ID, IP address, timestamp, purpose of access, and record IDs of the affected records. Temporary access is granted on a time-bound basis through secured tokens to allow specialists to give reviews or second opinions. Unauthorized access attempts are recognized and escalated to the Security Officer for investigation.

Data classification policies are established along content sensitivity: Public (e.g., anonymized health statistics), Internal (e.g., clinical notes), Confidential (e.g., prescriptions), and Restricted (e.g., Personal Health Information). Each level determines access protocols, encryption protocols, retention procedures, and audit procedures. Sensitive data fields are protected through Transparent Data Encryption (TDE) in SQL and field encryption in MongoDB. All communications are encrypted with TLS 1.3, and APIs are authenticated using JWT with role-level scoping.

When it comes to data quality management, the platform performs data profiling at the ingestion level to identify incomplete or invalid inputs, form-level validation for user inputs (e.g., date pickers, dropdowns, required fields), and cleanup scripts are scheduled to be executed periodically to remove duplicates and deprecated references. A handful of KPIs, such as >95% completeness for patient records, are tracked and reported monthly.

Master Data Management ensures centralized control over major entities-patients, doctors, hospitals, clinics, pharmacies, and medicines-between modules like `patients_registration`, `doctors_registration`, and `pharmacy_registration`. The master records are

versioned for synchronization across services to remain consistent with one another and avoid duplication.

A very strict retention and archiving policy protects the platform: active clinical data is retained for a minimum of 10 years; inactive records undergo archival after two years of inactivity in cold storage via MongoDB, with metadata references retained for system purposes. Requests for deletions under PHIPA (such as right to be forgotten) are met with respect to non-essential clinical notes. All guidelines pertaining to archivals and deletions are formalized in the logging system to render them auditable.

User identity, operation type, record identities, affected tables, statuses of successful or unsuccessful operations, and timestamps are all included in the log entry. Every operation, such as UI actions, API calls, or raw database queries, is completely logged in comprehensive audit trails. Any potentially sensitive operation, such as deletion or edits to PHI records, goes through a multi-tier authorization. Logs residing in hot storage are retained for three years, while archived logs sit for an additional seven years. Anomaly scans run weekly to detect suspicious access patterns and bring such incidents to manual review.

Moving on to ensure performance and scalability, strategic indexing is applied to primary key fields, foreign key fields, and fields used in queries most frequently. Composite indexes are brought into play only after a performance test. Obsolete or redundant indexes are brought to the bulletin and deleted periodically. Data synchronization is carried out with external entities through HL7 FHIR protocols such that hospitals, public health systems, and insurance bodies can operate on a common basis. Interoperable APIs have versioning, security, and scoping in place and are limited to fields necessary for maintaining patient anonymity where so required.

Database governance is looked after by a committee that is cross-functional, with members entering from each group. The committee consists of the Database Administrator (Chairman), Security Officer, Clinical Compliance Officer, Backend Engineering Lead, and External Auditor. The group convenes quarterly or, more importantly, meeting on the occurrence of incidents; with each session given topics spanning review on schema evolution, compliance

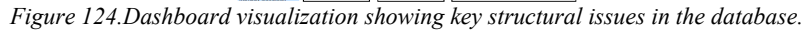
metrics, and emerging threats to data security. Their remit also entails issuing recommendations, instructing corrective actions, as well as taking charge of formal audits.

Strategic decisions create a culture of accountability and heightened awareness by way of an annual mandatory training offered to all users. The training deals with issues of PHIPA compliance, handling consent for data, breach response procedures, and composite data classification policies. This provides assurance that all players, from the developer to the clinician, approach the database with responsibility and ethics, thereby assuring safe, efficient, as well as compliant digital health operations.

5.3.9 Modernized Healthcare Database Architecture for e-Hospital [1]

The database for the e-Hospital platform was meticulously redesigned and documented by Khushi Anil Patel to transform the system into a secure, scalable, and regulation-compliant data infrastructure supporting modern healthcare operations. This relational SQL schema was fully refactored to ensure referential integrity, metadata consistency, and alignment with PHIPA and PROMIS standards. Over the course of the Summer 2025 term, more than 25 tables were created or updated across modular domains such as clinical services, AI diagnostics, PROMIS outcome tracking, billing and insurance, and data governance. Each table was normalized to eliminate redundancy and equipped with metadata attributes like `created_on`, `last_updated`, `status`, and `auth_level` to support lifecycle management and auditability. Foreign key constraints were enforced across over 50 relationships, linking critical entities such as patients, doctors, prescriptions, and billing records. Key enhancements included the addition of `users`, `audit_log`, and `password_resets` tables to introduce secure authentication and accountability, while AI-related modules such as `stroke_prediction`, `diabetes_analysis`, and `heart_disease_analysis` were expanded with fields for model versioning and clinical traceability. The PROMIS module was designed to enable quantitative tracking of patient-reported outcomes through structured domains, measures, and results tables. Additionally, a chatbot-integrated billing workflow was introduced through the creation of `services_catalog`, `billing_records`, and `billing_services` tables, enabling real-time invoice generation for insured and uninsured patients. The entire schema was deployed on AWS MySQL, ensuring high availability, disaster recovery, and seamless integration with backend APIs. Governance-driven documentation was maintained throughout the redesign, establishing version-controlled schema definitions, standardized naming

Feature	Count of Values containing Null Values	Count of Values containing NonNull Values	Count of Values containing Null Values	
chick_race	21	50	21: chick_race 50: chick_race 4: chick_race 1: chick_race 20: chick_race 60: chick_race	
chick_sex	28	22		
chick_age	4	32		Null Value 20
chick_weight				
chick_length				



To conclude this technical report, it is essential to take a moment to reflect on the advanced features of the e-Hospital website and its potential relevance and importance when implemented at various hospitals across Canada and possibly the US. This report summarizes all the features and functionalities, including every detail and button, across the various sections and portals within the e-Hospital. This work will be pivotal to future developers and groups working on the e-Hospital, as it contains all the necessary information they may need, as well as convenient references to previously written reports that detail how each section was developed. This will save a lot of time and allow for the focus to be towards improving the site rather than spending months figuring out what is missing and what information may be needed.

As mentioned at the beginning of this document, this technical report serves as a comprehensive reference to all the features and functionalities of the e-Hospital, encompassing frontend documentation, backend documentation, and database documentation. Since the e-hospital is a medical website that is always open for improvements and added features, there will always be more information to include in such a technical report. For future work, we

strongly recommend that groups continue to update this document to make it a valuable resource for future teams. Other suggestions would be to separate the portals into their own technical reports where each feature would have its own section with all three of the frontend, backend, and database documentations in one location to facilitate the understanding of the functionality of each feature in each portal and reduce the amount of effort needed for future designers and developers to find the necessary information prior to making any change or improvement to a section of the e-Hospital.

8. References

- [1] 'UI_UX eHospital Final report Template.docx'.
- [2] 'ELG5109_Final report_Ayushi.docx'.
- [3] 'ELG5901 Electrical Engineering Project - Final Report Renfei Gao.docx'.
- [4] 'IEEE_Report_ELG_6131_ehospital_Mobile_APP.docx'.
- [5] 'Final Report_EHospital_Summer2025[86].docx'.
- [7] 'APIs Documentation.docx'.
- [7] 'DTI 6997 Research Project Final Report EHospital Aurovindhya Srinivasan.docx'.
- [8] 'ELG5111_FinalReport_Group18.docx'.
- [9] 'Final Report Medical Diagnostics.docx'.
- [10] 'Final_Report AI Treatment Plan.docx'.
- [11] 'ML Model Evaluation Final Report (1).docx'.
- [12] 'BMG5109_xml (1).docx'.
- [13] 'Chatbot for eHospital Diagnosis Module.docx'.
- [14] 'ELG-6131_Data_Science_IEEE_report.docx'.
- [15] 'ELG6131 Final Report - Group 16.docx'.
- [16] 'ELG6131_Group7_Report (1).docx'.
- [17] 'ELG6131-Final Report-Group12.docx'.
- [18] 'FeedbackManagementSystem-Group13.docx'.
- [19] 'Final report (1).docx'.

- [20] 'Final report RAG.docx'.
- [21] 'Final report-medicine management.docx'.
- [22] 'Final_report-1-1.docx'.
- [23] 'MongoDB Documentation.docx'.
- [24] 'MySQL Documentation.docx'.
- [25] 'Vishwadath Ayyappan - Final Report.docx'.
- [26] ' Final Report db Hoa.docx'.
- [27] ' E-Hospital_Data_Governance_Policy[33].pdf'.
- [28] 'Project Report IEEE format (AutoRecovered).docx'.
- [29] 'DTI6302_Final report.docx'.
- [30] 'DTI6302_Group6_Final_Report.docx'.
- [31] 'DTI 6302 Final Report AS_KZ.docx'
- [32] 'GNG5300 Intelligent Health Informatics Final Project Report.docx'.
- [33] 'Project Report IEEE format (AutoRecovered).docx'.
- [34] 'Project Final Report Group 9.docx'.
- [35] '6302 Final.docx'.
- [36] 'Team3_Report_LLM-based Scheduling System for e-Hospital.docx'.
- [37] 'Medical Prescription Parsing Report.docx.'
- [38] 'RS Final Report IEEE.pdf'.
- [39] 'Aynaz_Final_Report (1).docx'.
- [40] 'elg7118-surgical-robotics-final-paper-jonathan-horton-evelyn-wattenbarger (1).docx'.
- [41] 'Robotics_Surgery_final_report (1).docx'.

Enhancing the e-Hospital Platform: Product Management and Quality Assurance Testing

1. INTRODUCTION

Electronic Medical Record systems have transformed contemporary healthcare with up-to-date real-time data access, streamlined documentation, and enhanced care coordination [1]. e-Hospital is an electronic health platform capable of managing simple functionalities like scheduling of appointments, video consults from remote locations, and remote prescription, with provider and patient remote access within a secure platform [2]. All these functionalities are augmented with eHealth innovations like telemedicine and digital workflows, where patient data can be accessed by clinical teams anywhere, at any time [2].

While e-Hospital has a good base for electronic care, it can be enhanced to the level of such a top platform as TELUS Health. With a doctor-guided tour of TELUS Health Portal and comparative review, there would be three high-priority things to enhance that needed to be added: (1) there is no predictive search engine that can discourage clinical navigation; (2) there are no dynamic clinical note templates to quickly and standardize documentation; and (3) there are AI-suggested treatment plans with lacking explanatory logic—preventing physician trust and clinical clarity. This report shows product management and quality assurance (QA) work that was performed to drive these important functionalities of the e-Hospital site. Conducting manual testing, usability inspection, and like-for-like comparison with TELUS Health, this project defines an agenda to enhance clinical appropriateness, efficiency, and usability of the site based on real-world health requirements.

2. BACKGROUND

With healthcare institutions moving towards digitalization of medical services, programs like e-Hospital aim to streamline clinical processes by integrating appointment booking, video consultations, and electronic prescribing into one digital system [3]. Such systems have the

potential to increase efficiency, expand access, and enable greater communication between healthcare professionals and patients through distant interaction and centralized data management [4]. However, to effectively cater to clinical needs, such platforms should be rigorously examined and continuously upgraded through structured quality assurance (QA) processes and evidence-based product management practices [5].

Strong QA for healthcare IT is not only a matter of discovering software bugs and usability issues, but also of assuring safety, reliability, and end-user satisfaction ahead of broader deployment [6]. Concurrently, activities around product management must prioritize the exploration of actual user needs in the real world and benchmarking against current EMR platforms to establish areas of differentiation and improvement [1]. This approach ensures platform development to be data-driven and workflow-aligned for providers.

The current study provided an in-depth competitive analysis of TELUS Health, a top Canadian electronic medical record (EMR) system [7]. This analysis was guided by a direct interview of an Ottawa, Ontario-based practicing physician who offered pertinent insights into the daily use of the TELUS Health physician portal. The physician emphasized several features that could either be duplicated or improved within the e-Hospital platform, including built-in prescription safety reminder notifications (including alerts in the form of yellow or red flags to indicate contraindications attached to prescribed medication), customizable clinical note templates (including referral letters, recording of red flags, sick leave records, and tracking of adverse effects), and efficient predictive search features that facilitate querying of large datasets. One particular area that needs improvement relates to the treatment recommendation module based on artificial intelligence. Compared to TELUS Health, which at times incorporates transparent reasoning or alerts to augment clinical decision-making, the current AI-based suggestions in the e-Hospital system lack transparency, thus compromising their reliability in a critical environment [7], [8]. The inclusion of descriptive tooltips or decision trees in this feature would significantly improve its clinical relevance.

One such perennial problem found relates to the speech-to-text function. Both TELUS Health and e-Hospital offer this functionality, but this feature presented inconsistencies in medical terminology interpretation (e.g., incorrect writing of “metformin”), leading to delays and

higher documentation error likelihoods [7], [8]. The physician stressed domain-specific speech modeling as well as real-time correction functionality to improve the usefulness of this tool within clinical settings.

Results were such that quality assurance tests carried out for this project underlined not only basic system operations but also higher-level attributes. The tests considered platform responsiveness, user interface, and accuracy in important areas such as prescription writing, clinical reporting, and underlying artificial intelligence module reasoning. The feedback mechanism that operated continuously—inclusion of manual evaluation along with physician-guided criteria—allowed for issues to be pinpointed that need to be corrected, thus augmenting usability, workflow quality, and clinical safety as part of the e-Hospital scenario.

3. SOLUTION & DESIGN

3.1 Product Management Role: Analysis and Prioritization of Feature Differences

Following an analysis of competitors and a direct gathering of information from a family doctor based in Ottawa, Ontario, several weaknesses were found in the e-hospital physician portal. The existing functionalities provided by TELUS Health were considered to be potential leaders for improvement because they have the ability to streamline clinical workflows, minimize administrative tasks, and enhance patient care outcomes.

3.2 Clinical Note Templates [9], [10], [11]

In continuation with the feature enhancement activities, a clinical interview was performed with a family physician practicing in Ottawa, who might be considered a daily end user of the TELUS Health platform. The walkthrough included demonstrations of various note templates used for multiple clinical functions by the physician (see Figure 3.1). The templates were intended to facilitate rapid documentation, provide uniformity in patient records, and lessen manual entry during periods of high clinic operational load. With consideration of such professional input, four major note templates became the focal point for integration into the e-Hospital platform as a means of furthering workflow efficiency and documentation quality.

- **Referral Letter:** This template serves to assist a physician to rapidly generate a detailed referral note for forwarding a patient to a specialist. As explained by the physician, it has standardized fields for patient history, relevant clinical findings, and current medications. Integration of this template within e-Hospital would aid in further continuity of care and save time for physicians who were otherwise repeating information in the referral letter.
- **Sick Note:** The physician explained this template is commonly used to issue work or school absence documents. It creates a standard format for such documentation and maintains clarity for recipients from third parties. The inclusion of this template facility within e-Hospital can better ensure a frequently requested service and hence make physician visit completion smoother.
- **RedFlag:** This template is to highlight urgent clinical warnings, such as suicidal ideation, abnormal neurological findings, or severe chest pains. The doctor said it is a vital tool during triage and safety documentation that marks high-risk patients and fast-tracks their management. Using this template within e-Hospital would improve clinical prioritization and risk communication.
- **Side Effects:** This template is to report an untoward reaction to given medication. The doctor stated that such entries are routinely looked at also during follow-ups and thus prevent future prescription errors. The implementation of this feature within the e-Hospital would go a long way in ensuring pharmacovigilance and patient safety.

These four templates represent a sturdy starting point with which to enhance documentation capabilities.

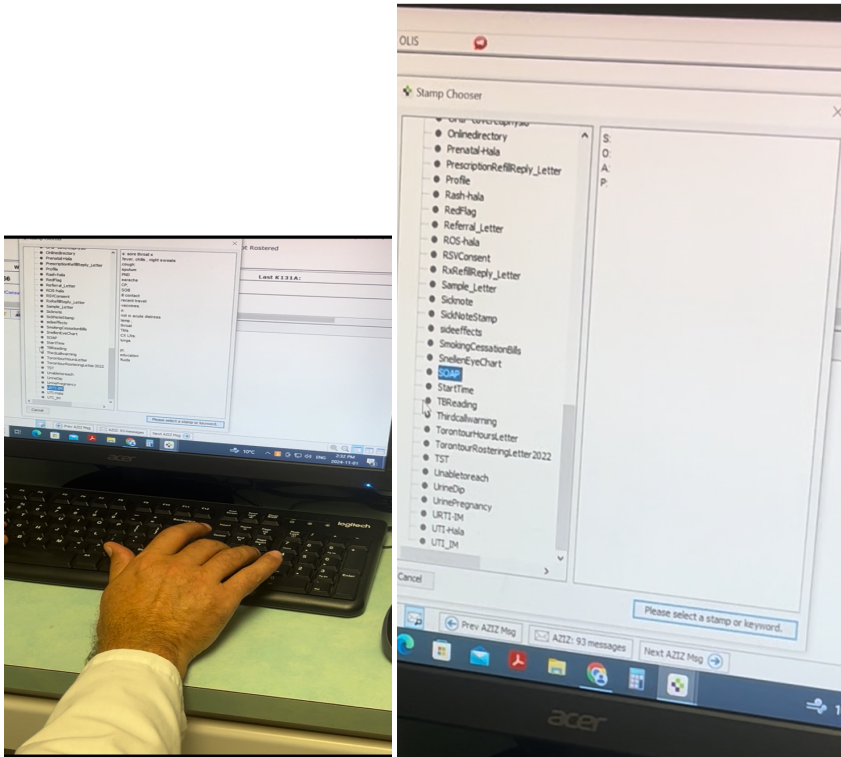


Figure 3.1. Clinical note templates used in TELUS Health platform, as demonstrated by the physician during the interview.

3.3 Predictive Search Functionality [12], [13]

The introduction of predictive search functionality into the e-Hospital platform was expected to reduce clinical inefficiency and input errors. After the user has typed in at least three characters, suggestions are provided in real time for medications, diagnoses, or other clinical terms. The clinician pointed out that this function is among the most effective and time-saving features in TELUS Health during the clinical interview and is especially efficient in daily practice when looking up medication names, selecting dosages, or looking up diagnostic codes.

Thus, in the TELUS Health interface, when a user types "Cele," the system pops up with a list of relevant medicine names like "Celebrex," accompanied by strength and formulation details. Likewise, partial diagnostic terms bring standardized diagnostic codes and their descriptions for precise selection and minimizing errors brought about by the typos or clinical documentations. The physician commented that the function has dramatically lessened the mental load involved in manual data entry and, since then, has greatly fostered standardized documentation across patient records.

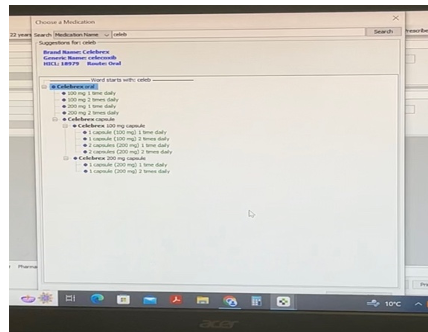


Figure 3.2. TELUS Health search engine functionality showcasing predictive text capabilities, where typing "Cele" generates a list of medications, including Celebrex with various dosage options, enhancing efficiency in clinical workflows.

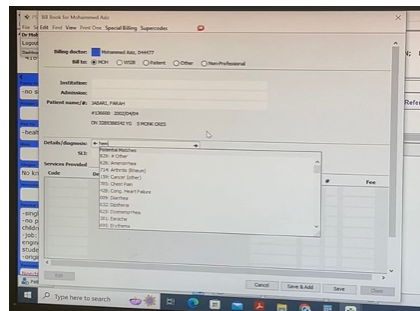


Figure 3.3. TELUS Health search engine displaying diagnostic options and their corresponding codes when partial text is entered, streamlining the process of selecting accurate diagnoses for clinical documentation.

It assists physicians to locate patient history, clinical guidelines, and laboratory tests easily from a single location. It frees up time for doctors by allowing them to search valuable information with fewer pages or menus, and hence, their work becomes less cumbersome and they attend to patient visits better.

Therefore, implementing predictive search with the e-Hospital platform would considerably enhance the system's usability. This practice would place it at the forefront of the industry and from the physicians' standpoint, simply expedite their clinical task at hand and guarantee accurate data entry every time. From a quality assurance viewpoint, this goes to reduce possible delays, to say nothing of the fact that it promotes patient safety by way of increased accuracy and puts e-Hospital's user experience on par with well-established systems such as TELUS Health. By proven capability, predictive search integration falls immediately behind another high-priority call for enhancement in user satisfaction and clinical performance.

3.4 AI Treatment Plan [7], [8], [14]

The AI treatment plan module in the e-Hospital platform acts as a decision support tool, by generating recommendations for medications and treatments that align with the patient's symptoms and history. This part of the module can make a significant difference in clinical workflow and allow for more rapid data-driven decision-making. The module generated useful pathways for appropriate treatment pathways and was especially helpful for more common conditions, during the quality assurance testing. The main usefulness of the module will come from how the proposed suggestions will enhance physician decision-making process with the inclusion of diagnostic considerations, medication history and clinical guidelines.

However, during the clinical interview, the physician indicated that in TELUS Health, a similar system has indicators that were transparent in terms of safety - red and yellow flags. A red flag is shown any time there is a contraindication found between a new prescription and the existing prescriptions and allergies of a patient. In the case of a red flag, the physician cannot proceed without overriding the system explicitly and stating their reasons. A yellow flag is shown whenever the prescription of two medications with overlapping active ingredients are prescribed and it acts as a prompt for a safety check for the physician to follow before proceeding with a conditional warning. Overall, this transparency and the level of clinical safety

was identified by the physician as a necessary component of the overall decision-making process in TELUS Health.

Figure 3.4 illustrates how TELUS Health presents these safety alerts in a graphical manner, providing immediate and clear feedback to the physician. These alerts not only enhance patient safety but also enhance physician trust in the clinical support tools of the platform.

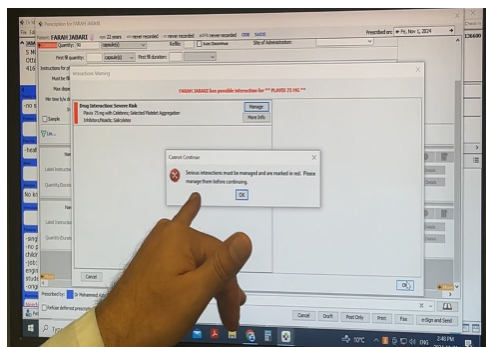


Figure 3.4 (a). The figure identifies where the system flags a red alert if two prescribed medications are known to interact and thus the physician needs to manually review and rectify it.

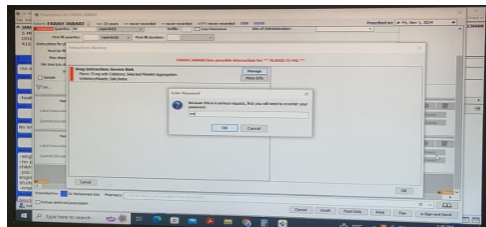


Figure 3.4 (b). TELUS Health interface prompts the physician to enter a justification code as part of addressing flagged drug interactions to ensure careful review and rationale documentation.

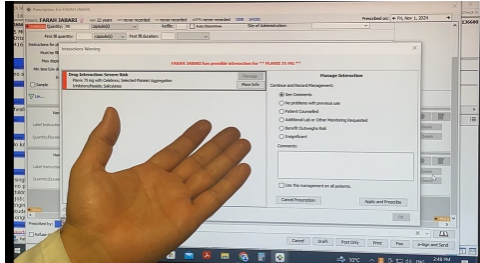


Figure 3.4 (c). TELUS Health interface prompts requiring the physician to key in a detailed explanation and justification for proceeding with flagged drug combinations, increasing safety by forcing careful consideration.

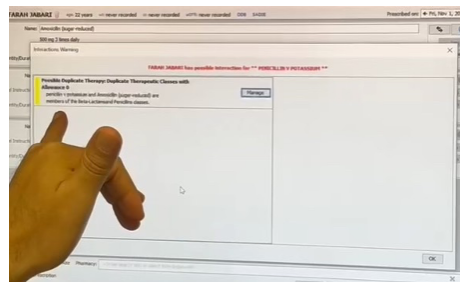


Figure 3.5. TELUS Health system showing a yellow alert indicating to the physician that there are similar medications ordered and the physician must justify why they are not substituting one medication for another. This feature supports patient safety through careful consideration of medications prescribed.

Conversely, although the AI recommendations of the e-Hospital platform are usually correct and helpful, the logic and safety checks behind them are not transparent to the physician. The system, for instance, might internally adjust for contraindications or redundant drug interactions, but the information is not displayed in a way that is easy to see at the point of care. Therefore, the physician has to rely on the AI without knowing the reason for the recommendation. In terms of usability and safety, this can be enhanced.

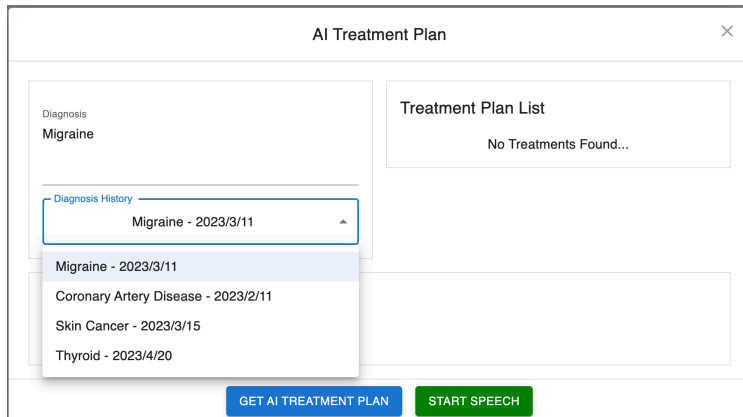


Figure 3.6. AI treatment plan of the e-hospital portal.

To facilitate clinical transparency and build confidence in the AI module, we suggest incorporating expandable tooltips or hover-based popups describing the rationale for each recommendation. These may comprise evidence-based guidelines, condition-to-treatment logic, patient-specific considerations (e.g., allergies, history), as well as interactions that were considered. This improvement will not only reinforce the physician's trust in the AI module but also enable better-informed and safer clinical decisions.

This enhancement sets e-Hospital to surpass TELUS Health by merging strong backend safety checks with transparent, interactive explanations—establishing a more complete and intuitive AI assistant for doctors.

3.5 Speech-To-Text Functionality [15], [16]

Testing the speech-to-text component embedded in the e-Hospital platform applied via the chatbot was one of the most crucial features during the quality assurance process. The purpose of the speech-to-text feature built into the e-Hospital platform through the chatbot is to ease the documentation burden off of physicians by allowing them to speak aloud clinical notes, prescriptions or patient instructions in one or two general commands. During the hands-on testing sessions it was clear that the tool was not currently optimized for medical needs. The tool regularly misrecognized medical terminology, for example, the most commonly prescribed drug

acetaminophen, was consistently misidentified as the drug acetamophol. Some diagnostic terminology like migraine or gastroenteritis were not recorded as intended, consistently misidentified, or altogether omitted. These misidentifications highlight the limitations of the current system and raise the concern for clinical risk if these misidentifications are not resolved.

When we followed up with a family physician who uses TELUS Health in his day-to-day clinical practice, he was able to corroborate these findings. He indicated that TELUS has speech-to-text capacity but rarely uses it since it has difficulty consistently with complex medical language and structured clinical input. Therefore, physicians will sometimes opt out of using speech-to-text tools, and will, instead, manually write in data to avoid transcription errors that could impact patient safety or requiring additional time to correct. As physicians face similar challenges across platforms this exemplifies a bigger industry need for a speech engine with a health-care context.

Considering the correctness issues we observed, the existing speech-to-text system in e-Hospital is poorly suited for clinical environments. To help enhance its appropriateness, we would recommend switching or accompanying the current speech-to-text system with a model trained on medical domain speech. An example is OpenAI's Whisper model, which has been fine-tuned with clinical datasets, while another example is Google's Healthcare Speech-to-Text API. Usability testing should also take place utilizing actual clinical scripts, as well as having real-time inputs from a physician while using speech-to-text input in patients' charts to ensure operational reliability of the feature in practice. Real time data error detection and prompting the physician for confirmation may also enhance the safety and trust around this feature. With these adjustments and enhancements speech and text can be used as a value add to improve efficiency and ultimately enhancing physician well-being by addressing cognitive load.

3.6 Unified Interface Prototype Design

In order to integrate the suggested enhancements of the e-Hospital doctor dashboard such as clinical note templates, predictive search, and AI treatment plan transparency, an interface prototype was created. With the purpose of improving the overall experience, the interface was designed to better integrate with the workflows of clinical users as physicians and was informed by clinical user feedback, as well as comparisons with TELUS Health and similar systems. As

seen in the TELUS Health interface, the prototype incorporates an intuitive, well-organized, and reflexive dashboard to systemized streamlining key functions which underscore the system's core value to user systems.

A search bar for predictive medication lookup was added to the top right corner of the interface. With this feature, clinicians can search for medications, diagnostic codes, clinical notes, and even past patient records using only a few characters. One of the interviewing physicians pointed out that this feature, along with many others, greatly enhances the ease of entry for documentation and retrieval of patient data. By ensuring e-Hospital users have faster, error-free documentation, the system will be easier to navigate and provide standard data port entry.

There is now a Show Plan Logic button in the AI Treatment Plan section beneath the search bar. This addition supports the physician's need to understand the basis behind the AI recommendation, thereby providing reasoning for the disclosure. TELUS Health, for example, displays safeguarding alerts like substantively duplicate ingredient alerts and, more importantly, harmful drug interaction alerts. e-Hospital's AI reasoning is, however, hidden at the moment. If e-Hospital would like to reach and surpass industry peers in clinician decision support, the addition of reasoning and guideline tooltips for treatments would be a starting point to e-Hospital logic explanation transparency.

On the right side of the treatment plan is the Notes section. This section contains a list of templates which are titled SOAP, Referral Letter, SickNote, SideEffects, and RedFlag. This list is by no means exhaustive but captures template fields that doctors within the interviewed clinic frequently used. This directly augments documentation within the clinic, by decreasing time to write and enter data, enhancing interoperability within the shared clinical systems.

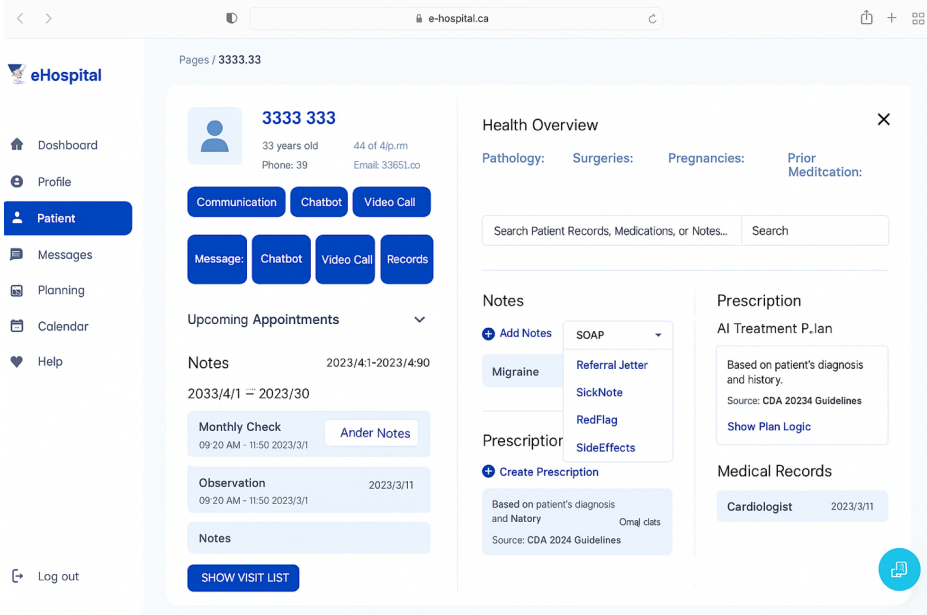


Figure 3.7. Unified Interface Prototype Design of the e-Hospital Platform Showing Integrated Clinical Note Templates, Predictive Search, and AI Treatment Plan Logic Functionality.

4. RESULTS [8], [9], [10], [11], [12], [13], [14], [15], [16]

The testing for quality assurance, as well as user feedback analysis offered both interesting, but also illuminating insights. While the e-Hospital platform has basic functionality for appointment scheduling, video consultations and prescription writing, there are a number of clinical areas for improvement that can improve its utility and usability for providers. The clearest insights were from comparing it directly to TELUS Health, which was summarized by a practicing family physician following a clinical walkthrough interview.

Predictive search was one of the most well-known features that came out of the comparison. As the physician demonstrated with TELUS Health, as soon as three letters were entered, e.g., “Met”, a dropdown with all of the appropriate drug names or diagnosis codes would populate. This means that doctors can quickly select the medication or condition without

having to fully type often long or complicated terms. The physician highlighted how this allows them to save time and limit typing strains, especially at busy clinical hours. To contrast, with e-Hospital the physician was required to manually enter each term and there was no autocomplete or intelligent suggestion features. This delivered inefficiencies and introduced the risk of making a typo. Implementing a similar functionality to a predictive engine within e-Hospital would streamline logging, increase fidelity of records, and increase user experience.

Another significant opportunity for improvement is the clinical note template system. During the interview the physician reviewed a variety of built-in documentation templates on TELUS Health. There were structured templates titled Referral Letter, Sick Note, RedFlag, and Side Effects. The physician described the use of each of these templates and how the templates could be used in day-to-day practice, for example, utilizing the RedFlag template to document the urgent symptoms requiring escalation, or for recording Side Effects to document any adverse reaction to medication. Due to this feedback we planned to select these four templates for e-Hospital to correspond to common clinical documentation workflows; by using the proposed templates we would not only increase our standardization of records but also help limit the repetitive manual formatting during the documentation processes.

A third significant observation pertains to the oversight regarding the AI Treatment Plan interface. Although e-Hospital's artificial intelligence model does provide automated recommendations for either diagnosis or medications, the provided recommendations do not have the rationale or description regarding their options, decision-making, or evidence base. Reliability can be better than a simple recommendation that e-Hospital suggested. The physician acknowledged that in TELUS Health, the AI model recognizes drug conflicts in real time through the use of flags—a red flag if the drug interaction was contraindicated, a yellow flag if the drugs had duplicate active ingredients across many prescriptions. Critically, TELUS also did not allow a physician to prescribe until they dispelled the drug conflict warning, in essence providing another layer of protection against safety risks. In contrast, e-Hospital's AI seems to be by design avoiding these types of blockages, reducing some transparency. In the original prototype of the e-Hospital, a best upgrade was planned to provide a “Show Plan Logic” button, which would design tool tips, or expanders, that would show the logic behind each AI recommendation. This would enhance trust and decision making for the clinician by providing

the rationale or evidence behind their suggested action such as the condition-match, treatment rationale, or evidence source.

Lastly, speech-to-text testing on the e-Hospital chatbot interface was conducted, incorporating actual clinical language such as medication names (e.g., acetaminophen) and common diagnoses. The system frequently misrecognized, generating things like acetaminophen as acetamophol. The doctor shared that even TELUS Health's system has this problem, and as a result, is not used much in clinical practice, as it is not accurately transcribing the information needed to improve integrated management. Given these limitations, it seems that real-world applications of healthcare must be addressed with models that are trained using medical data specifically. Currently, the e-Hospital speech recognition module is underutilized for clinical contexts. The next step that is suggested is to replace or add to the engine with a model trained in the medical domain, and conduct usability testing using live scripts in order to enhance its reliability and accuracy in transcription and context.

The results clearly indicate that e-Hospital can support the day-to-day functional needs of clinical practitioners having regarded some key improvements—predictive search, standardized templates, transparency in AI (and other integrations), and speech recognition—meaning it could be a viable EMR if improved upon to achieve improved overall performance.

5. DISCUSSION

The results from the quality assurance exposure indicate that the e-Hospital platform displays core functional stability, especially in key feature modules, however, continued revisions are needed to ensure better clinician workflows. During this work, four enhancements were identified, as follows, as they are impactful: note templates, improvement of speech -to-text accuracy, predictive search, and AI transparency to provide explainable logic.

First, it was discovered that, intended for streamlined documentation, speech-to-text transcription, performed poorly in both e-Hospital and TELUS Health [7], [8]. In the QA testing of e-Hospital, this system often misidentified medical terminology like "acetaminophen" and even misclassified standard diagnoses [8]. These are similar to the physician's comments that in

TELUS Health they actively avoided using the TELUS Health speech tool due to these problems [7]. Studies following W-T-E showed that general ASR (Automatic Speech Recognition) engines have difficulties with clinical specific vocabulary and exhibit average word error rates (WER) for medical use cases unless trained specifically [17]. To improve this situation in e-Hospital we recommend changing the existing engine and model to a specific domain-trained model (e.g., a fine-tuned Whisper or Google Healthcare API) and conduct usability testing using clinical scripts [18], [19]. These engines and models are performing well below expectations for clinical use—the recommended action is to replace or enhance the existing engine with a medical-domain-trained engine and usability test using actual US institutional clinical scripts [17].

Secondly, the predictive search function - not implemented within e-Hospital - was described by the physician as essentially the most powerful tool available from TELUS Health [7]. By typing just three letters, like "Met," the predictive search produces live suggestions for medications, such as Metformin, dosage selections for the medication, and diagnostic codes [12], [13]. Positive impacts of predictive search on lookup time, accuracy and cognitive load are impressive [12]. While there is growing evidence that transformer architecture powered predictive search systems (e.g. BERT or Med-BERT) can demonstrate increased documentation speed and standardization in EHRs, including e-Hospital will benefit the overall user experience and improve engagement with clinical workflow [13]. There is also potential for this function to be used more broadly - not just for medications or diagnoses, but for quickly finding patient notes, lab tests and clinical guidelines [13].

Third, AI treatment plan transparency was an important concern. As it stands, the e-Hospital platform provides AI generated recommendations that are not supported by any visible justification [8]. TELUS Health, on the other hand, has embedded red and yellow flags to which alerts prescribing concerns; for example, a red flag indicates a drug contraindication and a yellow flag show duplicated active ingredients [7]. The presence of flags improves prescribing safety and has doctors manually verify and/or overwrite flagged recommendations, therefore adding much-needed clinical validation [7]. The e-Hospital AI clinical decision support module may provide efficacious recommendations, but physicians' trust is limited when reasoning is invisible [8]. To facilitate usability and transparency, we proposed a tooltip called "Show Plan

Logic," which when clicked, produces the rationale for each recommendation—this rationale may include diagnosis match, evidence type, or previous patient history [8], [14]. This function will not only support physician understanding of recommendations but will also support adherence to clinical decision-making standard compliance [14].

Finally, the most significant enhancement is the clinical note templates. During the TELUS Health demonstration, the physician highlighted the many templates that can be used to improve the documentation process [7]. We chose four templates from the list that were commonly used and clinically relevant for e-Hospital:

- Referral Letter: Enables quick structured letter writing for specialist referrals, which saves time and increases standardization [9].
- Sick Note: A document that patients often request for work or school absence, with frequent use in primary care settings [10].
- RedFlag: Provides quick notation of serious or urgent symptoms, thereby providing a better and safer process for triaging the patient [20].
- Side Effects: Identifies and documents an adverse drug reaction, which can support follow up and reporting [11].

All of these templates were identified by the physician as essential to everyday life. Incorporating them into the e-Hospital would immediately increase clinical utility and lessen the burden of the administrative workload on providers [8]. Research on template-driven documentation found improved completeness of documentation and less variance among different users. Our design of the interface (see Figure 3.7) has placed these templates into a drop-down menu in the physician's notes section- representing a proven workflow and reducing writing time for notes.

6. FUTURE WORK & CONCLUSIONS

This report has shown that the e-Hospital platform is operationally functional and technically robust across its core modules; however, a number of feature enhancements are required to bring the system in line with clinician expectations using more sophisticated EMR systems such as TELUS Health [1], [7], [8]. Enhancements such as predictive search functionality, template based clinical note documentation, and explainable AI treatment plans served as core to this term's product improvement plan as they were significantly prompted by a clinical interview with a practicing physician and development testing for quality assurance through system workflows.

In future work, enhancing speech-to-text capabilities is certainly a priority [15], [16]. Our tests revealed inaccuracies in recognizing medications and diagnoses, as did feedback from physician participants. Using ASR engines that are trained on medical domains, such as Google's Medical Speech API or Whisper-Med, could improve transcription accuracy and usability in practice [17], [18], [19].

Next, we should enhance and improve the predictive search engine. By employing NLP models, such as BioBERT, to auto-suggest diagnoses and medications, the system could potentially provide suggestions after a few letters are typed [12], [13]. This would drastically reduce the time taken to search, and potential errors, as demonstrated in TELUS Health [12].

We would also like to improve the AI treatment plan module by creating transparent logic [8]. Even though the system suggests valid recommendations, it gave no feedback with explanations [8]. To address this concern, the addition of a "Show Plan Logic" button would allow physicians to see the rationale and evidence behind the AI-generated recommendations and increase clinical trust [14]. The TELUS Health safety flags (red indicating contraindications and yellow indicating duplicate ingredients, etc.) serve as an important model for implementing similar safeguards [7].

Lastly, clinical note templates are important as well. This semester, we focused on four high-use templates: Referral Letters, Sick Notes, RedFlag alerts, and Side Effects [7], [9], [10], [11]. More templates can be created in the future for follow-up visits and chronic care, for example [20]. We know that structured documentation can reduce variability and improve the completeness of the clinical note from studies [9].

In closing, strategic improvements to search solutions, documentation, and AI logic; along with regular QA testing and a user-centered design will allow e-Hospital to evolve into a clinician-friendly, efficient, and safe digital health solutions platform.

7. REFERENCES

- [1] M. Lee, K. Kim, Y. Shin, Y. Lee, and T.-J. Kim, “Advancements in electronic medical records for clinical trials: Enhancing Data Management and Research efficiency,” *Cancers*, vol. 17, no. 9, p. 1552, May 2025. doi:10.3390/cancers17091552
- [2] “EHealth Solutions,” eHealth Solutions - Siemens Healthineers Canada, <https://www.siemens-healthineers.com/en-ca/digital-health-solutions/ehealth-solutions> (accessed May 30, 2025).
- [3] R. Agarwal, G. (Gordon) Gao, C. DesRoches, and A. K. Jha, “research commentary—the digital transformation of healthcare: Current status and the road ahead,” *Information Systems Research*, vol. 21, no. 4, pp. 796–809, Dec. 2010. doi:10.1287/isre.1100.0327
- [4] M. B. Buntin, M. F. Burke, M. C. Hoaglin, and D. Blumenthal, “The benefits of Health Information Technology: A review of the recent literature shows predominantly positive results,” *Health Affairs*, vol. 30, no. 3, pp. 464–471, Mar. 2011. doi:10.1377/hlthaff.2011.0178
- [5] C. S. Kruse, C. Kristof, B. Jones, E. Mitchell, and A. Martinez, “Barriers to electronic health record adoption: A systematic literature review,” *Journal of Medical Systems*, vol. 40, no. 12, Oct. 2016. doi:10.1007/s10916-016-0628-9
- [6] E. M. Borycki and A. W. Kushniruk, “Health Technology, quality and safety in a learning health system,” *Healthcare Management Forum*, vol. 36, no. 2, pp. 79–85, Dec. 2022. doi:10.1177/08404704221139383
- [7] “Health and Wellbeing Solutions: Telus health,” TELUS, <https://www.telus.com/en/health> (accessed May. 31, 2025).
- [8] “EHospital,” eHospital, <https://www.e-hospital.ca/> (accessed Jul. 31, 2025).

- [9] G. Scaioli et al., “Communication between General Practitioners and medical specialists in the referral process: A cross-sectional survey in 34 countries,” *BMC Family Practice*, vol. 21, no. 1, Mar. 2020. doi:10.1186/s12875-020-01124-x
- [10] L. A. Baumann, J. Baker, and A. G. Elshaug, “The impact of electronic health record systems on Clinical Documentation Times: A systematic review,” *Health Policy*, vol. 122, no. 8, pp. 827–836, Aug. 2018. doi:10.1016/j.healthpol.2018.05.014
- [11] J. J. Coleman and S. K. Pontefract, “Adverse drug reactions,” *Clinical Medicine*, vol. 16, no. 5, pp. 481–485, Oct. 2016. doi:10.7861/clinmedicine.16-5-481
- [12] MLR, <http://proceedings.mlr.press/v126/gopinath20a/gopinath20a.pdf> (accessed Apr. 5, 2025).
- [13] K. Kawamoto, C. A. Houlihan, E. A. Balas, and D. F. Lobach, “Improving clinical practice using clinical decision support systems: A systematic review of trials to identify features critical to success,” *BMJ*, vol. 330, no. 7494, p. 765, Mar. 2005. doi:10.1136/bmj.38398.500764.8f
- [14] R. T. Sutton et al., “An overview of clinical decision support systems: Benefits, risks, and strategies for Success,” *npj Digital Medicine*, vol. 3, no. 1, Feb. 2020. doi:10.1038/s41746-020-0221-y
- [15] L. Zhou et al., “Analysis of errors in dictated clinical documents assisted by speech recognition software and professional transcriptionists,” *JAMA Network Open*, vol. 1, no. 3, Jul. 2018. doi:10.1001/jamanetworkopen.2018.0530
- [16] “Master dictation spelling: Tips & Techniques for accuracy - SCRIBEMD - automate your medical notes,” ScribeMD, <https://blog.scribemd.ai/master-dictation-spelling-tips-techniques-for-accuracy/> (accessed May. 20, 2025).
- [17] M. Johnson et al., “A systematic review of Speech Recognition Technology in health care,” *BMC Medical Informatics and Decision Making*, vol. 14, no. 1, Oct. 2014. doi:10.1186/1472-6947-14-94

[18] J. Suh, I. Na, and W. Jung, “Improving domain-specific ASR with LLM-generated contextual descriptions,” *Interspeech 2024*, pp. 1255–1259, Sep. 2024. doi:10.21437/interspeech.2024-377

[19] A. Radford, J. W. Kim, T. Xu, and G. Brockman, “Robust speech recognition via large-scale weak supervision,” *arXiv preprint arXiv:2212.04356*, Dec. 2022. [Online]. Available: <https://doi.org/10.48550/arXiv.2212.04356>

[20] A. Downie et al., “Red flags to screen for malignancy and fracture in patients with low back pain: Systematic review,” *BMJ*, vol. 347, no. dec11 1, Dec. 2013. doi:10.1136/bmj.f7095